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DOI:10.5055/jem.0855

Dear Reader:

We are excited to bring forth this special issue titled ***Climate Change and Sustainability in Emergency Management - Research and Applied Science in a Changing World***. Our thanks to the authors and researchers who have brought forth life-changing research that can be applied today across the globe. Emergency managers are at the nexus of the climate change-sustainability debate, whether they are aware of it or not! A blurring of the emergency managers job description and responsibilities has been in process for years, much like moss growing on the north side of trees, the change has been slow but persistent.

Climate change poses unprecedented challenges to sustainability and emergency management, demanding a paradigm shift in how we address environmental crises. As global temperatures rise and extreme weather events become more frequent and severe, the interconnectedness of climate change, sustainability, and emergency management becomes increasingly apparent. Sustainability, traditionally focused on preserving resources for future generations, now encompasses resilience-building efforts aimed at adapting to and mitigating the impacts of climate change. Emergency management, tasked with responding to disasters, must evolve to address the root causes of these disasters, including climate change-induced hazards.

Emergency managers play a pivotal role in navigating the intersection of climate change and sustainability by adopting proactive strategies to reduce vulnerability and enhance resilience. This entails integrating climate change considerations into disaster risk reduction planning, preparedness, response, and recovery efforts. By leveraging tools such as risk assessments, scenario planning, and community engagement, emergency managers can identify climate-related risks and vulnerabilities and develop tailored mitigation and adaptation strategies. Moreover, fostering collaboration among stakeholders, including government agencies, non-governmental organizations, and the private sector, is essential for building a comprehensive and coordinated response to climate-related emergencies.

The changing role of emergency managers in addressing climate change extends beyond disaster response to encompass long-term sustainability initiatives. This includes promoting sustainable development practices, such as renewable energy adoption, green infrastructure investment, and ecosystem restoration, to mitigate greenhouse gas emissions and enhance community resilience. Additionally, integrating climate change considerations into land use planning, building codes, and infrastructure development can help reduce vulnerability to climate-related hazards and enhance adaptive capacity. By adopting a proactive and holistic approach to sustainability, emergency managers can minimize the likelihood and severity of climate-induced disasters while promoting long-term environmental stewardship.

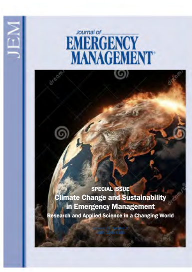
Furthermore, emergency managers must prioritize equity and social justice in climate change adaptation and mitigation efforts to ensure that vulnerable communities are not disproportionately impacted. This requires engaging with marginalized and underserved populations to understand their unique needs and vulnerabilities and incorporating their perspectives into decision-making processes. By adopting an inclusive and community-centered approach, emergency managers can build trust, empower local stakeholders, and enhance the effectiveness and equity of climate resilience efforts.

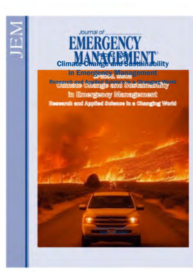
Addressing the intersection of climate change, sustainability, and emergency management requires a multifaceted and collaborative approach that integrates risk reduction, adaptation, and mitigation strategies. Emergency managers must adapt to the evolving challenges posed by climate change by embracing proactive and holistic sustainability initiatives, fostering collaboration among stakeholders, and prioritizing equity and social justice. By leveraging their expertise and partnerships, emergency managers can play a critical role in building resilient and sustainable communities capable of weathering the impacts of climate change.

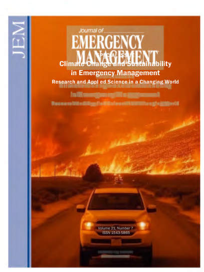
I invite your review of this special issue! Additional papers are in the final stages of peer review. As these papers are accepted for publication, they will be added to this special issue so check back frequently. The Call for Papers will remain open as we are just at the tip of the iceberg on this topic! I extend my heartfelt thanks to Deloitte for their unwavering support of this special issue. Visit their website as they have some impressive capabilities to address climate change and sustainability. *https://www2.deloitte.com/us/en/services/sustainability-climate-equity.html*

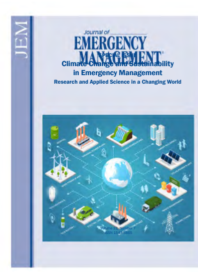
Finally, it is often difficult to pick a front cover especially for this challenging topic. As a behind the scenes view, please see several that we debated internally. We decided to have a more positive cover compared to the fire and brimstone betting on the creativity and ingenuity of humans to solve these challenges.

Very truly yours,
Richard A. DeVito, Jr.
Publisher
Journal of Emergency Management









Special Issue on Climate Change and Sustainability in Emergency Management

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Leveraging technology in emergency management: An opportunity to improve compounding and cascading hazards linked to climate change

Attila J. Hertelendy, PhD, Special Issue Editor

The Intergovernmental Panel on Climate (IPCC) Sixth Assessment report concluded that we will see an increase in frequency of extreme environmental events around the world including, hurricanes, droughts, and wildfires.[1] The report further describes cascading hazards when one hazard triggers another in a series such as extreme heat triggering a collapse of the power grid. The IPCC also discusses compounding hazards as multiple disasters occur at the same time for example a hurricane occurring at the same time as COVID-19 and a mass casualty event prompting a Urban Search & Rescue (USAR) response such as the Surfside and the Florida condo collapse.[2] Studies suggest that there are gaps relating to Hazard Mitigation Plans (HMP) in addressing cascading events.[3,4]

Extreme weather events linked to climate change require an all-of society approach, wherein mitigation efforts will increasingly require a multidisciplinary approach involving collaboration from numerous professions.[4-6] Emergency management (EM) professionals, climate scientists, and sustainability advocates are all trying to make the world a better and safer place, however, efforts to achieve this are often hampered by siloed approaches to decision making, community outreach, and fundraising.

While this special issue didn't exclusively address technology, herein lies the opportunity for additional research in emergency management both in terms of scientific advancement within the discipline and the practical application of technologies. Emergency managers should be aware of how technology can significantly augment their mitigation, preparedness, response and recovery efforts. The use of large complex datasets—big data and the advances in information systems and computing can support more effective approaches to EM and response to climate-driven extreme-weather events. On March 18, 2024 NVIDIA announced its Earth-2 climate digital twin cloud platform for simulating and visualizing weather and climate at an unprecedented scale.[7] Utilizing the power of artifical intelligence (AI) and supercomputing can potentially become a game changer for emergency managers and the field of EM overall.[8] Following are some of the ways that currently have practical use cases:

1. **Advanced forecasting and early warning systems:** High-performance computing enables the development and utilization of advanced forecasting models that can accurately predict the onset, intensity, and trajectory of extreme weather events such as hurricanes, floods, and wildfires. These models incorporate various data sources, including meteorological data, satellite imagery, and climate models, to provide timely warnings to communities and emergency responders. Early warning systems, supported by technologies such as mobile apps, text alerts, and sirens, can help evacuate populations from high-risk areas and mitigate the impacts of disasters.[9]

2. **Remote sensing technologies for risk assessment:** Remote sensing technologies, including satellite imagery and drones, provide valuable data for assessing environmental risks associated with climate-related extreme-weather events. These technologies can be used to monitor changes in land use, vegetation cover, and

sea levels, as well as to identify vulnerable areas and critical infrastructure. By mapping these risks, emergency managers can develop targeted mitigation strategies and allocate resources more effectively.[9] When combined with proprietary data owned by companies, applications like NVIDIA's Earth-2 can help deliver warning and updated forecasts in seconds compared to the minutes or hours in traditional central processing unit (CPU)-driven modeling.[7,10]

3. Communication and information systems: Technology facilitates communication and information sharing during emergencies, enabling stakeholders to coordinate response efforts and disseminate critical information to the public. Communication systems such as emergency radio networks, social media platforms, and mobile apps allow emergency managers to reach a wide audience quickly and efficiently. These platforms also provide opportunities for community engagement and education, empowering individuals to take proactive measures to protect themselves and their communities.[9]

4. Data analytics and decision-support tools: Big-data analytics and decision-support tools enable emergency managers to analyze large volumes of data to identify trends, patterns, and vulnerabilities associated with climate-related extreme weather events. Machine learning can help prioritize mitigation strategies, allocate resources, and make informed decisions in real-time. By integrating data from various sources, including weather sensors, social media feeds, and geographic information systems (GIS), emergency managers can gain a comprehensive understanding of the evolving situation and adapt their response accordingly.[11,12]

Cross-sector collaboration and innovation technology fosters collaboration and innovation across sectors, bringing together government agencies, academic institutions, private sector companies, and civil society organizations to address the complex challenges posed by climate-related extreme weather events. Collaborative initiatives, such as public-private partnerships and open data initiatives, leverage the expertise and resources of multiple stakeholders to develop innovative solutions and best practices for EM mitigation. Much of the research and development however, remains siloed at tech companies, and by computer and information systems scientists. Emergency managers will need to be proactive and take the initiative to leverage expected effectiveness and efficiency derived from technology.

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Emergency management and sustainability: Understanding the link between disaster and citizen participation for sustainability efforts and climate change

Brian Don Williams, PhD

ABSTRACT

The goal of this study is to examine how disaster experience influences local government views on citizen participation in addressing issues of sustainability, such as climate change. This study considers concepts such as wicked problems, the social order, the environment, economic development, and citizen participation where sustainability can be considered a solution to help manage and solve the challenges of disaster, like climate change. The data are taken from a 2015 International City/County Management Association national survey that examines the link between disaster and sustainability. The results show that more than half of the respondents do not view public participation as having much of an impact on sustainability; however, we can expect public participation to increasingly impact sustainability efforts as communities experience more disaster. This suggests that emergency management needs to understand public pressures regarding wicked problems, such as climate change, to collectively address the global influence of environmental, economic, and social issues that have local effects on their communities.

Key words: sustainability, emergency management, disaster experience, public participation; DOI: 10.5055/jem.0824

INTRODUCTION

In the United States (US), emergency management has evolved into a system that functions through intergovernmental, intersectoral, and intercommunity relationships to reduce the impact of hazards.[1-3] As the world becomes more connected, we find an emergency management system that must address challenges that are very difficult, if not impossible, to deal with for a single organization. Recent times have brought a spotlight to issues such as climate change as a *wicked problem* that demands inclusivity of multilevel government relationships as well as multiple organizations and community involvement, ie, government and nongovernment, through intersectoral relationships.

The purpose of this research is to better understand the influence of disaster experience on how local government understands public participation to impact the shaping of sustainability efforts. This study incorporates the three legs of sustainability, ie, environmental protection, social equity, and economic development, as these issues are directly related to the concerns of climate change with sustainability proposed as a possible solution for addressing climate change. To accomplish this goal, this research relies upon the 2015 International City/County Management Association (ICMA) data that Homsy et al.[4] partnered with ICMA to examine the links between sustainability/climate change and disaster.

The sections that follow will provide a review of the literature to define the wicked problem and the importance of the social order. This will be followed by the theoretical considerations of sustainability and climate change, disaster linkages to sustainability, and an examination of sustainability efforts with the connection to citizen participation before providing the methods and data, findings, discussion, and concluding comments. The findings here suggest that increased disaster experience is significant to increasing the impact that local governments report citizen participation to have on shaping sustainability efforts. This would suggest that the more a community experiences disaster, the more salient climate change and sustainability become to the public.

THEORETICAL CONSIDERATIONS

The wicked problem

The concept of the *wicked problem* emerged in the 1960s in the policy literature.[5] The wicked problem refers to challenges that are difficult for a single organization to solve on its own, if not impossible. Rittel and Webber[6] further define the wicked problem to embody complexities that include a basis of good or bad rather than true or false by the stakeholders based upon a lack of public tolerance for policy failure. Over the past decade, the emergency management community has embraced the term "wicked problem." It is important to note that this definition does not mean that a single organization cannot have an influence, just that the problem requires multiple organizations to act toward a solution.

Emergency management deals predominantly with problems that require an interdependent relationship among multiple organizations to access adequate resources. Recent events reveal an increase in the need for emergency management to involve multiple organizations that include the public and volunteers in a network to solve the complex problems they face in a high consequence–high impact environment.[7] This pervasiveness is evident from the proceedings of the 23rd Annual Emergency Management Higher Education Virtual Symposium where the term "wicked problems" is mentioned multiple times by representatives of the Emergency Management Institute, the International Association of Emergency Managers, and the Federal Emergency Management Agency (Region 8).[8]

Abou-Bakr,[9] more than a decade ago, emphasizes that approximately 85 percent of the critical infrastructure in the US, at that time, was owned and or operated by the private sector. This marks a visible shift from government (military and law enforcement) having complete responsibility for emergency management functions. This suggests that government in the US must dependent upon nongovernment organizations in order to effectively address issues of disaster down to the local level through the social order.

The social order

Tierney[10] proposes that rather than the impacts of disaster lying in natural or technological sources, it is the social order that produces disaster through everyday social, political, economic, and cultural realities that form the social construction of disaster. This proposal asserts that if the impacts of disaster are socially constructed and embedded in everyday life, then societies and communities can cope and sustain in the face of disaster through the social order, to absorb an impact and continue to function and recover. This gets to the core concern that Williams and Webb[11] find emergency management professionals to also define vulnerability in the social, political, and economic policies that create vulnerability and can result in disaster. Additionally, Tierney[10] proposes that social forces have the ability to determine what will happen based upon the dominant groups' perspective, which leaves less powerful groups at risk.

In the conceptualization of democratic theory, Kinzer[12] reminds us that including citizens in the planning and implementation process helps to build trust and buy-in, which in turn increases the possibility of adopting and implementing policy. This concept is also found in the emergency management realm by Williams and Webb,[7] where getting buy-in not only from elected officials but also from the community as well is important but is also a challenge for emergency managers in addressing the needs of vulnerable populations. This highlights the need to operate through the social order, which includes local industry and citizens in a community approach, to address wicked problems.

One wicked problem that cannot be solved, or at least not solved easily by a single organization, and requires consideration of the social order is that of climate change. Reid[13] proposes that, globally, communities are affected by issues such as environmental degradation and climate change. While it may not be viewed by some that a single local government has the ability to influence a global environment, collective action through sustainability policy is proposed to be a solution to address the effects of climate change. In fact, Reid[13] calls attention to the reality that, globally, national governments fail to address unsustainable practices in a comprehensive manner, while local governments have predominantly defined sustainable best practices. Other scholars, such as Svara et al.,[14] focus on the environment and climate change to address the issue of sustainability being a global concern that is pursued at the local level. The next section will discuss climate change with sustainability serving as a possible solution.

Climate change and the three legs of sustainability

To define sustainability, Wang et al.[15] work from Adams[16] and WCED as "a set of practices that address the social, economic, and environmental needs of present and future generations."[17(p842)] The study of sustainability has moved from the case study and basic understanding of sustainability to understanding the extent that local governments embrace sustainability. For example, Saha and Paterson[18] investigate the extent that local government engages in all three tenants of sustainability, ie, environment, economy, equity, to determine the level of commitment to a sustainability plan or program. This is based upon a sustainability plan or program that is comprehensive, ecologically sensitive, economically sound, and equitable for all populations rather than simply engaging in a piecemeal of activities that would have no real substantive outcomes.

Saha and Paterson[18] find that environmental aspects take precedence with a glaring absence of mention to issues of equity when discussing sustainability planning. While most cities do include sustainability principles as part of the overall disaster planning objectives, few report developing a sustainability plan. And while most plans do include sustainability activities, most local governments do not perceive the activities to be sustainability activities. While the three elements of sustainability planning and development are considered to be interdependent, Saha and Paterson[18] propose that the term sustainability remains a buzz word in practice with terms like energy conservation and green building dominating the narrative. In this narrative, little attention is given to the politics of who gets what, when, and how and policy decisions tend to result in the neglect of disadvantaged populations.

Svara et al.[14] also investigate sustainability based upon the three main avenues of environment, economy, and social equity but through the lens of socioeconomic characteristics. While heterogeneous populations of White and highly educated residents are proposed to be more likely to engage in sustainability policy, communities with higher income and increased home ownership show a reduction of engagement in sustainability policy. Overall, the findings do not support the idea of an association between engagement in sustainability policy and the municipality's socioeconomic characteristics such as race, class, or income.

Looking specifically at climate change policy, Head[19] examines Indigenous disadvantage in Australia. Head[19] proposes that Indigenous disadvantage comes overwhelmingly from economic inequalities because of a lack of ability on the part of government to agree upon how to solve the complexities of the social and economic differences. With many complexities in climate change policy, which includes issues of equity in sharing burden, Head[19] proposes that the issues do not reach resolution and only continue to be debated and renegotiated. The overall suggestion is that intergovernmental and intersectoral collaboration requires organizational learning and cultural change to develop new approaches to address these wicked problems.

Pitt[20] investigates the issues of collaboration through the influence of government networking with community interest and neighborhood jurisdictions on energy and climate change mitigation planning. The findings suggest that cities adopting a large number of energy and climate change policy actions report a collaboration that includes an elected official, a dedicated staff member, and a community representative and results in successful adoption and implementation of climate change policy. The results are suggested to support and align with previous reports such as Corbet and Hayden[21] and suggest that local governments engage not only with the community but also with adjacent communities that have active climate change mitigation programs.

Looking specifically at the types of community representatives and interacting economic characteristics, Sharp et al.[22] find that organized interest groups have an influence on adoption and implementation, while economic constraints can provide roadblocks for climate change mitigation implementation. The results suggest that political pressures are a contingent factor in organized interest group influence. While cities with low budgetary capacity are more likely to engage with a network, the budgetary restraints continue to provide major barriers to policy change. Overall, the findings suggest that community and nonprofit engagement do increase city capacity to engage in climate change mitigation activity. The next sections examine the linkages between sustainability and citizen participation in dealing with disaster.

Sustainability and citizen participation

While the bulk of the literature proposes a normative suggestion of a connection between sustainability policy and citizen participation, Portney and Berry[23] find that, historically, few studies have investigated why participation is necessary for sustainability. Portney and Berry[23] ground this idea in the democratic theory that a society comes closer to realizing its democratic values when citizens are included in the process, where the governed have a say in how they will be governed. Geczi[24] proposes that while citizen involvement has been advocated for on a global scale since 1987, the key element of inclusion has been absent, even in societies based on democratic values, such as in the US. Geczi[24] relies upon Dahl,[25] Lindblom,[26] and Dahl[27] to make the argument that, while citizen participation is a vital aspect to creating and implementing sustainability policy, the market economy in a democratic society can result in exclusion of social and economically disadvantaged groups as more powerful interest groups leverage capital to influence everyday policy. This gets to the core issue of capacity.

Wang et al.[15] propose that citizen involvement is vital to gain support to finance sustainability. Rather than focusing on the motivations to engage in and implement sustainability plans, at center stage is the concept of organizational capacity building. Capacity building in this case refers to human capital and buy-in. Wang et al.[15] find that while cities have developed a high level of political capacity, this political capacity is found at the executive levels of government rather than with citizens or community stakeholders. Additionally, there is no significant or direct association between political capacity and sustainability initiatives. However, political capacity at the community (citizen) stakeholder level is proposed to be important for local government managers to gather information, thus making stakeholder engagement one of the most important factors in implementing sustainability plans.

Kinzer[12] proposes that there is a connection between planning literature and public participation literature. However, until 2016, these literatures are spread across a multitude of journals and books. A second issue is that the planning literature is so vast that a single article is incapable of containing the body. Therefore, Kinzer[28,29] focuses on the connection between public participation and sustainability policy implementation. Kinzer[28] finds that public participation, through the form of a citizen advisory committee, does have a positive impact on sustainability outcomes during implementation. However, Kinzer[29] finds that citizen action committees actually have a negative impact on the speed of implementation when a citizen action committee is formed for the purpose of sustainability planning.

The findings of Kinzer[28,29] suggest that the citizen action committee that serves during the adoption process should be terminated, and a separate citizen action committee should be formed for implementation of sustainability policy. One reason given is that volunteer fatigue could result when no end is foreseeable. This is supported by the finding that the presence of a citizen action committee in the planning phase negatively impacts the speed of implementation, while the number of tasks given to the citizen action committee has a positive impact on speed of implementation.

Foss[30] focuses on how participation is structured rather than the presence or absence of citizen participation. This study also focuses specifically on climate change as the center piece of the sustainability plan. The results suggest that when citizen participation takes the form of public forums, where all citizens can provide input throughout the process, special interest groups with conflicting ideologies can derail the implementation of sustainability plans. Rather, a citizen's steering committee made up of community stakeholders provided input, and when the plan was adopted, the plan was then brought to the public at small meetings throughout the city rather than at one city wide meeting. While both cases of the Foss[30] study display a desire to protect the vast natural resources of the area, neither city included climate change in the discussion of the sustainability plan development.

This supports the findings of Portney and Berry[23] that suggest that cities that engage in sustainability have higher levels of participation. This is also supported by the Okubo[31] study that finds that local governments report the need to be an example for the community by engaging in actions that show the community that government is serious about sustainability. The connection is that sustainability policy and disaster policy, or lack thereof, can result in populations being disadvantaged and harmed by the policy decisions that exist in daily life, thus forming a linkage between sustainability policy and disaster policy.

Sustainability and disaster linkages

Homsy et al.[4] look at sustainability through disaster planning, two separate but related concepts that are considered to be wicked problems. Both sustainability and disaster planning function through a multiple organization structure in which each organization operates independently. The proposal is that disaster planning will lead to sustainability planning and action. However, the findings suggest that while disaster experience is indicative of taking sustainability action, the presence of disaster is not indicative of engaging in a sustainability plan.

While Homsy et al.[4] find that the creation of a disaster plan does increase the chances of adopting sustainability policy, the results agree with previous sustainability studies that find a connection between community involvement and sustainability but are not significant to disaster plans. This results in sustainability being a more comprehensive process than disaster plans. The takeaway is that disaster plans are found to be significant to taking sustainability actions and could serve as an alternate avenue to achieve sustainability when sustainability plans are not directly successful. This suggests a direct relationship between sustainability and disaster planning though they tend to function separately in practice and research.

ICMA data and sustainability

Binghamton University and Cornell University teamed up with the American Planning Association and the US Department of Agriculture through the ICMA for the main purpose of testing for the presence of sustainability as well as disaster response and mitigation plans with adoption of sustainability actions.[4] From the 2015 ICMA study, Homsy et al.[32] report that among the three sustainability foci, environmental protection, economic development, and equity (social), when sustainability plans are adopted, there is a better balance between the three legs; however, the equity portion remained the lowest priority with or without a sustainability plan. While only 31 percent of the responding governments report having a sustainability plan, 87 percent report having either a hazard mitigation plan or an emergency evacuation plan; of which, 69 percent include actions that are considered issues of social equity.

While Homsy et al.[4] focus on the impact of disaster plans in creating a sustainability plan versus simply taking sustainability actions, one finding of interest is that disaster experience has a direct and positive influence on sustainability actions as well as an indirect impact on sustainability actions through the presence of a disaster plan. However, disaster experience does not have a significant relationship with the presence of a sustainability plan. This is understandable since previous studies have found that most local governments do not have a sustainability plan but do engage in sustainable actions, even though the local government does not consider the actions as sustainability actions. This sets the stage for future research to understand the relationship between disaster experience and public participation for sustainability.

Hypotheses

John et al.[33] rely upon the Simon[34] idea of bounded rationality, as proposed by Egidi et al.,[35] to explain the need to increase citizen participation in policy creation. The idea is that the human mind is bounded and can only conceive a certain amount of information. Therefore, the more people who can be persuaded to provide input, a more beneficial outcome can be produced. This results in the need to approach each community based upon the specialized needs of different groups in the community to avert the negative effects of bounded rationality.

It is this social order that provides focus for the theoretical aspects of this study. While previous studies have focused on the direct influence of disaster experience on sustainability action or sustainability plans, this study focuses on the social order based upon how local government understands community participation to influence sustainability efforts. The following hypotheses consider the aforementioned previous sustainability research findings combined with the idea of the social order to better understand the linkages among disaster, social capacity, and sustainability efforts.

> H1: When environmental protection is a priority for local government, there will be a positive and significant increase in how local government reports citizen participation to impact sustainability efforts.
>
> H2: When social equity is a priority for local government, there will be a positive and significant increase in how local government reports citizen participation to impact sustainability efforts.
>
> H3: When economic development is a priority for local government, there will be a positive and significant increase in how local government reports citizen participation to impact sustainability efforts.
>
> H4: As disaster experience increases for local government, there will be a positive and significant increase in how local government views public participation and sustainability efforts.

METHODS

The data for this study are secondary data taken from a 2015 International City/ICMA survey on local government sustainability practices, which was funded by a US Department of Agriculture grant.* The original study surveyed 8,562 local governments across the US and received 1,899 responses and focuses on local government sustainability practices. The data for this current research are drawn from the original studies 1,899-respondent sample.

Sample framework

This study requires analyzing the sample of 1,899 responses for missing data and outliers based upon the dependent and independent variables. A listmiss and a Mahalanobis outlier analysis are performed, resulting in 421 cases dropped for missing data and outliers. This process resulted in 1,478 observations in the final sample size.

Dependent variable

The dependent variable measures local government understanding of public participation concerning sustainability efforts. Respondents were asked how much impact public participation has had in shaping sustainability plans and strategies in your community. The original survey received 1,762 responses with 58.6 percent saying little to no impact. The question allows four responses ranging from 1 = no impact, 2 = a little impact, 3 = some impact, and 4 = a lot of impact. The descriptive statistics for this study resulted in similar result with 57.7 percent responding that public participation has had little to no impact on sustainability plans and strategies. Table 1 displays the comparative results for the dependent variable descriptive.

Independent variables

The main independent variable of interest is the number of disasters the local government has

*It is important to note that the investigator(s) on this study did not participate in the original study and are not associated the International City/County Management Association, the US Department of Agriculture, Binghamton University, Cornell University, or the American Planning Association.

Table 1. Impact of public participation on shaping sustainability efforts			
	Frequency	Percent	Cumulative percent
No impact	471	31.9	31.9
A little impact	381	25.8	57.6
Some impact	412	27.9	85.5
A lot of impact	214	14.5	100.0
Total	1,478	100.0	

responded to over the past 15 years. The time frame covers the year 2000 to 2015, when the International City/County Managers Association conducted the local government sustainability practices survey. The list of disasters included hurricane, earthquake, tornado, wildfire, flood, drought, blizzard or ice storm, toxic spill, and other. The number of responses is counted for each respondent, with the resulting ranging being from zero to seven disaster experiences for the sample. The largest response was 32 percent responding to 1 major disaster from 2000 to 2015 and the lowest response of 0.2 percent for seven major disasters. Table 2 displays the descriptive results for the main independent variable.

Secondary independent variables

The secondary independent variables cover the three legs of sustainability: environmental protection, economic development, and social equity. Respondents were asked if each of the three legs is a priority in their jurisdiction. The original survey focused on how having a sustainability plan influenced the balance of priority between the three legs. When the jurisdiction has a sustainability plan, the priority for each leg increased, and the gap between each was reduced. However, for this study, the regression is switched.

Of interest for this study is the association that the priority of each leg has on citizen influence for sustainability efforts. The idea is that when local government places priority on the three legs of sustainability, then public participation in sustainability efforts will increase. Each of the three variables are

Table 2. Number of major disasters experienced in the last 15 years			
	Frequency	Percent	Cumulative percent
0	359	24.3	24.3
1	473	32.0	56.3
2	391	26.5	82.7
3	168	11.4	94.1
4	69	4.7	98.8
5	10	0.7	99.5
6	5	0.3	99.8
7	3	0.2	100.0
Total	1,478	100.0	

Table 3. Environmental protection, social equity, and economic development as a priority (N = 1,478)			
		Frequency	Percent
Environmental protection	No	747	50.5
	Yes	731	49.5
Social equity	No	1,070	72.4
	Yes	408	27.6
Economic development	No	131	8.9
	Yes	1,347	91.1

measured with a yes or no and recorded as 1 for yes and 0 for no. Environmental protection was reported to be a priority by 49.5 percent. Social equity and economic development resulted in opposite priorities with only 27.6 percent reporting social equity as a priority and 91.1 percent reporting economic development as a priority. Table 3 displays the descriptive data for the secondary independent variables.

Control variables

This study includes a number of variables to control for individuals who motivate or hinder sustainability as well as the number of sustainable actions taken, the number of energy actions taken in the last 5 years, and population size. The original research asked respondents how significant the following factors are in motivating sustainability efforts by your local government. The variables include leadership of local elected officials, pressure from residents, and pressure from advocacy groups. The variables are measured as not significant, limited significance, significant, and very significant. The variables are coded from 1 for not significant to 4 for very significant.

Respondents were asked how significant the following factors are in hindering sustainability efforts by your local government. Of interest are opposition of elected officials and lack of community support/resident support. Respondents were not asked about advocacy groups hindering sustainability efforts. The variables are measured as not significant, limited significance, significant, and very significant. The variables are coded from 1 for not significant to 4 for very significant. Table 4 displays the descriptive data for factors motivating and factors hindering sustainability efforts.

The number of sustainability actions taken is important to control for because there is a connection between disaster plans and sustainability actions, even in the absence of a sustainability plan. The actions include a dedicated budget, a climate mitigation plan, a climate adaptation plan, a greenhouse gas inventory of local government operations, a greenhouse gas inventory of the community, greenhouse gas reduction targets for local government, and greenhouse gas reduction targets for the community. A count is performed with 70 percent reporting none of these actions and only 13.4 percent reporting one of these actions, while 0.9 percent reported taking seven of these actions.

Respondents were asked about 15 possible energy actions, and the responses were counted. This resulted in only 10.1 percent reporting none of the energy actions and 0.1 percent reporting 15 of the energy actions. Over 75 percent of the respondents reported from 1 to 8 energy actions taken. The number of reported sustainability actions deals with climate

Table 4. Factors motivating and hindering sustainability efforts

	Frequency	Percent	Cumulative percent
Elected officials motivating			
0	84	5.7	5.7
1	185	12.5	18.2
2	525	35.5	53.7
3	684	46.3	100.0
Total	1,478	100.0	
Residents motivating			
0	218	14.7	14.7
1	495	33.5	48.2
2	563	38.1	86.3
3	202	13.7	100.0
Total	1,478	100.0	
Groups motivating			
0	298	20.2	20.2
1	721	48.8	68.9
2	377	25.5	94.5
3	82	5.5	100.0
Total	1,478	100.0	
Elected officials hindering			
0	335	22.7	22.7
1	426	28.8	51.5
2	373	25.2	76.7
3	344	23.3	100.0
Total	1,478	100.0	
Residents hindering			
0	285	19.3	19.3
1	471	31.9	51.2
2	470	31.8	82.9
3	252	17.1	100.0
Total	1,478	100.0	

change and is found to have a moderate correlation with the number of energy actions taken (0.546, $p < 0.01$). This suggests that the number of energy actions can be considered in the category of sustainability actions for local government. Table 5 displays the descriptive data for sustainability actions and energy actions taken.

Table 5. Number of sustainable and energy actions			
	Frequency	Percent	Cumulative percent
Number of sustainable actions			
0	1,035	70.0	70.0
1	198	13.4	83.4
2	67	4.5	88.0
3	54	3.7	91.6
4	36	2.4	94.0
5	41	2.8	96.8
6	34	2.3	99.1
7	13	0.9	100.0
Total	1,478	100.0	
Number of energy actions			
0	150	10.1	10.1
1	134	9.1	19.2
2	158	10.7	29.9
3	189	12.8	42.7
4	177	12.0	54.7
5	169	11.4	66.1
6	146	9.9	76.0
7	108	7.3	83.3
8	69	4.7	88.0
9	59	4.0	91.9
10	47	3.2	95.1
11	35	2.4	97.5
12	15	1.0	98.5
13	14	0.9	99.5
14	6	0.4	99.9
15	2	0.1	100.0
Total	1,478	100.0	

Modeling

A regression analysis is conducted using SPSS to regress the number of disasters experienced by the local government in the last 15 years on the impact that the local government reports public participation to have on shaping sustainability plans and strategies. The model is significant and produces a Model F of 59.711 ($p < 0.001$). The model explains 32.3 percent of the variance in how local government reports public participation to impact the shaping of sustainability plans and strategies.

FINDINGS

Table 6 displays the findings of the regression model. The findings suggest that the number of disasters experienced by the local government's jurisdiction in the last 15 years has a positive and significant relationship with the impact that local government reports public participation to have on shaping sustainability plans and strategies (0.056, $p < 0.01$). The secondary variables reveal that when environmental protection is a priority (0.133, $p < 0.05$) and when social equity is a priority (0.191, $p < 0.01$), there is significant and positive relationship with the impact that local government reports public participation to have on shaping sustainability plans and strategies. However, economic development being a priority (–0.74, $p = 0.360$) is not a significant indicator of, but does show a negative relationship with, the impact that local government reports public participation to have on shaping sustainability plans and strategies.

The control variables all show a significant relationship in the expected direction, either positive or negative except for population. Leadership of elected officials motivating (0.186, $p < 0.001$), pressure from residents (0.139, $p < 0.001$), and pressure

Table 6. Impact of public participation on shaping sustainability efforts (N = 1,478)			
Variable	**B**	**SE**	**β**
Number of disasters in last 15 years	0.056	(0.019)	0.063**
Environmental protection is a priority	0.133	(0.054)	0.063*
Social equity is a priority	0.191	(0.062)	0.081**
Economic development is a priority	–0.074	(0.081)	–0.020
Elected officials motivate sustainability	0.186	(0.031)	0.154***
Residents motivate sustainability	0.139	(0.037)	0.119***
Advocacy groups motivate sustainability	0.162	(0.040)	0.124***
Elected officials hinder sustainability	–0.061	(0.025)	–0.062*
Residents hindering sustainability	–0.089	(0.028)	–0.083***
Number of sustainable actions	0.151	(0.018)	0.222***
Number of energy actions in last 5 years	0.038	(0.009)	0.114***
Population	0.029	(0.017)	0.049
R^2	0.328		
Adjusted R^2	0.323		
S_e	0.869		
Model F	59.711***		

Note: B(SE) = unstandardized estimate of the regression coefficient (and its standard error). β = standardized estimate of the regression coefficient.
***p ≤ 0.001, **p ≤ 0.01, *p ≤ 0.05 (two-tailed tests).

from advocacy groups (0.162, $p < 0.001$) all have a positive and significant relationship with the impact that local government reports public participation to have on shaping sustainability plans and strategies. Opposition of elected officials (–0.061, $p < 0.05$) and lack of community/resident support (–0.089, $p < 0.001$) have an expected negative and significant relationship with the impact that local government reports public participation to have on shaping sustainability plans and strategies. The number of sustainable actions (0.151, $p < 0.001$) and the number of energy actions in the last 5 years (0.038, $p < 0.001$) each have a positive and significant relationship with the impact that local government reports public participation to have on shaping sustainability plans and strategies. The next section will provide a discussion of the findings and their significance to this research.

DISCUSSION

The findings suggests that there is a relationship with how local government officials understand public participation to impact sustainability efforts based on increased community experiences with disaster. However, we cannot reliably interpret what this impact looks like, other than to say that the impact of public participation on sustainability efforts increases. Yet, by looking at the priority of the jurisdiction, we can better understand the impact of public participation on sustainability efforts. The hypotheses are listed here for reference.

H1: When environmental protection is a priority for local government, there will be a positive and significant increase in how local government reports citizen participation to impact sustainability efforts.

H2: When social equity is a priority for local government, there will be a positive and significant increase in how local government reports citizen participation to impact sustainability efforts.

H3: When economic development is a priority for local government, there will be a positive and significant increase in how local government reports citizen participation to impact sustainability efforts.

H4: As disaster experience increases for local government, there will be a positive and significant increase in how local government views public participation and sustainability efforts.

All of the hypotheses proposed are supported by the findings, with the exception of the third hypothesis. Hypothesis 1 (environmental protection) and hypothesis 2 (social equity) are significant with social equity providing a more significant and grander impact than environmental protection. However, hypothesis 3 is not supported, as a priority on economic development does not show to have a statistically significant relationship with how local government understands the impact of public participation on sustainability efforts. While not statistically significant, a priority on economic development does show a negative relationship with public participation and sustainability efforts.

This negative relationship between economic development and public participation in sustainability efforts was not expected and runs opposite to the original research that collected the data used for this study. Homsy et al.[4] propose that when local governments engage in sustainability planning, the priority placed upon each of the three legs of sustainability (environmental protection, social equity, and economic development) is expected to increase and reduce the gap in the priority placed upon each of the legs. Homsy et al.[4] report that 30.8 percent of their sample report has a sustainability plan, which supports previous research that sustainability plans are rare. However, of those with a sustainability plan, 93 percent report economic development as a priority, while only 62 percent report not having a sustainability plan and say economic development is a priority.

The finding in this study shows that 91.1 percent of the samples report economic development as a priority, which can explain a lack of variation and significance of economic development priority to the model. However, economic development as a priority reveals a negative relationship with how local government views the impact of public participation in sustainability efforts. The results, while not statistically significant for economic development, do suggest that when economic development is a priority for a community, local government sees a reduction in the impact public participation has on shaping sustainability efforts.

The economic piece can be explained by Geczi[24] and Sharp et al.[22] who proposed that market economies can reduce citizen participation as more powerful interest groups leverage political capital to influence everyday policy. This lack of significance can also be explained by an increased fiduciary and physical capacity of local industry directly and local government indirectly through increased revenue, which may negate the need for a citizens political capacity and buy-in to influence and fund sustainability efforts as proposed by Wang et al.[15]

While the number of energy actions and sustainable actions are significant and have a positive relationship in the model, their purpose is as control variables for the influence of climate change as a slow onset disaster. The sustainable actions consist of climate change and greenhouse gas factors. An interesting finding is that while 70 percent of the sample reports taking no sustainable actions, ie climate change and greenhouse gas, the number of sustainable actions is highly significant to increasing the impact that local government says public participation has on sustainability efforts. This suggests one of two possible interpretations.

First, this can be interpreted as a push by the public to engage in climate change efforts as the number of disaster experiences increases with local government resisting climate change or greenhouse gas actions. This interpretation would suggest that local government has a negative view of the impact of public participation. The second interpretation is that the public, for the majority of those local governments surveyed, does not view climate change action as a priority to advocate for, while approximately only half of the local governments responding report environmental protection as a priority. However, the number of energy actions shows more promise in explaining the impact of public participation for sustainability actions, as most energy actions can be considered from an economic perspective, especially when combined with most respondents, saying that economic development is a priority.

CONCLUSION

The purpose of this research is to better understand the influence of disaster experience on how local government understands public participation to impact the shaping of sustainability efforts. As such, the study is conducted through a lens where citizen participation can be a tool that can help local government solve wicked problems through sustainability in a democracy that depends on citizen participation to function in the best interest of the public. The idea is that while research has proposed that climate change should be addressed at the local level on a global basis, communities may not participate in sustainability efforts because communities do not experience disaster at the same levels.

A community includes many stakeholders who can be useful in solving wicked problems such as climate change through sustainability efforts. While citizens are one piece of the community sustainability solution model, based upon these findings, most local governments do not see citizen participation as having much of an impact on sustainability efforts. However, the more a community experiences major disaster, citizen participation can be expected to have a greater impact on sustainability efforts.

The findings do suggest a need for local government to have a solid system that can accommodate increased citizen participation pressures as community experiences more major disasters. Whether citizen participation has a detrimental or supportive impact on sustainability efforts, increased public involvement can strain local system and have an impact on the community's political environment. This gets to the issue of local government and its emergency management capacity to effectively communicate with and address the unique needs of its community.

For future research, the findings suggest that we still know little about what the impact of citizen participation looks like. For example, previous research suggests that priority placed on economic development increases when a local government has a sustainability plan, but this current research suggests that a priority on economic development decreases the impact of citizen participation on sustainability efforts. One explanation could be that as economic development increases, the community becomes more sustainable through financial capacity and reduces the need for public political capacity to influence sustainability efforts.

This lack of significance of economic development priority and the significance of equity and environmental protection can be explained by the Tierney[10] proposal of the social order influence through advocacy groups and citizen participation in how a community defines and deals with disaster. While climate change is proposed to be a slow onset disaster, little attention is given to actions that address climate change by most of the respondents in this study. This lack of attention to slow onset disasters such as climate change can be explained by the overwhelming literature on emergency management that proposes a lack of attention, and finance is given to something that may never happen or that is not considered a true hazard to the individual community.

The findings here raise the question about how future research can help local government better understand how to create and maintain a sustainable community based upon the unique needs of each community. For example, can research determine if local government believes that climate change is an issue that can be solved by local governments working on a global scale? This can help to identify the realities and true local government concerns about harnessing citizen participation and investing in sustainability to address the issues of climate change.

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Identifying and assessing corporate employment variables that influence community resilience: A novel model

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ABSTRACT

Quantifying the concept of disaster resilience on a local level is becoming more critical as vulnerable communities face more frequent and intense disasters due to climate change. In the United States (US), corporations are often evaluated using social justice or environmental sustainability matrices for financial investment consideration. However, there are few tools available to measure a corporation's contribution to disaster resilience on a local level. This study includes a focused literature review of employment variables that contribute to community resilience and a national survey that asked US emergency managers to rank the variables they believe have the greatest influence on individual resilience. A novel corporate community resilience model that ranks corporate contributions to disaster resilience in the communities where they operate was developed and then tested against data from five employment sectors from the same area. This model can be used by stakeholders to better understand how corporations can most efficiently contribute to county- and subcounty-level disaster resilience. The metrics used in this study are universal and translative, and thus, the development of this resilience model has global disaster resilience implications.

Key words: community resilience, corporate social responsibility, disasters, emergency management; DOI: 10.5055/jem.0808

INTRODUCTION

Vulnerability indices have evolved over the last several decades in an attempt to better quantify the subjective nature of social vulnerability. Community leaders, particularly those with an interest in disaster mitigation, have employed popular social vulnerability indices, eg, SVI or SoVI,[1,2] to meet federal disaster funding requirements while attempting to distribute resources more equitably to those who need them the most. This speaks to the overarching community goal of increased resilience, which is defined here as "the ability of individuals, communities, organizations or countries exposed to disasters, crises and underlying vulnerabilities to anticipate, prepare for, reduce the impact of, cope with and recover from the effects of shocks and stresses without compromising their long-term prospects."[3] In recent years, the extent to which these complex indices like SVI and SoVI can be left unchecked as the sole driver of mitigation policy and practice has been questioned.[4-8] It is clear that having other accessible means to assess vulnerability and promote local community resilience is necessary—particularly for lower-income areas.

During the same period that SoVI and SVI were developed, there has been an evolution of the profit-driven culture of corporations toward a culture of shared values in the present day.[9] Charles Schwab & Company broadly defines socially responsible investing (SRI) as "any general investing strategy that considers environmental, social, and corporate governance factors."[10] However, the mainstreaming of terms like corporate social responsibility (CSR) has not been accompanied by a greater understanding of what impact corporations have on local community disaster resilience. There even appears to be a disconnect between those shared value concepts and practical core business functions.[9] "CSR seems to be lacking specific research with regards to how to address the core business activities through CSR and seems to point out a reason why CSR can be implemented only partially and even may raise questions about its potential benefits."[9] More precise measurements of CSR are needed to quantify the value of CSR both internally and externally.[11-13] A study by Carroll and Brown[14] lists the common components of CSR, including "business ethics, stakeholder management, sustainability, corporate citizenship, conscious capitalism, creating shared value, and purpose-driven organizations." Therefore, corporations pursuing CSR in their mission statement would be highly valued partners according to guiding disaster management documents like the Federal Emergency Management Agency (FEMA)'s whole community guidelines[15] or the Sendai Framework, which lists a priority for action as "to increase business resilience and protection of livelihoods and productive assets throughout the supply chains, ensure continuity of services and integrate disaster risk management into business models and practices."[16]

Today, sophisticated models exist, which measure CSR for investors to decide where to put their investment dollars. Some studies suggest that investors are willing to forgo strict objectives of financial gain in order to be associated with socially responsible corporations.[17] In response, more organizations are pursuing B Corp Certification, which indicates a level of environmental and social sustainability with added levels of transparency.[18,19] In this context, this study proposes a means to assess a corporation's contribution to the community's overall resilience where it conducts business. This corporate community resilience model (CCRM) is not proposed as a punitive tool to catch organizations acting irresponsibly. Instead, this study proposes a model that all community stakeholders (including corporations) can use to define the most efficient and symbiotic way that employers can actively participate in local disaster resilience efforts. To this end, this study is guided by the main research question: *How can communities measure the impact that corporations have on their local disaster management-related resilience?* This study answers this question from a mitigation perspective with a focused review of the existing and relevant literature and a national survey of United States (US) emergency managers (EMs) that provides input on factors that impact individual resilience. Results from both of these data sources are analyzed, and the CCRM is developed. Sample data are inputted into the CCRM, and synthesis takes place in the final portions of this study.

METHODOLOGY

This study needed to establish a qualitative foundation in the existing literature while also gaining additional quantitative data to help confirm or challenge the existing literature. According to Creswell and Creswell,[20] this constitutes a mix-method approach, which is common in research when attempting to find a gap in the existing literature or when suggesting new models or avenues of research.

Focused literature review methodology

Employing a focused review is common in peer-reviewed research as it can establish a plausible foundation for novel concepts while informing and directing avenues for future research.[21,22] The intent was to examine this topic through a planning and mitigation lens and not a response-centric perspective that, for example, might evaluate performance based on the arbitrary donation of in-kind goods, eg, bottles of water, to locals in need. The data from this focused literature review were used to plausibly establish variables that would contribute to the study's survey questions and help create a CCRM.

Search terms were used to explore and identify potential CCRM variables and develop them further in the focused review. For example, primary search terms like *income + resilience + community* were associated with secondary terms like *corporate compensation + employee pay* in varying Boolean arrangements. Each article that was considered even "slightly relevant" or "likely relevant" was read completely for further assessment. Initially, 53 abstracts and conclusions of potentially related articles were reviewed. Concerning Table 1, a total of 26 articles supported three dominant themes or variables that contribute to employment-derived resilience. Each was ranked from most prevalent to less prevalent with a minimum of five citations to qualify as a variable.[48] This ranking was developed concurrently with the survey results.

Table 1. Variables used to construct corporate community resilience model (CCRM)			
Variable (top three variables in order of ranking)	**Weighting (total of all variables = 1.0)**	**Justification (core citations)**	**Subvariables for future study**
Income (amount of take-home money or pay scale of the individual)	0.727	Morrow,[23] Eriksen and O'brien,[24] Klein,[25] Leichenko and Silva,[26] Tierney,[27] Hallegatte et al.,[28] Hallegatte et al.,[29] Islam and Winkel,[30] Glynn and Casey,[31] PEW,[53] UNDRR,[32] Wood and Frazier,[33] USHUD[34]	Job type (permanent vs temporary)
Access to healthcare benefits (access to and quality of healthcare)	0.173	Stiehm,[35] Chandra et al.,[36] Runkle et al.,[37] Cox and Hamlen,[38] Hornor,[39] Orentlicher,[40] Karaye and Horney,[41] Ruck et al.[42]	Quality of benefits Access to mental health benefits
Job benefits (five additional benefits: paid education/ professional development, paid maternity/paternity leave, paid time off, flexible (remote and hybrid), retirement plan)	0.100	Baird and Reynolds,[43] Spangler et al.,[44] Bardoel et al.,[45] Ali et al.,[46] Hite and McDonald,[47] Dominant themes from survey Q2	Break down this variable into subvariables, eg, isolate *maternity leave* as a subvariable Contract employee vs permanent

Survey methodology

Public agency EMs working at the US state, county, or subcounty levels were asked to participate in a short, anonymous email survey. This group of vital, frontline practitioners in the US has experience and training, which has the potential to connect concepts of community resilience more directly to meaningful practice and policy.[49] Email contacts from all 50 states were harvested with the criteria that the individual either carried the title of EM, emergency services director, or similar derivation—but not that of administrative staff. This contact list was originally assembled by this author group, and additional information on the database criteria can be found in Wood et al.[8] The 3,001 existing contacts were filtered by Qualtrics software as currently active respondents. This resulted in 2,593 successfully sent emails. The anonymous, nonpersonal nature of the single-question survey (Figure 1) resulted in a low-risk ethical determination when considering the institutional review board's (IRB's) standards.[50] The goal of this survey was to gather a simple ranking from these national, public-sector EMs of the three most predominant individual resilience variables as suggested by the existing literature. The second, open-ended question was included as a means to evaluate other variables—either not previously considered in this study or as subvariables to be considered for future research.

FOCUSED LITERATURE REVIEW

It is common to use analysis of existing research to determine "the extent to which a specific research area reveals any interpretable trends or patterns."[51] This focused review of the literature produced three dominant trends or variables that can influence individual resilience from an employment perspective. These include income, access to healthcare, and job benefits. Since community resilience is an aggregate of individual resilience,[36] these leading variables directly contribute to whole community resilience. It could be argued that these factors are also indicators of vulnerability. However, for this study, these three variables are primarily considered as means to cope with and recover from existing shocks per the IFRC[31] definition offered in the "Introduction" section. "While vulnerability speaks to the conditions that make communities susceptible to harm, resilience refers to coping with and recovering from a hazard that has already occurred."[52]

The purpose of this 30-second survey is to ask you to rank the variables listed in order of how much they contribute to an individual's resiliency opposite disasters and based on your experience serving your community as an emergency manager.

Q1. Based on your experience, please rank the factors listed below in order of highest importance (1) to less important (3) as related to individual resiliency within a community.

	1	2	3
Job benefits (Additional benefits to individual — like paying for education, maternity leave, or paid time off)	○	○	○
Income (Amount of take-home money or pay scale of the individual)	○	○	○
Access to healthcare benefits (Does their employer offer a healthcare plan?)	○	○	○

Q2. Optional question: Please list any individual resiliency variable that you think ranks *higher* than the three listed in the first question.

Figure 1. National survey sent to emergency managers.

Income

Since Morrow[23] stated that "we most often think in terms of economic resources—the association between poverty and vulnerability is easy to make," many subsequent studies have reinforced this correlation.[24-30,33] When rental costs exceed one-third of employee wages, there is a sharp rise in homelessness,[31] eviction rates, and use of social safety-net programs.[53] There is no location in US where minimum wage workers can afford a two-bedroom apartment rental.[54] A chief economist at Freddie Mac states that "if your income is $500,000 a year, you can pay 40 percent [towards rent] and still have money left. But if your income is $20,000 a year, it will be hard to make ends meet if you're paying 30 percent of your income on rent."[34] Not having enough money is a cause and effect of disaster risk where people can only afford to live in unsafe conditions, eg, poorly built housing located in a floodplain, which, in turn, increases their exposure to a hazard or shock.[32] Exposure of a population increases risk and is inversely related to community resilience.[55]

Access to healthcare

Lower-income is directly related to poorer health and access to healthcare.[35,40] This cause-and-effect relationship with wages makes access to healthcare an important driver of local resilience. An individual making the lowest wage cannot prepare for, reduce the impact of, cope with, or recover from routine medical costs let alone an unexpected medical bill. Nearly two-thirds of rent-burdened families have less than $400 in savings to direct toward unexpected shocks.[53] Ruck et al.[42] noted that access to healthcare and existing health conditions was among the key socioeconomic variables that could predict the spread of coronavirus disease 2019 (COVID-19). Indeed, existing socially vulnerable populations, under any circumstance, are predisposed to greater healthcare challenges.[38,41] "Vulnerable populations with prior health disparities because of race, poverty, and less than optimal health care coverage often have chronic conditions characteristic of their sociopolitical standing, leaving these at-risk individuals in greater peril following a disaster than are other members of the population with better health and health care coverage."[37] It is generally agreed upon that the potential gap in local resilience created by lack of access to healthcare also includes access to mental healthcare.[37,39] A Rand Corporation technical report and literature review identified key levers that contributed to community resilience, highlighting access to healthcare and income.[36] However, this same report noted that "we have limited understanding about the components that can be changed or the 'levers' for action that enable communities to recover more quickly."[36]

Job benefits

Job benefits, beyond access to healthcare, like paid education, paid maternity/paternity leave, or paid time off, can add to an individual's resilience without putting that financial burden on the individual.[45,46] While employees, in general, may not ask for these types of benefits, they are more likely to still use them once they are offered by the employer.[43,56] Spangler et al.[44] found that providing resources and benefits to employees was one of the most effective means of reducing workplace stress and increasing individual employee resilience. "Cultural practices as simple as free lunch were mentioned by individuals from two separate organizations. One participant said that the free lunch her family-owned employer still provided was a benefit she valued because it reduced distress from daily time conflicts."[44] The COVID-19 endemic has caused employers to rethink job benefits beyond wages and healthcare packages.[47] "As systems begin to ramp up work again, many already are reconsidering protocols and processes, making this an ideal time for HRD [human resource development] to encourage change."[47]

SURVEY RESULTS

The short survey sent to the 2,593 active domestic EMs' contacts garnered 178 replies, with a response rate of 6.9 percent. This response rate is understandable among a group of professionals who are currently extremely busy servicing public health, seasonal hazards, and other local stressors. Based on this response rate, the survey classification was changed to that of a "pilot survey," which has been shown to have merit in advancing the body of research while helping to guide possible areas of future research.[57,58]

In question 1 of the survey, EMs were asked to comparatively rank three individual resilience factors within a community: income, access to healthcare, and job benefits (Figure 2). The results shown in Figure 2 are summarized as follows:

- 66 percent of respondents indicated *income* was the #1 factor;
- 52 percent of respondents indicated *access to healthcare* was the #2 factor;
- 52 percent of respondents indicated *job benefits* was the #3 factor.

The open-ended and optional second question received 65 replies (or 36 percent of total replies) that

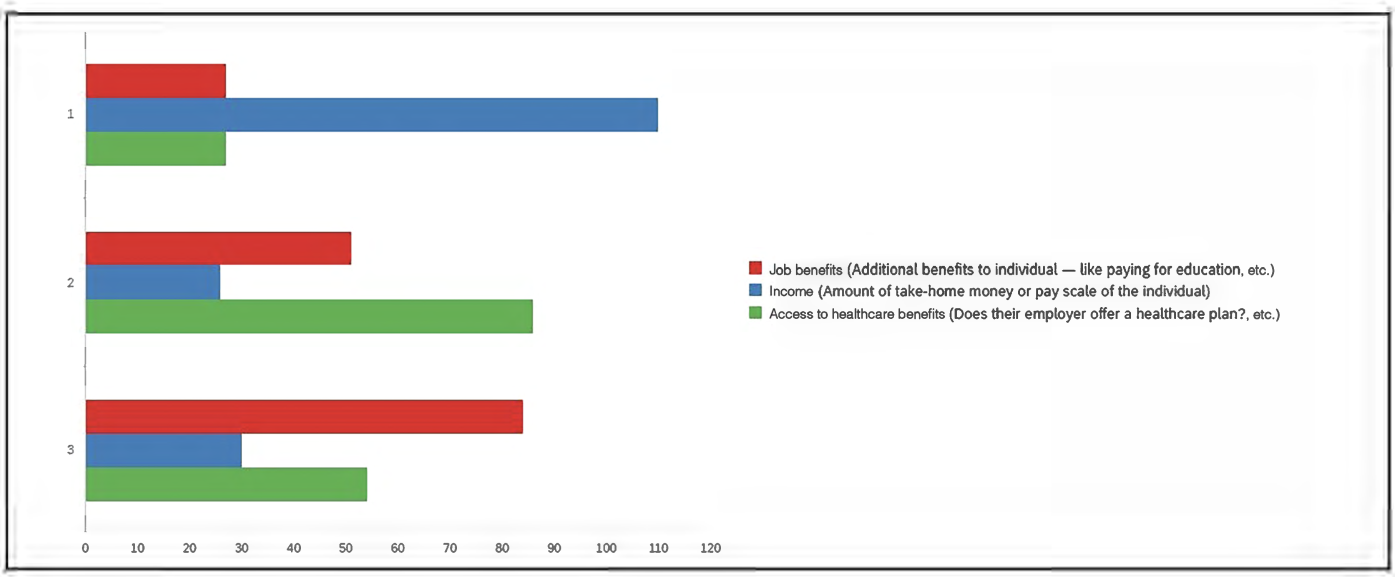

Figure 2. Color-coded ranking of survey respondents.

were analyzed to find common themes and subthemes. This response rate was not robust enough to include a fourth variable to this initial CCRM. However, subthemes including types of job benefits, social fabric, leadership, and type of employment helped bolster the credibility of the *job benefits* variable while also identifying possible avenues for future research.

ANALYSIS/DEVELOPMENT OF CCRM

The weighting of the three core CCRM variables was based on the 178 survey replies and substantiated by the results of the focused literature review (Table 1). It was clear from the survey of EMs that *income* was the dominant variable. The survey results showed that *income* was 4.2 times more prevalent than *access to healthcare*. To create an accurate weighting of these three factors normalized on a scale of 0 to 1, *income* received a multiplier of 0.727 and *access to healthcare* a factor of 0.173, which divides into 0.727, approximately 4.2 times. Next, 72 percent of survey respondents found *access to healthcare* more important than other *job benefits*, ie, paid education/professional development, paid maternity/paternity leave, paid time off, flexible (remote and hybrid), and retirement plan. As a result, the *access to healthcare* multiplier of 0.173 needed to represent a 72 percent higher weighting than *job benefits*, which was assigned the appropriate 0.100 weighting (Table 1). The results of the literature review were added to Table 1 to demonstrate qualitative support for each variable. While the literature overwhelmingly supports *income* as a dominant variable, this data source was primarily used to confirm, deny, or find any gaps in the survey result. The sum of all three multipliers is 1.0, which means that the highest score a company could receive (based on a 0-5-point scale) is 5.0.

The aim with assigning a company score to each of the three categories is to be geospatially consistent within the area being assessed. Furthermore, it is not the goal of this study to derive an arguably subjective standard, whereas, for example, an aggregate score higher than 3.0 means a company is contributing to local disaster resilience adequately, and a score less than 3.0 indicates some kind of failure on the company's part. Instead, the CCRM, in this first derivation, is intent upon simply providing a comparative analysis between companies that operate in the same region using consistent criteria for all scoring. It should be noted that any debate about the merits of this criteria is secondary to its consistency. In other words, stakeholders could employ the CCRM using, for instance, a different website source for healthcare rankings as long as the said source of agreed-upon data was consistently applied to each company.

Scoring income

Using the regional minimum wage (RMW)[59] as a baseline for a study area, the following tiered scoring system was adopted using 0.5 increments as income (wages and salary) increased beyond the RMW:

Income:

- 5.0 points: greater than 600 percent above RMW;
- 4.5 points: 400-600 percent above RMW;
- 4.0 points: 250-400 percent above RMW;
- 3.5 points: 125-250 percent greater than RMW;
- 3.0 points: 100-125 percent greater than RMW;
- 2.5 points: 60-100 percent greater than RMW;
- 2.0 points: 30-60 percent greater than RMW;
- 1.5 points: 10-30 percent greater than RMW;
- point: up to 10 percent above RMW.

Scoring access to healthcare

While the survey question about *access to healthcare* was intentionally more of a yes or no question, ie, is there healthcare offered or not, the scoring of healthcare needed to not only assign a value for no access to healthcare (score of 0) but also provide a ranked score between 1 and 5 based on the quality of healthcare being offered. To this end and for this initial CCRM development, this study adopted the five-point ranking system from the National Committee for Quality Assurance (NCQA). This organization appeared to provide neutral data about the quality of healthcare in each US state and claims to "use measurement, transparency, and accountability to highlight top performers and drive improvement."[60] Figure 3 shows an example of the data provided by this resource.

Scoring job benefits

Since the proportional value of one benefit opposite another in the context of community disaster resilience could support a standalone study, a simpler scoring methodology was adopted for the *job benefits* category. Based on the literature and the open-ended responses from the survey, the top five benefits were each given a single point value if they existed in that company's benefits package with a maximum, again, of five points. Below is a list of these benefit subcategories.

Job benefits: 1 point each, maximum of 5.0:

- Paid education/professional development
- Paid maternity/paternity leave
- Paid time off
- Flexible (remote and hybrid)
- Retirement plan

CASE STUDY RESULTS

To begin to move the CCRM concept toward a practical application in the field of disaster risk reduction (DRR), five anonymous employer groups from the same region (Los Angeles County in California) were evaluated. These included the following:

1. A national coffee retailer; Los Angeles retail location;

2. A local construction company, Los Angeles area;

Rating	Plan Name	States	Type	NCQA Accreditation	Consumer Satisfaction	Prevention	Treatment
4.5	Kaiser Foundation Health Plan Inc. - Southern California	CA	HMO	Yes	2.5	5.0	4.5
4.5	Kaiser Foundation Health Plan, Inc. - Northern California	CA	HMO	Yes	3.0	5.0	4.5
4.5	Sharp Health Plan	CA	HMO	Yes	4.0	4.0	4.0
3.5	Aetna Health of California Inc.	CA	HMO/POS	Yes	2.0	2.5	3.0
3.5	Anthem Blue Cross Life and Health Insurance Company	CA	PPO/EPO	Yes	2.5	3.0	3.0
3.5	Blue Cross of California dba Anthem Blue Cross	CA	HMO/POS	Yes	2.5	3.5	3.0
3.5	Blue Cross of California dba Anthem Blue Cross	CA	PPO/EPO	Yes	2.5	3.0	3.0
3.5	Blue Shield of California	CA	HMO/POS	Yes	2.5	3.5	3.0
3.5	Chinese Community Health Plan	CA	HMO	No	2.0	3.5	I
3.5	Health Net Life Insurance Company - California	CA	PPO/EPO	Yes	2.5	3.5	3.0
3.5	Health Net of California, Inc.	CA	HMO/POS	Yes	2.5	3.5	3.0
3.5	United HealthCare Services, Inc. (California)	CA	PPO	Yes	2.0	3.0	3.0
3.5	UnitedHealthcare Insurance Company (California)	CA	PPO	Yes	2.0	3.0	3.0
3.5	UnitedHealthcare of California (formerly PacifiCare of California)	CA	HMO	Yes	2.5	3.5	2.5
3.5	Western Health Advantage	CA	HMO	Yes	2.5	3.5	3.0
3.0	Aetna Life Insurance Company (California)	CA	PPO/EPO	Yes	2.0	2.5	3.0
3.0	Blue Shield of California	CA	PPO	Yes	I	2.5	2.5
3.0	Blue Shield of California Life & Health Insurance Company	CA	PPO	Yes	I	2.5	2.5
3.0	Cigna Health and Life Insurance Company - California	CA	PPO/EPO	Yes	2.0	3.0	3.0

Figure 3. Sample data provided by the National Committee for Quality Assurance (NCQA).

3. The national law firm's staff (nonlawyers); Los Angeles area;

4. The same national law firm's associates and partners; Los Angeles area;

5. A public school teacher; Los Angeles Unified School District.

Publicly available data from each sector were evaluated, scored, and entered into the CCRM, as shown in Table 2. Figure 4 demonstrates the results from Table 2 using a simple bar graph comparison. This type of aggregate comparison does not distinguish between employment sectors due largely to the small sample size. With a larger sample size, it may be appropriate to break the results down by employment sectors like food services, construction, and professional services. However, the aggregate comparison does present meaningful insights for EMs into what employed populations may be most resilient in the community. In Figure 4, for example, it is relevant to community-level resilience efforts to note the multisectoral range of CCRM scores from a coffee retailer to a practicing attorney.

DISCUSSION

In order to better understand and evaluate the role corporations play in our local community disaster resilience efforts, this study developed a novel CCRM

Table 2. CCRM scores of employment entities 1 through 5

	#1 raw score	Weighted score	#2 raw score	Weighted score	#3 raw score	Weighted score	#4 raw score	Weighted score	#5 raw score	Weighted score
Income	1	0.727	2	1.454	3	2.181	5	3.635	3.5	2.545
Healthcare	5	0.865	3	0.519	5	0.865	5	0.865	5	0.865
Benefits	5	0.500	1	0.1	4	0.400	4	0.400	4	0.400
Aggregate score		2.092		2.073		3.446		4.900		3.810

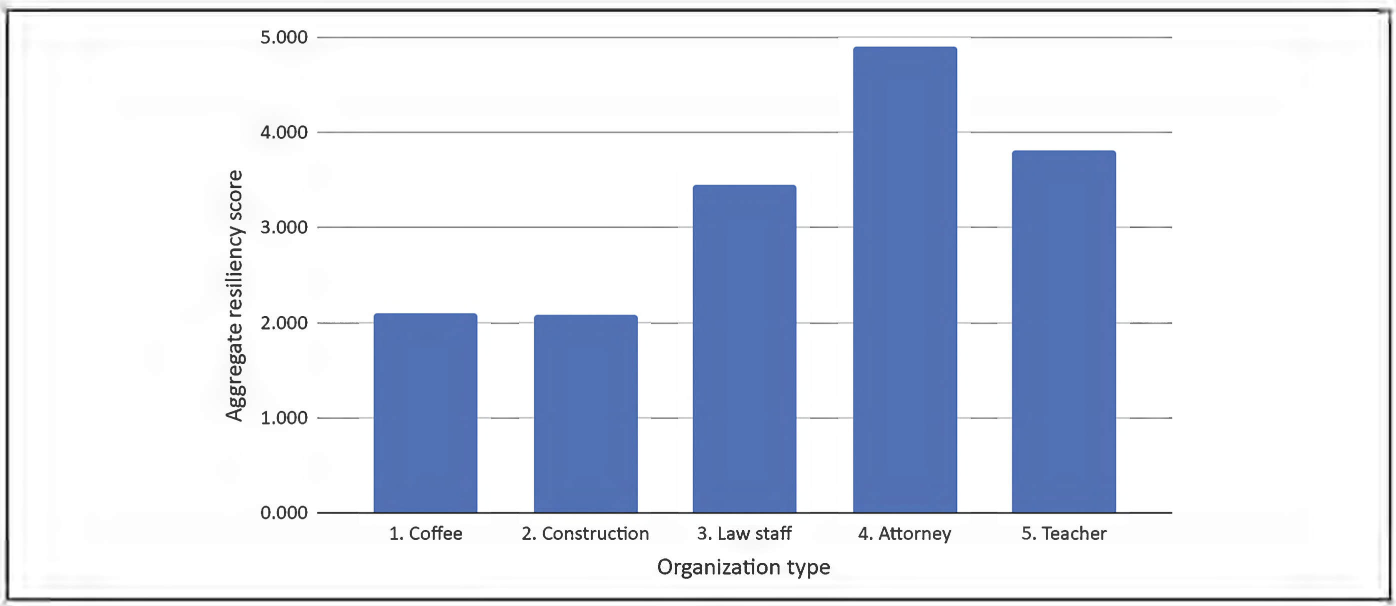

Figure 4. CCRM comparative results for five Los Angeles County employment groups.

based on a focused literature review, a nationwide pilot survey of EMs, and a test run of case study data from several employment groups.

The literature review and the survey identified three leading employment-related resilience variables within a community, developed an initial method to weight those factors, and a means to calculate an aggregate CCRM score. More robust data from, for example, a larger national survey followed-up by semistructured interviews might slightly alter the weighting of the three variables (*income*, *healthcare*, and *job benefits*) shown in Table 1. As with any model, updating data can adjust this type of weighting. However, the calculus derived from the pilot survey is also supported by the qualitative findings in the literature and provides a plausible starting point for the CCRM to be tested further. One probable reason that *income* will be more heavily weighted opposite *healthcare* and *job benefits* in any comparison is that having extra income can buy or subsidize both healthcare and other employment benefits. Cash can buy better healthcare. Cash can buy time off or midcareer professional development. And as seen in more recent studies on cash transfer programs, cash is flexible, allows the individual to define their needs, and brings the individual dignity, choice, and empowerment.[61-63] Additionally, extra pay leads to higher rates of homeownership, which, compared to renters, improves an individual's resilience.[64] Less take-home pay relative to rental rates leads to drastic increases in community vulnerability as well. "Income growth has not kept pace with rents, leading to an affordability crunch with cascading effects that, for people on the bottom economic rung, increases the risk of homelessness."[31]

It is obvious from the CCRM formula and the company comparison seen in Table 2 that *income* is the easiest way to improve a company's aggregate score. How then does a traditionally lower-wage industry "compete" with a traditionally higher wage sector from the same area? The simplest answer, as more data are collected, would be to divide the comparison categories into sectors like food services, construction, and professional services. The methodology should remain one where the unique data from the study area drive the criteria. An EM may, for example, acknowledge that the community they serve has such a large presence of government sector employees that they decide to add a CCRM "government" category similar to the public-school teacher data in Table 2. This would allow them to compare those employers both within their own sector and within the larger pool of community employers. Sectoral comparisons are also a means to see the two lower weighted variables (*healthcare* and *job benefits*) through a more granular and influential lens. For example, a stakeholder within a lower-wage sector that pays the state minimum wage would be able to move their CCRM sector score beyond their competition by providing better *healthcare* and more types of *job benefits*. This type of practical application of the CCRM aligns with the literature and the survey results, which prioritizes income but demonstrates the value of healthcare and benefits (Table 1).

This same literature tends to find that sustainable resilience building efforts consistently occur at the local level.[65,66] Therefore, it is worth reemphasizing here that the CCRM was developed for comparative analysis at the county to the subcounty level. With any derivation of the CCRM, it is important to compare data from companies in the same area, so that all DRR stakeholders from that city or county can gain insights into how employers contribute to individual and community resilience. For example, a local store of a national coffee chain might receive a high ranking in Aroostook County, Maine, but the same local store in Los Angeles County, California, may not fare as well, which demonstrates that this is not intended to be used as a national or even regional metric. CCRM comparison, evaluation, and competition all happen locally. To some degree, this could be considered a system that grades on a curve, relying upon relative contributions to determine value and possibly foster "resilience competition" among community actors. Furthermore, the CCRM outputs are not intended to be derogatory or punitive. Instead, the intention is to raise awareness among all DRR actors, including corporate stakeholders, nonprofits, and local government employers. This represents the collaborative planning and mitigation concepts set forth in FEMA's whole-community framework[15] and the Sendai Framework.[16] "Community approach attempts to engage the full capacity of the private and nonprofit sectors, including businesses, faith-based and disability organizations, and the general public."[15]

CONCLUSION

Employers are often called upon to donate money and in-kind goods once a community has already been impacted by a disaster. However, within the county or subcounty area, the CCRM score can demonstrate to EMs how their corporate community partners might be part of a proactive employment mitigation strategy and not a reactive partner in the post-disaster response and recovery phase. The CCRM should be seen as a means to improve community resilience and not an end of community resilience efforts in itself. DRR community partners might use the simple and accessible scoring system to create a tiered system of recognition, eg, silver, gold, and platinum, like other business frameworks use. The first step is to acknowledge this largely ignored link to community resilience by populating a CCRM spreadsheet with local employer data and calculating an overall CCRM score. Since the model is simple to present and interpret, more stakeholders will be able to weigh in on how that score is best used in their communities. This simplicity and accessibility of the CCRM has the potential to connect academic research to the practice of emergency management and community risk reduction.

Future research/limitations

The limitations to this research study center around small datasets for the pilot survey and the first CCRM scores, which only included five employment entities (Table 2). Results that include this type of limitation have been shown to successfully promote future research and/or define meaningful gaps in the existing literature. The dataset represented in Table 2 demonstrates the plug and play simplicity of this model that avoids the "black box" nature of complicated algorithms or indices, which, in turn, makes it more accessible to a busy group of DRR professionals.

A second limitation was the decision to not include a fourth variable—both in the nationwide survey and in the first derivation of the CCRM. There is some evidence that a fourth (and possibly fifth) variable may be worth evaluating in future studies. Factors such as social fabric and community leadership that may fall under the heading of corporate community philanthropy were seen in the literature[67-72] and some of the survey replies. Additionally, other factors like community power dynamics and digital access to information need to be included in the overall body of research to avoid oversimplifying or overemphasizing a single factor like income. Finally, this research was focused on corporate employment variables that impact individual resilience, thereby impacting community resilience. In this regard, corporate philanthropy was viewed as a mechanism that jumps to community resilience and is not necessarily connected to the individual.

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JEM

A Hegelian approach to resilient communities

Richard A. Buck, PhD

ABSTRACT

This theoretical study draws on the insights of Georg Wilhelm Friedrich Hegel to suggest certain aspects of communities and other groups that would tend to make them more resilient in the face of climate change. While Hegel addresses resilient dimensions at the societal level, this study interprets Hegel's work to derive aspects of groups within society that would tend to make them resilient.

Key words: climate change, communities and groups, Hegel, resilient, theoretical; DOI: 10.5055/jem.0811

INTRODUCTION

This study draws on the insights of Georg Wilhelm Friedrich Hegel to suggest certain aspects of communities and other groups that would tend to make them more resilient to climate change.

Sylves[1] cites *resilience* as "the ability of individuals, communities, organizations, and states to adapt to and recover from hazards, shocks, or stresses without compromising long term prospects for development." *Recovery* connotes the re-establishment of normalcy and *adaption* signifies making the changes required by the new reality brought on by climate change.

The author comes to the question of resilience mainly from the experience of a 30-year career with the United States (US) government's disaster relief programs. The author worked with governments and communities throughout the US and its territories and witnessed that successful disaster response and recovery in communities often was related to strong internal relationships and good ties to their broader environment. Granted, climate change is different from disasters, although more disasters are likely to result from climate change. Unlike disasters, climate change continues for many years, and its effects may be gradual or even debatable. Yet, resilience to disasters and climate change has two aspects in common: communities strive to return to normalcy, and they must adapt to changes in their environment. Since the retirement from the Federal Emergency Management Agency, the author has pursued a scholarly career studying philosophy and the transformation of consciousness and societies, and therefore, the author can also bring a theoretical perspective to the examination of resiliency.

The author's observation of the importance of good internal relationships and outside ties is confirmed by empirical studies. Aldrich[2,3] investigated five disasters: the 1923 Tokyo earthquake, the 1995 Kobe earthquake, the 2004 Indian Ocean tsunami, Hurricane Katrina in 2005, and Japan's 2011 earthquake, tsunami, and nuclear meltdown; he concluded that social capital is more important to resilience than factors such as economics, resources, assistance from the government, and levels of damage. Social capital for Aldrich consists of good internal networks among people in a group, as well as good networking with outside groups. Crisman et al.[4] analyzed resilience in Puerto Rico after Hurricane Maria. The authors found government performance in restoration of services, and providing emergency assistance was poor; however, the people of some towns and villages worked together to put themselves on the road to recovery. These communities were resilient not because they had accumulated resources or had special capabilities; rather, they initiated recovery because of strong internal relationships and external bonds with people and groups in their broader geographic area.

What is missing from the author's observations and the conclusions drawn from the two empirical studies cited is a theoretical understanding of how internal and external relationships affect resilience. A robust theoretical foundation for community resilience would be useful to practitioners working with communities facing the effects of climate change and would be a basis for empirical research that would further our understanding of community resilience. This study turns to Hegel because of his insights on what constitutes stable and effective civil societies and states, which are robust enough to withstand disruption and change.[5,6] While Hegel draws his conclusions at a societal level, this study examines the question of whether dimensions of Hegel's robust society have relevance to communities and other groups of people within societies. The research question is: Can Hegel's societal dimensions be translated into community or group aspects that would promote their resilience?

METHOD

The method is theoretical and is employed to interpret Hegel in ways that will permit application of his ideas to groups within societies. Other thinkers are referenced to enhance the argument, and empirical studies are consulted that shed light on this extension to groups and communities. The term "group" will be used in referring to groups and communities internal to societies because the study is intended to apply to both communities and groups that would not normally be named "communities," such as businesses and government agencies.

FINDINGS

The four Hegelian social dimensions are diversity, full participation, shared social goals, and external relations. The resilient group encourages diversity, taking advantage of the diverse talents and perspectives of its members. People are recognized for applying their skills toward the good of the group and are encouraged, likewise, to employ those skills to enhance their own lives. Everyone in the group is urged to participate actively in group activities. This includes decision-making, which includes a form of power sharing in which the person most capable in a particular endeavor is assigned the task of organizing and leading that endeavor. Receptivity to diversity and full participation can happen only if members share in social goals, which include concurrence on rights and duties within the group.

External relationships of group members can be supportive or detrimental to group resilience. Since Hegel addresses only state-to-state relationships, he does not offer guidance on the external relationships of people and groups within society. This is an area that needs more theoretical development.

DISCUSSION

Hegel[7] distinguishes the state from civil society primarily by their different ultimate ends: civil society having to do with the happiness of individuals and the state having to do with the common interest. He confines his conclusions to the societal and institutional levels, so this study recasts Hegelian societal dimensions to permit their application to groups.

To focus on the group level, this study refers to Hegel's view of human nature, which premises all human relationships on the need for recognition. He begins his investigation of human relationships in the *Phenomenology of Spirit* with the master/slave dialectic.[5,8] The master achieves power over the slave because she overcomes him in combat. The slave achieves power over the master by making things for her, and the master now must recognize her dependency on the slave. The seeking of recognition and the fact of mutual dependency then become the basis for all social interactions, including interactions at the group level.

Diversity

Hegel[5] notes that individual diversity is given in the population of a state. Individuals have a diversity of worldviews, talents, and capabilities, and this variability results in a diversity of needs, interest, and desires. It is through *work* within institutions of civil society that the diversity of individual interest can be satisfied. To accommodate the *diverse needs of individuals*, the institutions of the state and civil society must be *diverse*. Hegel establishes in his civil society a variety of estates, institutions that perform the necessary economic and political functions of the state and civil society. Hegel intends that every individual will find a place in one of the estates that fits his or her capabilities and satisfies his or her needs.

A group within a society can likewise embrace diversity, utilizing individual diverse talents and capabilities to promote both the good of the individual and the good of the group. Rather than diverse institutions, it offers them diverse roles that fit their unique talents. This increases the likelihood of everyone being *recognized* for their contributions. The group must emphasize the granting of recognition, particularly for what might be viewed as minimal contributions and for unusual talents that might not be seen as important by many. What is key is that everyone be recognized for their group role, whether it is the exercise of a talent that is crucial to survival or one that may be thought of as trivial.

The importance of diversity in groups is supported by an economic model developed by Page.[9] His focus is mental diversity, that is, looking at diversity in ability to predict outcomes and solve problems. Page tests his model against a variety of empirical studies that examine the performance of diverse and nondiverse situations, ranging from the behavior of markets to small group experiments set up to test performance. Some of the empirical studies focus on identity diversity, and Page concludes that the results of such studies are mixed. Some show a positive correlation between identity and mental diversity; others do not; however, Page finds that identity diverse groups do perform better when they are unanimous in a desired outcome. Under Page's model, mental diversity trumps mental homogeneity because it can achieve the *global optimum* solution to a problem, the best possible solution. A homogeneous group arrives at a solution that it is incapable of improving upon, a local optimum. The group is stuck because all the members have similar local optima. In diverse groups, the members have numerous different local optima. So, Person A demonstrates that his solution is better than that of Person B, and Person C shows that her approach is better than those of A and B. Person D comes up with an even better method. Diverse groups get closer to the global optimum.

Full participation

Hegel[6] observes in *Elements of the Philosophy of Right* that every individual has the right to participate in civil society and to be *recognized* for his or her contribution. Individuals participate by working within their estates or the institutions of government.

In groups within societies, participation would be achieved by employing everyone's talents in execution of group projects and bringing everyone into the decision-making process. As with diversity, it is important that everyone is recognized for his or her contribution. If some people in the group are left out of group activities or decision-making or simply not recognized, they become a potential source of disharmony and possibly, rebellion. Social psychologist Dacher Keltner[10] proposes the idea of power sharing as a way of maximizing participation in a group. No one would be permanently in charge; rather, the person best able to assume leadership for a particular project would take leadership of that project and then relinquish leadership when other projects arise in which others would be better leaders.

Keltner[10] examines studies of other primates, anthropological investigations of indigenous hunter–gatherer groups, and conducts small group experiments using students. In the primate and hunter–gatherer data, he finds that often different individuals assume power in different circumstances. In his small group experiments, he discovers a variety of strategies that result in power going to specific individuals. Invariably, power is assumed by the individuals who are perceived as being able to do the most for the group. As circumstances change, different leaders emerge.

With a diverse and power-sharing group, resiliency stems from the fact of both diversity and power sharing. Diversity creates a greater variety of people with skills to take on specialized tasks, which will be needed to restore normalcy and to confront a changed environment. Power sharing results in a larger pool of people experienced in leadership in different kinds of situations. Furthermore, experience as a leader often makes a person a better follower; that is, the person understands what it is like to lead, knows a leader often acts based on uncertain information, and, therefore, is likely to give him or her the benefit of the doubt.

Shared social goals

For Hegel's state to exist as a stable entity, there must be general agreement on the direction of the state as a whole—the common interest. Pelczynski[11] concludes that, for Hegel, common national and cultural backgrounds are necessary for the proper functioning of civil society. National community results in the shared interest and goals required for public action, and civic culture brings citizens to agreement on their rights and duties.

A group within society is likewise affected by national community and civic culture. There must be agreement on the ultimate goals of the group and members' rights and duties as part of the group. Disagreement on goals and rights and duties would limit the extent of the group's ability to encourage diversity and promote participation. Page[9] sees divergence within national and civic culture as a limitation on diversity, which he names as *difference in fundamental preferences*. Whereas instrumental diversity (using different means) brings globally optimal approaches to problem solving, fundamental preference diversity (wanting different outcomes) can bring a group to a standstill.

External relations

Hegel's analysis of the external relations of states is an exercise in explicating international relations, since the state is a sovereign entity interacting with other sovereign entities. As such, it exists in a state of nature with other sovereigns.[12] Peace is maintained using diplomacy, but war may be the only way the state can achieve its ends.

A group within society does not face a state of nature with other groups and individuals. Members act within a social and legal environment, as do people in the other groups with whom they are dealing. The group is strengthened when individuals in the group have supportive relationships with those outside. It is weakened when the outside relationships threaten the cohesion of the group. An example of an outside relationship threatening the group would be family relationships that prevent individuals from devoting full attention to group matters. So, resiliency can be enhanced by supportive outside relationships and degraded with divisive ones. Crisman et al. find in their empirical investigations that outside supportive relationships are important to resilience.[2-4] Local people in the face of governmental failure reached out to their contacts for needed expertise, supplies, and equipment. A theoretical exploration of the resilience value of external relations should look at supportive and detrimental outside relationships, outside institutional affiliations of members, and a variety of other outside connections, including family, ethnic, religious, friendship, political, professional, business, governmental, and charitable.

CONCLUSIONS

Resilience of groups in society

Returning to the research question: Can Hegel's societal dimensions be translated into community or group aspects that would promote their resilience? The answer is in the affirmative for three of the dimensions: diversity, full participation, and commitment to shared goals. The fourth dimension, external relations, cannot be directly translated from Hegel.

A diverse group learns to tap into the full resources of the members of the group and has a good chance of arriving at the global optimum solution to both recovery and to dealing with changes in the environment.

The group that encourages full participation by all members in both action and decision-making can, by power sharing, call on the most capable members to lead projects for which they have special expertise. Leadership, then, becomes a function of capability in dealing with the current crises; leadership is changed when a new crisis demands a different expertise.

The smooth functioning of the diverse group fostering full participation is made possible by shared goals, which promotes the highest stage of group interaction, a common understanding of group rights and duties. At this point, the group gains further resiliency by the unification of goals between individuals and the group, so that there is no dissension when a path is selected to recover and address environmental change. This resiliency would be further enhanced by supportive external relationships between group members and people outside the group, but Hegel does not provide much guidance in this because he deals only with state-to-state external relations. For groups within society, external relationships are much more complicated, with different individuals and subgroups having external relationships which have a varying effect on the group. Empirical evidence suggests its value in resiliency, but further theoretical work and empirical research are needed.

Direct application and policy implications

This study provides a model for resilient groups that would serve as a resiliency training tool for communities and other groups that need to prepare themselves for problems resulting from climate change. It lends itself to a bottom-up approach. While it does not deal with the content of climate change issues, it has within it the building blocks for maximizing a group's ability to deal with any new crisis or situation that threatens it without bringing in big business or big government. The model uses the talents of the diverse group members to both do the work and make the needed decisions. It emphasizes full participation of all members, no matter how great or minimal their ability to contribute. Power sharing creates a leadership pool that extends to much of the group. Learning how to use diversity and full participation and how to expand external relationships would come from doing. It would start with everyday matters (such as repairing sidewalks) and expand to cover much of the activities of the group. With this concept, rather than hire a consultant to develop a plan for a climate change problem, the group itself would do the planning and potentially the physical work that might be needed to protect the group.

Practitioners who are attempting to revitalize failed institutions anywhere could replace those institutions with new groups built on the resilient model presented here or reform existing groups to be resilient using the four Hegelian dimensions. An example would be using such groups to revitalize failing Catholic parishes in Ireland. The parishes traditionally have done much of the social and educational work in Ireland, but many do not have enough priests and parishioners to continue being this vital force in the country. Lay people in Catholic groups could take over this work.

Future research

1. Investigate the prevalence of resilient groups with the characteristics developed here in various segments of society.

2. Find groups having the four Hegelian dimensions and examine the strength of each dimension and the statistical correlations among them. Then relate the findings to an objective evaluation of their level of resilience.

3. Conduct case studies of groups exhibiting the four Hegelian dimensions, examining how each dimension contributes to group resilience.

4. Develop a theoretical model of the effect of external relations on group resilience.

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Ocean state rising: Storm simulation and vulnerability mapping to predict hurricane impacts for Rhode Island's critical infrastructure

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ABSTRACT

Predicting the consequences of a major coastal storm is increasingly difficult as the result of global climate change and growing societal dependence on critical infrastructure (CI). Past storms are no longer a reliable predictor of future weather events, and the traditional approach to vulnerability assessment presents accumulated loss in largely quantitative terms that lack the specificity local emergency managers need to develop effective plans and mitigation strategies. The Rhode Island Coastal Hazards Modeling and Prediction (RI-CHAMP) system is a geographic information system (GIS)-based modeling tool that combines high-resolution storm simulations with geolocated vulnerability data to predict specific consequences based on local concerns about impacts to CI. This case study discusses implementing RI-CHAMP for the State of Rhode Island to predict impacts of wind and inundation on its CI during a hurricane, tropical storm, or nor'easter. This paper addresses the collection and field verification of vulnerability data, along with RI-CHAMP's process for integrating those data with storm models. The project deeply engaged end-users (emergency managers, facility managers, and other stakeholders) in developing RI-CHAMP's ArcGIS Online dashboard to ensure it provides specific, actionable data. The results of real and synthetic storm models are presented along with discussion of how the data in these simulations are being used by state and local emergency managers, facility owners, and others.

Key words: hazard impacts, storm consequences, coastal hazards, vulnerability assessment, risk management, decision support tools, participatory action research, implementation research; DOI: 10.5055/jem.0801

INTRODUCTION

Forecasting and predicting the consequences of major ocean storms (hurricanes, tropical storms, and nor'easters) has become more challenging due to progressive climate change, rising sea levels, growing coastal populations, and increasing societal dependence on critical infrastructure (CI) for daily functioning.[1-3] Emergency managers in coastal regions often have difficulty accessing the detailed, actionable data needed to anticipate impacts and consequences of major storms prior to landfall.[4] Many of the traditional tools local emergency managers rely on for this task are designed to estimate damage and loss in the aggregate and do not provide the specific information needed to support data-driven decisions about emergency planning and hazard mitigation.

We developed the Rhode Island Coastal Hazard Modeling and Prediction (RI-CHAMP) system to address this gap, leveraging recent advances in storm modeling[5] and computer simulation to provide state and local emergency managers with localized consequence data.[6,7] Our approach predicts specific potential storm consequences by combining high-resolution storm model outputs with expert-informed georeferenced data about CI facility and asset vulnerabilities.[8] The RI-CHAMP project is a participatory effort that engages stakeholders and subject matter experts to build a decision support tool meeting the identified needs of end users, primarily state and local emergency managers.

The RI-CHAMP tool incorporates our team's prior work developing Hazard Consequence Thresholds (HCTs) that provides (1) links of quantitative impact measures with qualitative consequence descriptions,[9] (2) a robust connection to storm models that provides quantitative input to trigger HCTs,[8-10] (3) spatial analysis and visual presentation of vulnerabilities and predicted storm consequences,[8] and (4) a data collection methodology for CI vulnerabilities using a Participatory Action Research approach.[6]

This paper describes the case study to help researchers and emergency managers build and deploy hazard consequence prediction systems as decision support tools that can minimize the impact of major storms for coastal communities and reduce unanticipated consequences. We address the research question: "How can the results of storm simulation and consequence prediction be delivered to Rhode Island's emergency managers, so the information helps them prepare for a storm's landfall and/or prioritize long-term mitigation and planning efforts?"

This implementation research uses a mixed methods approach with a sequential transformative design. Implementation research helps integrate research findings into real-world practice by examining how interventions function in the field without efforts to control or remove other causal effects.[11] This paper begins with a background on hazard consequence prediction in emergency management and then presents the results of a RI-CHAMP training workshop held with federal, state, and local emergency managers in January 2021. In the "Discussion" section, we apply these findings to present a framework for delivering consequence prediction results to end users. We conclude by discussing the implications of this work for continued development and implementation of hazard consequence prediction tools.

BACKGROUND

Traditional vulnerability assessment tools lack the resolution local emergency managers need

To be useful for hazard mitigation and emergency response planning, quantitative hazards data in a hazard vulnerability assessment (HVA) must be linked to qualitative vulnerability outcomes, including the mapping of vulnerability distribution based on multiple attributes.[12] Here, "useful" is defined as that which is relevant, pertinent, and able to guide decisions such as steps regarding disaster preparedness, response, and mitigation.[13,14] These data are based on comprehensive analysis of hazard information, exposure, vulnerability, and adaptive capacity; the relationships between these elements and a given population's capacity to manage these risks cannot be determined based on damage and loss data alone.[15]

Many emergency managers use tools like the Federal Emergency Management Agency (FEMA)'s HAZards United States (US) (HAZUS) and HURRcane EVACuation (HURREVAC) to predict the impacts of a given storm scenario. These tools are primarily designed to help state and federal agencies anticipate response capabilities and recovery needs; they estimate aggregate damage and loss, typically in economic terms, but do not identify specific impacts that local decision makers can act upon.[16,17] Without detailed storm impact predictions, local officials often rely on experience with past storms and develop emergency plans based on easily anticipated scenarios but do not address the potential for low-frequency, high-impact storms.[17] Consequently, many coastal communities ". . . drastically underestimate the extent of damage and loss of life and the economic and infrastructural implications" that may result from such events.[17] Failure to appreciate the full extent of local vulnerability is why storms like Hurricanes Katrina (2005), Sandy (2013), and Maria (2017) are known to produce what Parker[18(p207)] call a *policy surprise*: "an abrupt revelation—often after being victimized by a sudden disaster or unanticipated event—that one has been working with a faulty threat perception regarding an acute danger . . ."

Simulation and modeling help local decision makers plan and prioritize

Computer-based storm simulations, such as the National Weather Service's Sea Lake and Overland Surges from Hurricanes model (SLOSH), have long been recognized as useful for emergency preparedness and can help local decision makers evaluate the interaction of storm hazards with human populations and the built environment.[8,19,20] Simulation also helps stakeholders and decision-makers understand and communicate the complex risks presented by storm hazards and the cascading consequences that may be triggered during a major storm.[12,21]

Commonly used models, such as SLOSH, are designed to run quickly with minimal computational resources to rapidly provide actionable information. Their ease of use has helped to ensure wide adoption, and they play an important role in storm preparedness. However, these models do not have sufficient granularity to help emergency managers to identify and address specific local concerns. Advances in hydrodynamic storm modeling offer a more exacting and realistic output that can be used to simulate the complex interaction of flood waters with the land and built environment.[3,22,23] The Advanced CIRCulation (ADCIRC) and Simulating Waves Nearshore (SWAN) models used with RI-CHAMP for this study predict inundation and wind at a much higher resolution than earlier generations of storm models.[24] This advanced storm modeling helps predict storm consequences and outcomes when linked with the specific, localized vulnerabilities of human populations and communities.[3]

The RI-CHAMP approach to consequence prediction

The RI-CHAMP system provides state and local emergency managers with rich, detailed data about potential storm impacts for CI by combining high-resolution wind and inundation models with geolocated, qualitative data about CI vulnerabilities that are collected directly from subject matter experts.[8-10,25] The result is an actionable inventory of vulnerable CI with descriptions of consequences should that infrastructure be damaged or destroyed. Results can be accessed through an online dashboard or through printed reports (Figure 1). Users apply RI-CHAMP in either of two ways: to guide real-time storm preparations for an actual storm (a "current storm forecast") or to simulate various fictional storm scenarios ("reference storms") as a basis for long-term mitigation and planning.

CI is a well-established basis for measuring physical vulnerability because it is amenable to deterministic predictions—ie, HCTs—the specific points at which experts feel that assets may be impaired by wind or inundation. CI assets contribute to specific societal functions, so community impact often correlates with the extent to which functioning of the CI is impaired.[16,26] Other measures of vulnerability—eg, Social Vulnerability Index[27]—are useful for determining comparative vulnerability but are not conducive to deterministic predictions and cannot be easily tied to storm models due to the complexities of utilizing aggregated areal data.[28] The same is true of metrics based on economic loss, such as the FEMA damage curves often used with HAZUS and SLOSH; the damage curves predict damage to facilities as a percentage of replacement cost without reference to loss of function.

Implementing simulation-based hazard consequence prediction in emergency management

Despite their many advantages, hazard consequence prediction tools like RI-CHAMP have yet to enter mainstream use by emergency managers. Research-based tools often struggle with the transition from laboratory to real-world practice because it can be difficult to produce scientific information that is helpful to end users.[14] With RI-CHAMP, more work is needed to understand exactly what information is useful to emergency managers and describe how prediction data will inform local decision-making. Tools like RI-CHAMP are more likely to be widely adopted if decision-makers find them effective in reducing community vulnerability to major storms and achieve outcomes related to acceptability, adoption, appropriateness, feasibility, fidelity, implementation cost, coverage, and sustainability.[29]

METHODS

Multiple studies suggest that stakeholder involvement plays an important role in developing data

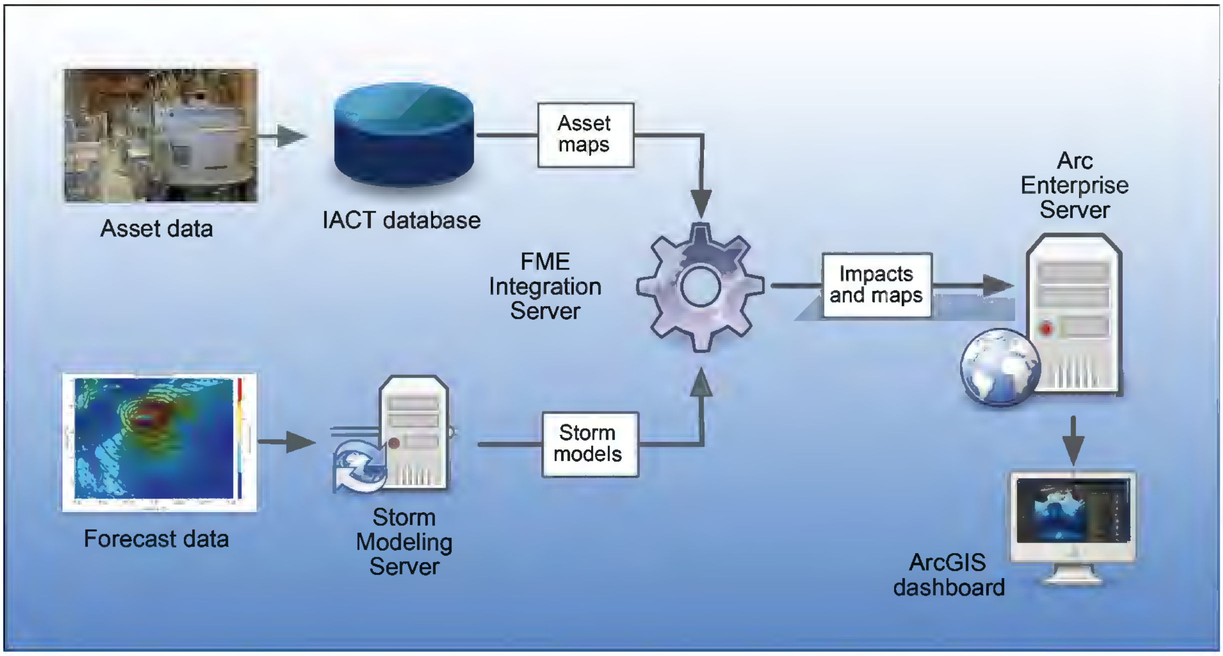

Figure 1. The RI-CHAMP workflow. The Infrastructure Asset Consequence Threshold (IACT) database generates asset vulnerability maps from consequence threshold data, while the Storm Modeling Server produces storm models from NOAA forecast data. These outputs are analyzed in the FME Integration Server to deliver impact predictions and maps to an Arc Enterprise Server, which renders the data in an ArcGIS Online dashboard available to end-users (image: authors).

systems used for decision support.[20,30] Creating tools that are easy to use and relevant to the intended audience facilitates uptake by real-world users.[20] The RI-CHAMP project is guided by an interagency steering committee that represents anticipated end-users, including federal, state, and local emergency management officials (Table 1). Involvement of trusted state and local agencies helps foster confidence and participation of privately held facilities that might otherwise be reluctant to furnish sensitive data about their CI.[6] The steering committee also helps develop CI datasets relevant to emergency management users and guides dashboard interface development based on end-users' information needs. Cantor[30] calls this approach "decision-driven data," whereby the data design reflects the decisions that will be made with that data.

Study site—Rhode Island CI in the coastal zone

Our study included HCT data collection for two CI subsectors using an online survey tool and developing a web-based dashboard for viewing the results of RI-CHAMP's hazard consequence predictions. We based our study in the State of Rhode Island (Figure 2). Narragansett Bay bisects the eastern portion of the state, extending from the Atlantic coast to the state capital of Providence. At Providence, the Woonasquatucket and Moshassuck Rivers feed the Providence River, which discharges into the head of Narragansett Bay at Fox Point. Here, a flood barrier can be closed to protect the city's downtown area from storm surge pushed up the bay by a major storm. When the Fox Point Hurricane Barrier is closed, three large outfall pumps discharge storm runoff from the Providence River and inner harbor into the bay. Providence has experienced significant amounts of flooding during historical storms such as the so-called Great New England Hurricane of 1938 and Hurricane Carol in 1954.[31] According to the US Army Corps of Engineers, Rhode Island's coastline is ". . . continuously affected and transformed by storms

Table 1. RI-CHAMP Steering Committee membership (2015-2022)		
Agency	**Title/role**	**Sector (subsector)**
City of Providence, Capital Asset Management & Maintenance	Director of Security	Government
City of Providence, Emergency Management Agency	Director	Emergency Services
City of Providence, Public Works Department	Public Property Coordinator	Government
City of Providence, Water Supply Board	Director of Engineering	Water & Wastewater
Lifespan	Director of Enterprise Business Continuity Planning	Health & Medical
Narragansett Bay Commission	Engineering Manager	Water & Wastewater
Rhode Island Emergency Management Agency	Critical Infrastructure Key Resources Manager	Emergency Services
Rhode Island Department of Environmental Management	Principal Engineer	Water & Wastewater Systems
Rhode Island Department of Health (RIDOH)	Deputy Chief, Center for Emergency Preparedness & Response	Emergency Services
RIDOH	Program Support Specialist	Healthcare & Public Health
Rhode Island Department of Transportation	Chief of Sustainability, Autonomous Vehicles, and Innovation	Transportation
Rhode Island Emergency Management Agency	Operations Section Chief	Emergency Services
Rhode Island Energy (formerly National Grid)	Senior Coordinator, Investment & Economic Development	Energy
Rhode Island Infrastructure Bank	Chief Resilience Officer	Government
US Coast Guard	Marine Transportation Recovery Specialist	Transportation

and tidal inundation . . .," including hurricanes and nor'easters.[31] With 21 of its 39 municipalities exposed to storm surge, large ocean storms are a significant feature of emergency management in the Ocean State.

Training and testing workshop

Our study included a training workshop for state and local emergency managers to evaluate the beta user interface and discuss how emergency managers will use information from RI-CHAMP's analysis and incorporate it into their existing workflows. Workshop participants (n = 14) served as a focus group to validate the dashboard's usefulness and included representatives from the emergency management organizations most likely to implement RI-CHAMP. These included the Rhode Island Emergency Management Agency, Providence Emergency Management Agency, Rhode Island Department of Environmental Management Wastewater Division, Rhode Island Department of Health, and others from the project steering committee (Table 1).

A 3-hour workshop was held virtually due to the ongoing coronavirus disease 2019 pandemic. The research team provided each participant access to the RI-CHAMP portal in ArcGIS Online with three scenario-specific dashboards prepared for the study (described below). The workshop included an instructional component followed by a "hands-on"

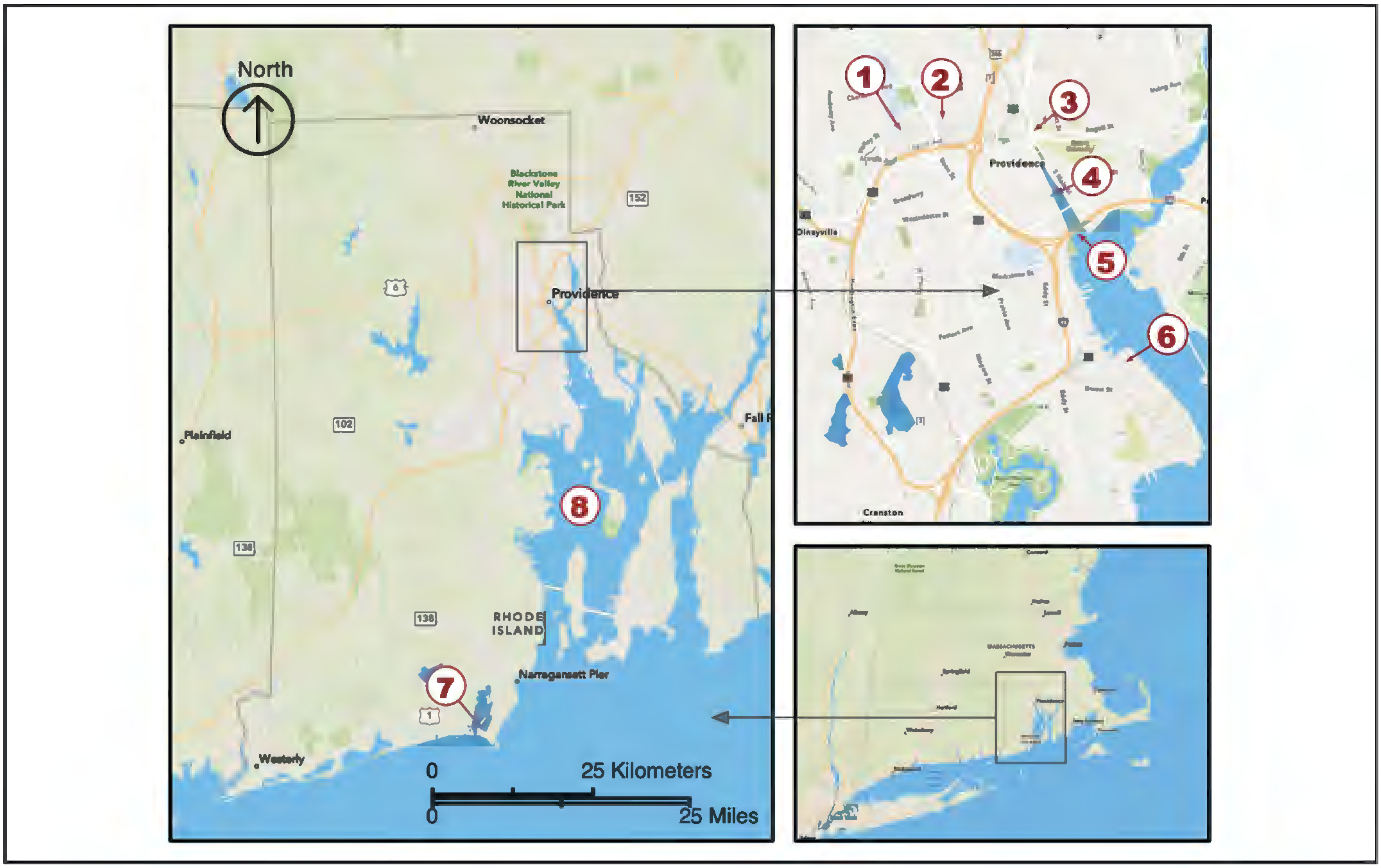

Figure 2. Map of the State of Rhode Island showing (1) Woonasquatucket River; (2) RI Blood Center; (3) Moshassuck River; (4) Providence River; (5) Fox Point Hurricane Barrier; (6) Port of Providence fuel terminals; (7) Block Island Ferry Terminal at Galilee; and (8) Narragansett Bay (image: authors).

practical exercise session, in which users logged into the RI-CHAMP portal and worked with several storm scenarios. At the conclusion of the workshop, learners were asked to describe at least two ways RI-CHAMP output can be integrated with Emergency Operations Center (EOC) tools they use regularly, eg, WebEOC, and identified ways RI-CHAMP outputs might inform preparedness and response activities in their respective agencies.

Although most learners were familiar with the RI-CHAMP project, this was their first time using the recently completed dashboard. We collected participant feedback by recording and transcribing the 3-hour workshop and then categorized comments as Positives, Concerns, Suggestions for Improving the System, and Suggestions for Improving the Training. We also asked learners to complete an anonymous online survey to capture their overall impressions (URI IRB Protocol # IRB1819-226).

Storm models used in RI-CHAMP training and testing workshop

We used wind models to establish wind fields to force hurricane simulations using the ADCIRC model and SWAN model to predict water levels and wave action. Riverine flooding was also incorporated for the Woonasquatucket River so learners could simulate functioning of the Fox Point Hurricane Barrier and its outflow pumps. ADCIRC simulates water levels using an unstructured grid with variable distances between nodes (locations where calculations for the physical model are performed). Nodes in the nearshore area may be adjusted to suit the topography and be as close as 20-40 m.[32,33] This allows much higher resolution than SLOSH and similar tools that are designed to predict inundation using regular grids with node spacings ranging from tens of meters to a kilometer or more.[10,34] The ADCIRC water level predictions are used with high resolution (~1 m) digital elevation models to establish inundation depth. With ADCIRC and associated wind models, users have access to inundation and wind speed on a block by block or even parcel by parcel scale at regular time increments (timesteps) in addition to maxima.[8,23]

Our modeling team developed three hypothetical, but physically plausible, storm scenarios for the workshop. Scenarios 1 and 2 used the fictitious "Hurricane Ram," a slow-moving Category 3 storm that stalls over the Providence area, depositing 18.5 in. of rain in a 41-hour period. Ram served as a "reference storm" that RI-CHAMP learners could use to answer long-term planning and mitigation questions. Because the Hurricane Ram simulation includes variables for the Fox Point Hurricane Barrier and storm water outfall pumps, modelers configured Scenario 1 with the barrier closed and all pumps operational. Scenario 2 was identical except that the overflow pumps were disabled, allowing learners to explore the consequences for downtown Providence should storm runoff be trapped behind the hurricane barrier (an unlikely but potentially devastating scenario). Scenario 3 represented a "current storm forecast" that learners could use to simulate real-time preparations for an actual storm. Scenario 3 used the fictitious "Hurricane Ishmael," a powerful and fast-moving storm based on the great Hurricane of 1938, a Category 3 storm with sustained wind speeds of 100+ mph and an 18- to 25-foot storm tide, modified with an additional 12 in. of sea level rise and an eastward shift to the storm track that brings the eye of the storm directly over Block Island and up Narragansett Bay. Modelers designed Ishmael to be a high-impact, low-probability storm that would encourage emergency managers to consider future worst-case scenario events beyond those they may have experienced previously.

Collecting CI vulnerability data

We worked with local facility managers to populate the Infrastructure Asset Consequence Threshold (IACT) database with HCTs for two CI subsectors statewide—wastewater treatment and maritime transport—using the participatory method described by Becker et al.[6] We also included existing IACT data from earlier pilot studies in Westerly and Providence, Rhode Island. Each HCT includes the asset name or other identifier, the specific location of the asset (X/Y coordinates), the flood depth and/or wind speed at which subject matter experts—eg, facility managers—estimate functioning of the asset would be compromised, the consequences of that asset's compromise, and likely restoration time. HCT entries also include emergency contact information for the facility in question and a photograph of the asset when possible. Each entry is associated with a CI sector based on the function it performs, an Emergency Support Function (ESF) based on who needs to know about potential compromise of that asset during a storm scenario, and a Community Lifeline indicating how the asset relates to societal functioning.

Facility managers worked with our team to enter HCT data in the field using a survey form we built for the ESRI ArcGIS Survey123 application. Submitted data were stored in a secure ESRI Arc Enterprise database where it could be reviewed and edited through an ESRI ArcGIS Online dashboard. We also developed an online training for RI-CHAMP's Survey123 app to help facility managers gather and submit their HCTs. For this study, the IACT database included 575 HCTs for 375 CI assets collected from facility managers statewide by our team[6,7] (Figure 3).

Consequence prediction

RI-CHAMP determines likely storm consequences by comparing storm model output against georeferenced consequence threshold data to identify assets that will likely exceed one or more of their thresholds. For example, "If the location of the backup generator has 36 in. of flooding, we expect to lose power to our network servers, resulting in a loss of communication for 50 percent of our network." HCTs from the IACT database are compared directly to outputs from wind and surge models using an automated post-processor written in Python. The direct comparison of consequence threshold points to the models allows for rapid analysis downscaling of data without delays and degradation of

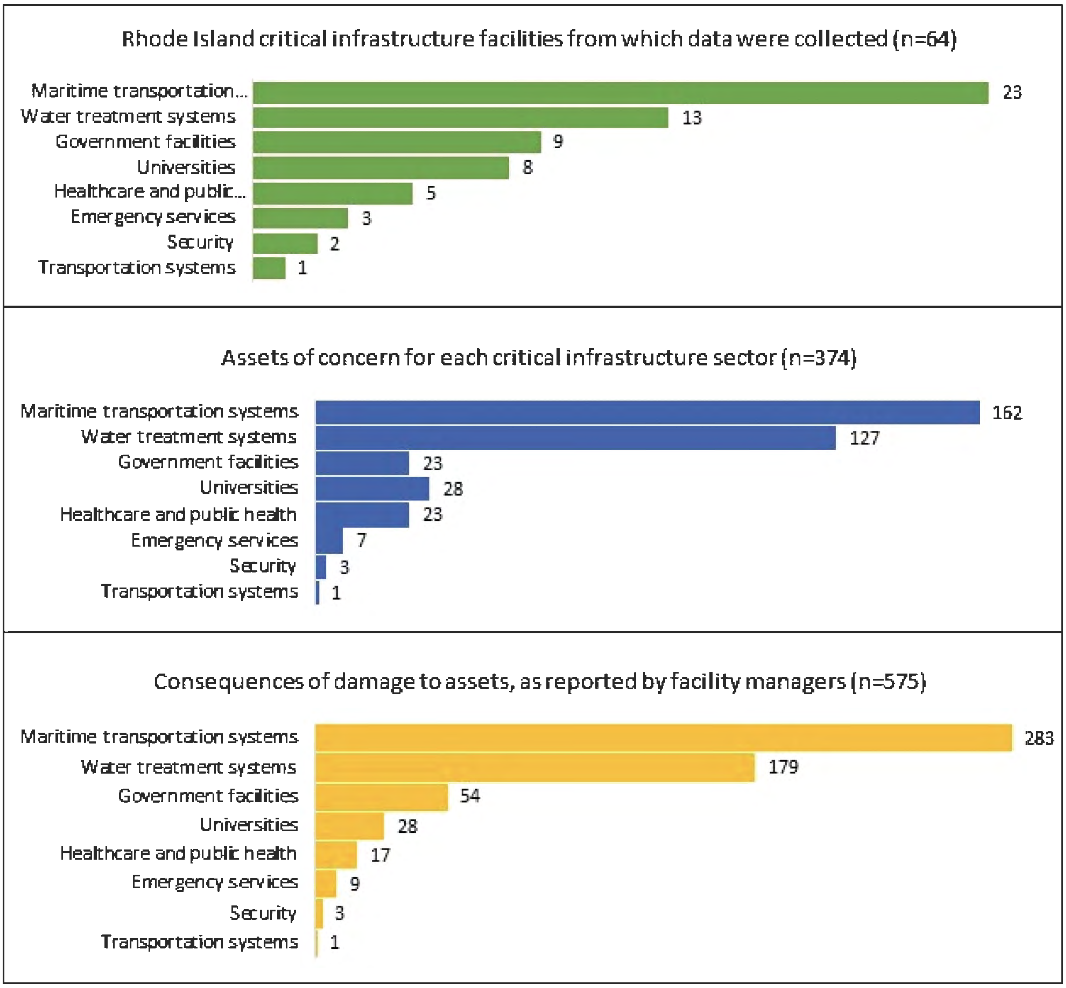

Figure 3. IACT entries used for the RI-CHAMP end-user workshop (image: authors).

resolution that can occur using more conventional geographic information system (GIS) workflows. Of equal importance, conducting analyses in a single step using a common programing language allows for scrutiny and avoids errors resulting from complex workflows.[8]

User interface

Outcomes are written back to the ArcEnterprise server, which populates an interactive RI-CHAMP dashboard that emergency managers and facility operators can use to filter and view consequence predictions (Figure 4). For an actual/real-time storm, the scenario is reprocessed, and the dashboard refreshed, each time the National Weather Service updates the storm forecast (typically every 6 hours). By default, the RI-CHAMP dashboard shows maximum inundation and wind speed in the scenario dashboard. If a storm simulation is time-enabled, a tab is added to the dashboard allowing users to scroll back and forth through the model's timeline and see the progression of wind and inundation over time, and the resulting impacts to CI facilities and assets.

RI-CHAMP presents simulation output—ie, the combination of a specific storm simulation with consequence threshold data—to end users visually using the web-based ArcGIS Online dashboards viewer. Real-time and near-real-time dashboards are rapidly increasing in popularity for disaster and emergency management because they provide multiple users in different locations the ability to view dynamically

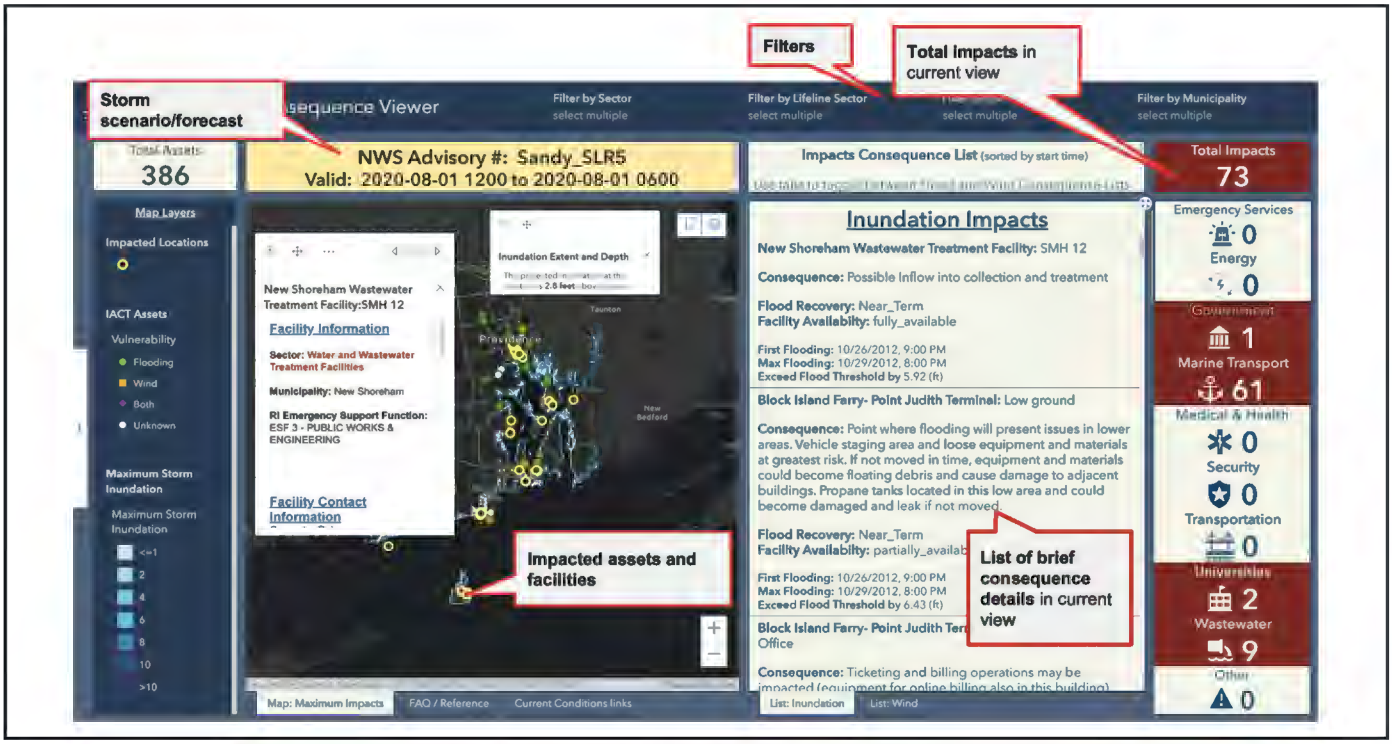

Figure 4. RI-CHAMP dashboard showing the results of a storm simulation with impacted CI highlighted on the center map. The impacts are also provided in a list view along with a breakdown of total impacts by CI sector (image: authors).

updated statistical data and geospatial information in a single interface.[35] Despite this increase in use, actual studies detailing their workflows, use, and design are limited.[8,35] Essential design considerations include delivering timely information to decision-makers without unneeded complexity or detail that could render the information unintelligible.[36,37] Other concerns include the careful management of uncertainty and the potential for detailed visualizations to mislead audiences through their seeming certitude (a concern mitigated in part by the expertise of RI-CHAMP's intended users).[34] We focused on delivering clear, actionable, and clearly summarized information and gave RI-CHAMP users the ability to filter data based on ESF and CI sector.

In addition to the interactive ArcGIS dashboard (Figure 4), RI-CHAMP can produce templated, time-stamped "prediction reports" in PDF form, summarizing anticipated consequences of the current storm scenario by facility, municipality, CI sector, or ESF. These reports can be used for record-keeping in the EOC and be shared with facility operators and local emergency management partners as needed.

In this study, researchers developed a unique dashboard for each of the three scenarios that combined IACT CI data with wind, wave, and surge data from the storm models to highlight predicted impacts. Because Scenarios 1 and 2 were time-enabled, users could also scroll through an hour-by-hour timeline to anticipate when specific impacts would need to be addressed.

Security and confidentiality

All IACT data are stored on a secure Arc Enterprise Server managed by the State of Rhode Island's Department of Internet Technology. To address concerns over handling of data, workshop participants signed a nondisclosure agreement.

Access controls are also being built into the IACT database itself. Once fully implemented, some IACT facility and asset data will be available to all RI-CHAMP users, while other data are restricted based on security considerations. RI-CHAMP users will have different levels of access to IACT data depending on where they work and what their responsibilities are. In the RI-CHAMP dashboard, permission to view and/or edit IACT data will be determined by a user's organization, facility, ESF role, CI sector, and jurisdiction.

WORKSHOP RESULTS

During the workshop, all participants successfully executed assigned tasks that involved loading specific storm scenarios and answering questions relevant to their role in an EOC or on a planning and mitigation team. Learners discussed how RI-CHAMP predictions would inform decision-making for long-term planning and real-time storm preparations and made response decisions based on information received from RI-CHAMP.

Workshop participant feedback

When asked to rate the extent to which the workshop accomplished its objectives, all respondents (n = 9) said they felt "good" or "very good" about their ability to run hazard scenarios and generate consequence prediction reports using the RI-CHAMP dashboard. Most respondents also felt "good" or "very good" about their ability to describe how RI-CHAMP outputs will inform preparedness and response activities, and the remainder said they felt it was "acceptable." With respect to demonstrating understanding of the RI-CHAMP tools, all but one respondent said they felt "good" or "very good." On the other hand, none of the respondents felt "very good" about their ability to teach others after completing the workshop, though all said they felt "good" or "acceptable."

Specific feedback captured during the workshop includes:

- The level of detail in RI-CHAMP prediction data will help emergency managers gain stakeholder buy-in, particularly from elected officials.
- The refinement in RI-CHAMP flood models will improve storm preparedness including the ability to better anticipate resource needs and make advance requests prior to landfall.
- RI-CHAMP data can help municipalities prioritize hazard mitigation projects.
- RI-CHAMP simulations can add realism to training and exercises, and one participant reported that their agency is already using RI-CHAMP for this purpose.
- Predisaster facility and asset imagery captured in the IACT database will aid post-storm recovery and reimbursement.
- Adding key dependencies for each CI asset—eg, electricity, natural gas, and internet—in the IACT database would increase usefulness.
- The RI-CHAMP system will be increasingly useful as more data are added; existing GIS datasets can be added easily—eg, evacuation maps, wastewater systems, radio communication towers, electrical transmission lines, historic structures.

Scenario-specific workshop products

Using the workshop's RI-CHAMP scenarios, learners successfully identified CI assets with at least one HCT exceeded by wind or inundation and described the consequences for preparedness, response, recovery, and/or mitigation. Examples include:

- In both "Hurricane Ram" scenarios, fuel terminals in the Port of Providence would likely be inoperable following the storm, impacting the availability of petroleum products for gas stations, aviation, home heating, and other needs.
- In the second "Hurricane Ram" scenario, failure of the Fox Point Hurricane Barrier's outfall pumps would lead to catastrophic

flooding in downtown Providence like what occurred during Hurricane Carol in 1954. Flooding of the Woonasquatucket and Moshassuck Rivers far inland of the hurricane barrier, would inundate CI facilities not normally considered vulnerable to hurricane-related flooding.

- Riverine flooding could inundate the Rhode Island Blood Center, located on the banks of the Woonasquatucket River approximately a mile and a half inland of the hurricane barrier. RI-CHAMP's time-enabled storm model showed that the main entrance of the blood bank would become inaccessible relatively early in the storm, but the rear entrance would flood later, allowing time for contingency management. The time-enabled view also showed that the timing of peak riverine flooding lagged peak surge flooding in the Port of Providence.

- In the near-real-time "Hurricane Ishmael" scenario, the Block Island Ferry terminal in the Port of Galilee (Narragansett, RI) would be inoperable following the storm due to flooding of the hydraulic equipment that operates the ramps needed to load vehicles and cargo on the ferry. This would have a significant impact on recovery for Block Island (New Shoreham, RI) because the ferry is the only means of moving bulk materiel and vehicles to and from the island.

Challenges identified

Participants also identified concerns and challenges related to their agencies implementing RI-CHAMP:

- The system requires time and training to use. Many local emergency management staff have limited capacity to take on additional tasks during a storm, and local emergency management agencies often lack sufficient technical support resources. This could be mitigated by having the State EOC use RI-CHAMP's "consequence prediction report" function to send PDF reports to local jurisdictions rather than requiring them to pull the information from RI-CHAMP themselves.

- Entering and maintaining IACT data will likely be an ongoing challenge. Greater clarity is needed about who will do this work and how data quality and currency will be ensured.

- Facilities—both public and private—are likely to have concerns over the security of vulnerability data they submit and with whom it is being shared through RI-CHAMP. A robust permissioning scheme will be essential.

DISCUSSION

During the workshop, emergency managers successfully used the RI-CHAMP dashboard to identify specific impacts and consequences to local CI for each of the three storm scenarios. They also applied RI-CHAMP's hazard consequence predictions by describing how the data would be used in the workflows of their respective agencies/organizations. User comments captured during the workshop and provided in the post-workshop survey show that participants found the information obtained from RI-CHAMP useful and accessible for decision support in the context of actual storm response and for longer-term planning and mitigation. They noted that RI-CHAMP's simulations allow decision-makers to evaluate the effects of various storm preparations such as having the Fox Point Hurricane Barrier open or closed during a particular storm, and, perhaps most importantly, the simulations provide a means to evaluate specific impacts of storm scenarios the region has yet to experience. This makes tools like RI-CHAMP invaluable for simulating the storm-related effects of climate change on a local scale.

These findings validate RI-CHAMP's participatory approach to developing consequence prediction tools based on the input of end-users and gathering consequence data directly from the individuals most knowledgeable about likely impacts to local CI.

A framework for storm hazard consequence prediction tools

This study has several key implications for building and deploying storm hazard consequence prediction tools. The need to provide local decision-makers with specific, actionable storm hazard consequence predictions is already established in the literature. We show how RI-CHAMP can be used to deliver such predictions to end-users by combining high-resolution storm models with detailed data about local CI vulnerabilities. The following points contribute to a framework for implementing hazard consequence prediction in the emergency management environment:

1. Data systems are most effective and useful as decision support tools when the data design is informed from the beginning by a thorough understanding of producers' and users' decision-making needs.[20,30] We developed the RI-CHAMP system with regular input from a steering committee made up of emergency managers and other officials likely the system. This helps steer the project as a whole and directly contributed to building the dashboard interface used in this study.

2. End-user participation in data collection ensures consequence predictions are directly relevant to emergency managers when allocating resources and preparing for an imminent storm. Participation of local experts with knowledge about their community and its infrastructure is particularly valuable; it improves the reliability and transparency of impact assessments and gives decision-makers a greater confidence in the results.[3,9,20] We worked directly with facility operators to collect consequence threshold data for this study using the survey techniques developed in this project.[6] This has ensured accurate, local data relevant to RI-CHAMP's end-users, as demonstrated by this study.

3. High-resolution storm models are essential to producing credible and accurate consequence predictions.[10,23] Some studies use more generalized models like HAZUS and SLOSH to make probabilistic predictions.[17,20] These are helpful for making broad comparisons between storms and for predicting overall impact, but they do not provide the detailed, deterministic predictions local emergency managers need to prioritize specific actions.[3,38] We built the simulations for this study using ADCIRC and SWAN because these tools calculate conditions on a mesh as fine as 20-40 m, allowing the system to evaluate consequence thresholds at each specific point on a vulnerability map. Although RI-CHAMP can utilize storm models from other sources, its full potential is realized only with storm simulations that match the resolution of its consequence threshold point data.

4. Building RI-CHAMP with widely used and readily available software reduces the learning curve for EOC personnel and facilitates integration of existing datasets such as local roads, hospitals, and evacuation routes.[20] A vendor-supported, "off-the-shelf" product contributes to long-term sustainability and autonomy, placing control in local hands for the continued development of the database and system functions with or without external support from the original product team. Proprietary, purpose-built software can be difficult to sustain over the long haul and often requires an ongoing relationship with the original development team. We built RI-CHAMP to run on ESRI's

suite of GIS products (ArcGIS Online, ArcGIS Dashboards, Arc Enterprise, etc.) so the RI-CHAMP solution can be easily replicated and implemented in other jurisdictions.

Challenges

Like any data-driven system, RI-CHAMP's predictions are only as good as the information upon which they are based. The precision of ADCIRC and SWAN storm models relies on the accuracy of the National Oceanographic and Atmospheric Administration (NOAA)'s storm track and other forecast data. This is of minor relevance with reference storms used for long-term planning and training, but unexpected deviations in storm track can significantly impact real-time storm predictions and response preparations. This potential can be offset by running multiple permutations of the current NOAA forecast but doing so greatly increases processing time.

The accuracy and relevance of RI-CHAMP's predictions are also highly dependent on the quality of vulnerability data supplied to the IACT database, with respect to both the quantitative data for consequence thresholds and qualitative consequence descriptions. Consequence predictions are also constrained by the completeness of vulnerability data, as predictions are limited to the infrastructure captured in the database—RI-CHAMP cannot offer predictions for infrastructure not included in its database. Furthermore, the consequences are constrained by the expertise of the facility manager or subject matter experts who provided the data. Thus, results must be communicated in such a way as to emphasize that these are the concerns voiced by facility managers and not a definitive prediction of what may happen in the event of a hazard threshold being exceeded. It naturally follows that HCT data are only useful if maintained; the IACT database must be regularly reviewed and updated as facilities modify their physical plants, and local jurisdictions carry out mitigation projects. Agencies implementing systems like RI-CHAMP must dedicate resources to ensure vulnerability data are reviewed and updated regularly, and the technology is maintained properly.

Processing and rendering time required for each simulation presents an additional challenge for real-time storm consequence prediction. Depending on the ADCIRC resolution used, we currently have a lag of 2-4 hours from the time an updated NOAA forecast is received until the results are reflected in the RI-CHAMP dashboard. This makes the data less useful for real-time storm preparations, especially when there is a significant change in the forecast. We are currently working to reduce this lag time by accelerating storm model processing and automating consequence mapping.

Next steps

The workshop discussed in this study was designed for RI-CHAMP's end-users (primarily emergency managers). Our next step is holding workshops for facility managers who have supplied infrastructure vulnerability data to the IACT database. These workshops will have participants review their data against RI-CHAMP's storm scenarios as a means of validating entries and updating HCT data as needed. Meanwhile, we are working with agencies in our stakeholder group to adapt RI-CHAMP to their specific needs and integrate its reporting capabilities into their various emergency management processes. We are also developing protocols to expand IACT by incorporating existing GIS datasets (currently fire stations and evacuation routes) by adding HCT information to those existing records. And, finally, we are working with the Rhode Island Emergency Management Agency to deploy RI-CHAMP on the State of Rhode Island's Arc Enterprise system with the goal of handling off long-term stewardship to their in-house team.

CONCLUSION

Emergency managers need access to specific, actionable storm consequence predictions at a local level but often have limited access to such data. This data gap is being made worse by the effects of global climate change, which limit the relevance of historic storms as planning benchmarks. Our study is intended to help emergency managers implement simulation-based consequence prediction tools as alternatives to existing heuristic approaches. Simulation can improve assessments of local vulnerability to storm hazards and reduce the likelihood of unanticipated outcomes. We produce local-scale information useful to state and local emergency managers by combining high-resolution storm models with a detailed, localized inventory of vulnerable infrastructure, delivering the local-scale information decision-makers need to evaluate their vulnerability to major ocean storms.

Our study demonstrates the value of a highly participatory process when developing hazard consequence prediction tools like RI-CHAMP. End-user needs and concerns are incorporated into every aspect of RI-CHAMP's development. Unlike tools that rely on economic measures of damage, often on a census tract scale, our approach integrates quantitative storm data with qualitative information about specific consequences for local communities. This ensures that RI-CHAMP's predictions are relevant to local decision-makers seeking to mitigate storm impacts. Our study also validates previous work on the visual presentation of this prediction data with a GIS interface that our workshop participants used successfully to identify specific impacts and consequences for multiple storm scenarios.

Simulation-based hazard consequence prediction is a powerful decision support tool that emergency managers can use to improve local storm HVAs. But more work is needed to identify the specific datasets most useful to emergency managers and to understand how these data will inform decision making for preparedness, response, recovery, and mitigation. The present study has focused on the process of producing and delivering consequence predictions to end users. Future work will need to address exactly how emergency managers can integrate this tool with existing processes.

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Notes

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COVID-19 and climate change concerns: Matters arising

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ABSTRACT

Until the outbreak of the coronavirus disease 2019 (COVID-19) pandemic, developing countries, especially countries in the African continent, battled with the impact of climate change on the food value-chain systems and general livelihood. In this study, we discuss climate change concerns post-COVID-19 and argue that the outbreak of the COVID-19 pandemic has exacerbated the vulnerabilities of most developing and emerging economies. This has heightened political tensions and unrest among such developing nations. We suggest enhancement and intensification of efficient and effective locally engineered adaptation strategies in the post-COVID-19 era for countries that have been susceptible to the impact of climate change and other recent shocks.

Key words: coronavirus disease 2019, pandemic, climate change, Africa, developing countries; DOI: 10.5055/jem.0802

INTRODUCTION

The root cause of climate change is undoubtedly global. However, the degree of greenhouse gas emissions varies by country and continent. This implies that some countries and continents emit more than others, albeit Africa is the most vulnerable continent. For example, least developed countries in 2019 contributed about 1.1 percent of total world CO_2 emissions, yet the associated impact on health, environment, food security, and economic growth in these regions has been devastating. Prior to the outbreak of the coronavirus disease 2019 (COVID-19) pandemic, experts predicted that the impact of climate change on developing countries, especially Africa, is likely to be severe*.[1] As if the impact of climate change on livelihoods is not enough, the outbreak of the COVID-19 pandemic has exacerbated the plight of the poor and their sources of livelihood. In this study, we acknowledge and discuss post-COVID-19 climate change issues and prescribe relevant adaptation and mitigation strategies for sustainable livelihood.

COVID-19, CLIMATE CHANGE, AND FOOD SECURITY

The onset of the COVID-19 pandemic sparked global shocks and disruptions in food systems, particularly in developing African economies. Given the extent of vulnerability of developing and poor countries to the adverse effects of climate change, such as floods, droughts, rising temperature, and erratic rainfall, the wave of the pandemic resulted in serious food insecurity. While existing evidence shows that global carbon emissions showed a steady decline during the wave of the pandemic,[2] nonetheless, the double hurdle of climate change and COVID-19 intensified food insecurity in developing regions of the world. Access to the required quantity and quality of food and the ability of households in developing regions to eat preferred food were seriously affected by the pandemic. The rise in food insecurity can be attributed largely to the restrictions imposed on human movement by governments across the globe, coupled with high rising prices of food commodities. In developed economies

*United Nations: Climate change is an increasing threat to Africa. 2020. Available at *https://unfccc.int/news/climate-change-is-an-increasing-threat-to-africa*. Accessed July 23, 2022.

and urban communities in developing countries, the uncertainties and fear created by the COVID-19 pandemic resulted in mad rush for food commodities, resulting in exorbitant high prices. Inversely, food insecurity in rural communities was associated with rising poverty posed by climate change and mass movement of people from urban communities to rural areas. Existing reports show that in the wake of the COVID-19 pandemic, which was mainly in urban communities, many people in India, Ghana, and other African countries moved from urban centers to rural communities as a measure to prevent contracting the virus.[3,4] However, such movements increased the burden in poor households in rural communities, further increasing food insecurity. Another dimension of the impact of climate change on food security during the wave of the pandemic is the high vulnerability and food insecurity of women. Globally and especially in developing economies, women have limited access to the productive and reproductive economy, which negatively affects their capacity to absorb and bounce back after shocks. Implicitly, COVID-19 has increased food insecurity and deteriorated the resilience of women, compared to men. Furthermore, the Russian–Ukraine war has exacerbated global food insecurity.[5] Indeed, Africa and other developing countries being grain and oil import dependent have experienced famine, rising food insecurity and upheaval as a result of the war.

COVID-19, CLIMATE CHANGE, AND ADAPTATION

Intensifying transformative (structural and functional) adaptation to climate change is critically needed in many parts of the world, with particular emphasis on developing economies in Africa.[†] Developing countries are bearing the brunt of climate change such as the drastic reduction in agricultural yields and productivity, which is the backbone of many of the economies. Also, food insecurity challenges and poverty levels are on the rise in many developing countries due to climate change and the COVID-19 pandemic.[6] The economic shocks caused by the pandemic have adversely affected gross domestic product and other economic growth and development indicators, thereby crippling the ability and resilience of developing economies to mitigate climate shocks and stressors. While adaptation strategies offer window of opportunities for developing economies to get back on track and advance their development, such efforts are highly limited due to the economic effect of the pandemic coupled with climate change, political instability, and low adoption of technologies. The situation is further worsened by the limited ability of developing economies to access global adaptation and development financing facilities as a result of the impact of the pandemic on the global economy. As such, employing homegrown climate-smart interventions that are less-resources intensive can cushion developing countries to adapt to hard times posed by climate change and the COVID-19 pandemic. Again, developing African economies should prioritize the implementation of fit-for-purpose and context-specific adaptation strategies that have high potential to increase agricultural productivity, boost adaptation, and adaptive capacities in vulnerable communities and households while minimizing the emission of greenhouse gases (GHGs) from the agricultural sector.

[†]IPCC: Climate change 2022: Impacts, adaptation, and vulnerability. 2022.

COVID-19, CLIMATE CHANGE, AND WATER

Global water supply has been a serious challenge all around the globe due to several factors, including climate change impact, poor management of water resources and supply systems, inadequate political will, financial commitments to infrastructural development for water resources and supply, etc. These challenges are not new but have existed for several decades before the emergence of COVID-19 pandemic, which further exacerbated the challenges. Since COVID-19 was officially declared a pandemic by the World Health Organization, there was an urgent need to increase water supply around the world, especially in Africa. This is because water supply for increased personal hand hygiene and sanitation has been identified as one of the major strategies to fight against the pandemic.[7] Water insecurity has been identified as a major factor contributing to the difficulties in containing the spread of the virus in most developing countries.[8,9] The panic response to ensure water security led to urgent financial support from global financial bodies like the World Bank, the International Monetary Fund, etc. to support the fight against COVID-19 in developing countries, especially in Africa. Unfortunately, while the financial burden of water supply was lessened by these international financial supports, supply of water for the fight against the pandemic suffered other major setbacks that included pollution and climate change-related extreme events, such as droughts and floods. In some places, droughts led to extreme difficulty in supply and access to water, especially in rural areas of developing countries. Floods resulted in increased pollution of water bodies and damaged water supply infrastructure, thus hindering adequate water supply to fight the pandemic. These reaffirm the claim that nearly everything is affected by climate change, and water resources and supply are among the most affected.

COVID-19, CLIMATE CHANGE, AND ENERGY

Climate change and COVID-19 together have not only just presented a daring challenge to global public health but also affected energy supply. Evidence from a study by Suryadi and Safrina[10] observe that beside fossil fuels, renewable energy sources have been equally hit. The observed impact on the renewable energy subsector happened because China has been the major global supplier of green energy components, and since production companies and shipments in China were closed down due to the pandemic, the renewable energy industry suffered a slowdown if not a halt in production. In a like manner, the observed tremendous effects on the fossil fuels industry by the pandemic were due to the drastic reduction in transportations (aviation and land) and consequently the downturn in demand for fossil products. While the negative effect on green energy projects has had an adverse influence on climate change actions, the disruptions on the fossil fuel industry have led to less GHGs emission and have had a positive effect on the climate. This could be worth considering by global stakeholders as a key strategy for ensuring energy transition and climate change mitigation, sustainable environmental management, and economic development.[10] Additionally, as a result of industrial lockdowns because of the pandemic, there was, generally, less consumptive use of energy including that from thermal power plants generation and reduction in GHG emissions, thus a positive effect on the climate. The COVID-19 pandemic brought about temporary disruptions in prepandemic energy use trends. Admittedly, it remains empirically unclear how post-pandemic recovery will influence the future trends and energy transition to renewable energy production and consumption.[11] The energy sector has therefore been a key sector affected by the COVID-19 pandemic due to considerable drop in both fossil fuel consumption and green energy projects expansion. Evidence from the pandemic containment measures that were employed in relation to energy and climate change would be informative to policy formulation with regard to future energy transition and usage.

COVID-19, CLIMATE CHANGE, AND IRRIGATION

Irrigation agriculture contributes about 40 percent of global food supplies and is a source of livelihood for the majority of the world's poor who are mostly found in rural areas. Any impact on irrigation directly poses a threat to food security, poverty alleviation, and climate resilience of people, hence impeding the achievements of Sustainable Development Goals 1, 2, and 13.[‡] Under irrigation, there is always increased investment in inputs such as fertilizer and pesticides, stemming from the reliability in water supply, and these culminate into higher productivity and increased farm incomes. However, climate change has phenomenal consequences on irrigation. Climate change contributes to reduction in water resources and increases the prevalence of extreme temperatures and outbreak of crop diseases, which hinder the full realization of the intended objective of irrigated agriculture. This is an imminent threat to the sustainability of irrigated agriculture. The threat of climate change on irrigated agriculture has been exacerbated by the COVID-19 pandemic, resulting in far-reaching consequences. This has escalated the

[‡]Sustainable Development Goals (SDGs) 1, 2 and 13 focus on addressing poverty; hunger and food security; and climate change, respectively. See the SDGs at *https://sdgs.un.org/goals*.

climate change challenge to irrigated agriculture in particular and the broader water-energy-food nexus in general. In responding to the pandemic, many nations implemented a number of measures to save the lives of their people such as travel restrictions and lockdowns. These measures have disrupted input and output markets in the agricultural sector, thereby weakening the resilience of irrigation systems to climate change. For instance, supply chain of agricultural inputs, machinery services, and spare parts were significantly affected both in terms of availability and price, which had direct impact on production volumes and productivity. On the other hand, commodity markets were also affected as physical marketing of agricultural products was no more effective, hence affecting farm incomes. Furthermore, inadequate labor availability due to lockdowns also significantly increased agricultural wages,[12] causing shortages of labor at critical stages of production. Moreover, many studies have shown the impact of remittances to rural economic activities. The COVID-19 pandemic resulted in significant reduction in remittances, causing decline in investments in the irrigated agricultural sector, which significantly lowered productivity and farm incomes.[13]

Thus, undoubtedly, though the measures adopted as response to COVID-19 across the globe were necessary, the negative impacts of climate change on irrigation development have exacerbated. Hence, particular attention must be given to the irrigation sector to restore the sector to a path of recovery. Alternative sources of inputs such as locally produced organic fertilizers should be explored while digital technologies could be employed in marketing of agricultural products.

COVID-19, CLIMATE CHANGE, AND HEALTH

Climate change has affected human health and healthcare delivery in diverse ways. Extreme temperatures in the tropics, especially in Sub-Saharan Africa, have often been associated with spinal diseases such as cerebrum spinal meningitis (CSM), while extreme flooding causes water-borne diseases such as bilharzia, cholera, typhoid, and malaria. Climate change also causes water scarcity, which increases competition for scarce water resources by humans and livestock, especially in rural areas, increasing the risk of disease transmission between animals and humans. Food shortages as a result of floods and drought also cause malnutrition among rural and poor communities. The advent of the novel COVID-19 pandemic posed a new challenge, which compounded the effects of climate change on health. Low- and middle-income countries (LMICs) have very limited capacity to manage pandemics due to weak health infrastructure. Efforts to contain this global pandemic amidst these weak systems in LMICs have worsened the climate change impacts on human health. In the wave of the pandemic, the already weak health systems in these countries were stretched to the limits.[14] As a result, other diseases that were triggered by climate change have worsened. In addition, the attention in terms of awareness creation, resource allocation, among others that were hitherto given to such diseases was relegated, while priority was given to the containment of the novel virus. For example, in Ghana, the outbreak of CSM during the dry seasons, especially in the hotter regions of the North, is not uncommon. However, in the wave of the pandemic, as a result of "too much attention" to COVID-19 containment, the deployment of a meningitis surveillance team and the rapid response team by the Ghana Health Service to tackle CSM outbreak was late, ineffective, and poorly resourced, causing much devastation in the outbreak areas. Reports from other parts of Africa have also shown that efforts toward containment of COVID-19 have weakened efforts to control other health problems that are endemic, such as malaria.[15] For example, there have been reports of a significant increase in malaria cases in Zimbabwe during the peak of the COVID-19 pandemic. We argue that improving health infrastructure in LMICs could improve resilience to climate change effects on human health, especially during pandemics.

POLICY IMPLICATIONS OF THE COVID-19 PANDEMIC AND CLIMATE CHANGE

The double hurdle of COVID-19 and climate change has serious implications on global development in general and on developing economies in particular. The COVID-19 pandemic and climate change could derail sustainable development gains particularly in the Global South. Rising poverty due to the pandemic and climate change could push many people in abject poverty, food insecurity, and high malnutrition-related challenges, resulting in low human capital development in vulnerable regions.[16] Although governments in the Global South with support from the Global North have instituted measures to remedy the effects of climate change and COVID-19, yet the high capital required to effectively tackle these duo challenges can push many economies into financial crunch, recession, and instability.[17] For instance, in Africa, governments require huge financial expenditure to address climate change and COVID-19, which is currently lacking. Public unrest due to the pandemic and governments' inability to meet the needs of citizens while mitigating the effects of climate change and COVID-19 could disrupt economic activities, resulting in social vises, survival of the fittest, and ultimately political instability. Thus, the COVID-19 and climate change could undermine democratic consolidation across the globe with emphasis on developing countries and worsen fragility in war-torn areas.

Essentially, developed economies and international financial/development institutions such as the Bretton Woods institutions should provide financial, technological, and technical support to economies battling to recover from the adverse effects of the pandemic and climate change. For instance, the pandemic has exposed the extent of fragility of healthcare systems in many developing countries. This implies that tackling future pandemic would require building robust healthcare infrastructure coupled with the enabling technologies. Undoubtedly, the devastating effects of the COVID-19 pandemic in developing countries could be partly attributed to limited technological capacities for contact tracing and sequencing of the genome of the virus. The same applies to climate change where low technological adoption and agricultural mechanization increase climate vulnerability and its associated adverse effects on developing economies. Hence, policy directions should aim at building the capacities of societies and people through education, research, and development, and the development of technologies and infrastructure needed to enhance resilience and robust recovery before and after shocks.

To ensure that communities have access to safe and reliable water supplies, governments need to invest in water infrastructure, especially in the areas of water treatment plants, maintaining aging and deteriorating infrastructure, and promoting water conservation measures and practices. The identified water insecurity amidst COVID-19 pandemic could lead to inequities in access to clean water, particularly in low-income and marginalized areas and, therefore, requires governments to ensure that everyone has access to safe and reliable water, regardless of their income or social status. The COVID-19 pandemic has shown that communities, especially in low-income and marginalized areas of developing countries, need to be resilient in the face of unexpected crises. Governments, therefore, need to invest in building such marginalized communities to be resilient and adapt to the impacts of climate change and other crises. Climate change and water issues, especially in times of unexpected crises like COVID-19, are global challenges that require international cooperation. To address these issues, governments need to cooperate for sharing of resources and knowledge. More importantly, governments need to cooperate in taking concrete and immediate steps to address climate change. This includes getting more committed to the various National Determined Contributions and setting achievable targets to reduce greenhouse gas emissions, promoting the use of renewable energy sources and investing in research and development to find new ways to mitigate the impacts of climate change.

Governments can take advantage of the pandemic to implement green recovery programs that boost economic growth, create decent jobs, and lower greenhouse gas emissions. Investments in sustainable mobility, energy efficiency, and renewable energy sources could be part of these strategies. Energy demand patterns have changed significantly as a result of the pandemic, with many people working from home and travel restrictions reduce energy use which is associated with transportation. While making plans for future energy systems and infrastructure, policymakers must take these changes into account. The COVID-19 pandemic has shown how crucial international cooperation is in addressing global crises. To combat climate change and make the transition to a sustainable, low-carbon energy system, governments must collaborate. Existing social and economic inequalities have been brought to light by the epidemic and must be addressed in climate and energy policies. It is important to ensure that the transition to a low-carbon economy is equitable and benefits all members of society, especially those who are most vulnerable to the impacts of climate change. The need to combat climate change remains urgent despite the pandemic's effects on the world economy and society. Governments must continue to prioritize climate action and invest in clean energy solutions.

As elicited in this study, the irrigation sector suffered labor, input, and output market challenges during the lock-down periods of the COVID-19 pandemic. Meanwhile, various actors in the agricultural value chain play significant roles in strengthening the resilience of irrigation and other agricultural systems against climate change. These findings implied that irrigation systems became weaker in responding to climate change during the pandemic as a result of the overreliance of the sector on external sourcing of inputs and low usage of hard and soft technologies. This calls for a rethinking in policy directions in the sector to more innovative approaches. Going forward, national policies on agriculture and irrigation should consider the domestication of input supplies through the acceleration of local production of essential inputs such as irrigation equipment, fertilizers, pesticides, and herbicides. Furthermore, in order to abate labor challenges, a radical mechanization of the sector should be prioritized, and future policies should be formulated in this regard. The use of digital technologies in marketing has also become imperative for sustainable market inclusion.

In the health sector, climate change is often associated with extreme climate events which have significant impacts on human health. Amidst these pressures on health systems by climate change, the COVID-19 pandemic came as a shock to the world and increased the already existing pressures facing the sector, especially in LMICs, such that attention to other diseases became relegated. COVID-19 cases were prioritized during competition for space, funds allocation, awareness creation, etc. These findings have significant policy implications. Future policy interventions should consider the expansion and improvement of healthcare infrastructure in order to boost the capacities of health systems in handling global crises and megatrends. Also, there is a need to promote the adoption of digital technologies in the health sector for remote healthcare delivery.

CONCLUSION

In line with the simultaneous effects of the COVID-19 pandemic and climate change, developing and emerging economies, particularly in Africa, have typically not been exempt from the devastating risks to human survival. The devastating effects of the COVID-19 pandemic in a world where the climate is changing are not a typical problem or, even better, a known problem that can be fixed with existing policies. Rather, a better presentation of the scenario that gives a better picture for policy reflection is critical. Based on the aforementioned discussions, we show the interplay and interlinkages between parallel themes such as COVID-19, climate change and food security; COVID-19, climate change and adaptation; COVID-19, climate change and water; COVID-19, climate change and energy; COVID-19, climate change and irrigation; and COVID-19, climate change and health. Under these somewhat broad themes, urgent concerns that deserve urgent attention have been raised and discussed, along with associated policy prescriptions.

Admittedly, it must be acknowledged that some of the policy directions discussed appear to be generic, and countries that matter in the discussions have to tailor them to fit their context. That is, wholesale adoption of prescriptions must be done with caution. Furthermore, country-specific studies are highly recommended by the study to unravel and tackle COVID-19, post-COVID-19, and climate change intricacies and nuances.

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Notes

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Designing user-centered decision support systems for climate disasters: What information do communities and rescue responders need during floods?

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ABSTRACT

Flooding events are the most common natural hazard globally, resulting in vast destruction and loss of life. An effective flood emergency response is necessary to lessen the negative impacts of flood disasters. However, disaster management and response efforts face a complex scenario. Simultaneously, regular citizens attempt to navigate the various sources of information being distributed and determine their best course of action. One thing is evident across all disaster scenarios: having accurate information and clear communication between citizens and rescue personnel is critical.

This research aims to identify the diverse needs of two groups, rescue operators and citizens, during flood disaster events by investigating the sources and types of information they rely on and information that would improve their responses in the future. This information can improve the design and implementation of existing and future spatial decision support systems (SDSSs) during flooding events. This research identifies information characteristics crucial for rescue operators and everyday citizens' response and possible evacuation to flooding events by qualitatively coding survey responses from rescue responders and the public. The results show that including local input in SDSS development is crucial for improving higher-resolution flood risk quantification models. Doing so democratizes data collection and analysis, creates transparency and trust between people and governments, and leads to transformative solutions for the broader scientific community.

Key words: spatial decision support systems, community knowledge base, disaster response, disaster rescue operators; DOI:10.5055/jem.0741

INTRODUCTION

Flooding events are the most common natural hazard globally and result in vast destruction and loss of life.[1,2] Recent studies have found that flood risk within the United States (US) is expected to increase, with a predicted 26.4 percent increase in flood risk by 2050 from climate change alone (RCP4.5)[3] and an increase in compound flooding events in major US cities.[3,4] An effective flood emergency response is necessary to lessen the negative impacts of flood disasters.[5] However, disaster management and response efforts face a complex scenario. A wide array of organizations with varying levels of expertise and resources attempt to consolidate and organize themselves within a short period to distribute available resources and perform critical tasks, such as rescue operations.[6,7]

Simultaneously, regular citizens attempt to navigate the various sources of information being distributed and determine their best course of action. One thing is evident across all disaster scenarios: having accurate information and clear communication between citizens and rescue personnel is critical. In flooding events or hurricanes, these scenarios can be further complicated by lack of power, barriers to travel, and communication difficulties. However, different groups have different needs, and those differences must be considered when developing decision support systems aimed reducing flood risks. This research aims to identify the diverse needs of two groups, rescue operators and citizens, during flood disaster events by investigating the sources and types of information they rely on and information that would improve their responses in the future. This information can improve the design and implementation of existing and future decision support systems during flooding events.

This research aims to identify information characteristics crucial for rescue operators and everyday citizens' response and possible evacuation to flooding events. This information will be used to develop a decision-support system that rescue operators and civilians can use to make risk reduction decisions in real-time. To do this, we deployed a survey questionnaire. Through qualitatively analyzing the closed-ended responses, we attempt to answer the following questions:

1. What types of information do rescue operators and everyday citizens currently rely on during a flooding event?
2. What types of information do these two groups wish they had access to during a flooding event?

LITERATURE REVIEW

Flooding and disaster management

When flood disasters are about to strike, numerous actors are put into motion. Citizens shore up their homes or plan to evacuate, agencies gather resources in preparation for emergency response, search and rescue teams gear up to conduct rescue ops; the list goes on and on. A flood disaster response requires the efforts of multiple agencies and organizations, resulting in a transition from autonomous organizations into an interdependent decision-making network to render aid and extract vulnerable populations.[8] Managing this complex situation requires three essential items: multiagency coordination and collaboration, accurate and effective communication, and resource and supply chain knowledge.

Disaster management is a product of diffuse, cooperative emergency management systems that come together to produce a highly interdependent system.[9,10] *Multiagency coordination and collaboration* have become paramount during disaster scenarios, as no single group or agency has the capacity or resources to handle disaster scenarios on its own.[6] Identifying each agency/organization's role and coordinating their efforts are vital to preventing the misallocation of resources and ensuring an effective disaster response.[11] *Accurate and effective communication* is critical for rapid, effective response and rescue operations[12] and to reduce losses and prevent the situation from worsening.[8,13,14] This requires effective avenues for sharing information during a disaster, eg, social media, apps, and mobile phones, and intelligent tools to provide real-time information, eg, sensors, satellite images, and UAVs.[15-17] Finally, *resource and supply chain knowledge* is necessary to coordinate and prioritize the distribution of materials and identify materials that need to be procured.[18] While these three concepts are necessary for any flood disaster, additional considerations must be made for the location of the disaster and local community capital.[19-21] A disaster response effort in a highly urban area will differ vastly from a rural community simply because of the reduced financial and human resources at their disposal.[21-24]

The flow of information

Disaster response actors can be defined as government agencies, nongovernmental organizations (NGOs), private entities, and volunteer citizens.[25,26] Having numerous actors present to manage the impacts of a disaster makes *coordination, collaboration*, and *communication* difficult.[8,23,27] There are two types of decision-making processes in disaster response: rapid decision-making and long-term decision-making.[28] Rapid decision-making occurs in emergency situations where first responders will make the decisions under time pressure based on how well the persons in charge of the saturation know the areas at risk. Long-term decision-making is used when the disaster response mission is not extremely urgent. Geographic Information Systems (GIS) have been often used to support disaster response to identify risk areas and available resources.[28-33]

Coordination refers to the alignment of goals and methods across multiple levels of disaster response. Research points out that better coordination can lead to more desirable outcomes, measured by more saved human lives, less property damage, and faster delivery of recovery and relief resources, eg, food, shelters, medical supplies, and less waste.[34] The literature argues that to create frameworks capable of better handling coordination challenges during disasters, actors must find the appropriate balance between top-down and bottom-up engagement.[35] However, it is unclear what this balance looks like in different regions or disaster types. The conditions and/or constraints under which this balance is likely to be reached depends on the number of actors involved. Actors range from governments (local, state, and federal in the US, for example) to NGOs, private stakeholders, community and faith-based organizations, and ordinary people and residents of disaster-affected areas.

There is evidence from previous disaster events, eg, Hurricane Katrina in the US (2004) or the Fukushima Daiichi nuclear accident in Japan (2011), that the cost/significance of a disaster could potentially reduce or increase the transaction costs associated with coordination.[36] Some researchers have even suggested that coordination in disaster response works best when it is "emergent," ie, through indirect, unplanned, and spontaneous collaboration.[37] Additionally, what complicates the matter is how government agencies and other key players tend to take credit (or blame) for the quality and timeliness of response under good or bad coordination circumstances.[37]

Collaboration among various disaster response actors prior to and after a disaster necessitates that several stakeholders come together and work across conventional organizational jurisdictions and professional boundaries. Research indicates that to be successful, such collaboration should be grounded in a mutual understanding of resources, constraints, and roles played by different actors.[38-40] Agranoff and McGuire[41] and Jung and Song[42] categorize these collaborative networks into vertical and horizontal structures. While the former describes the interactions between different levels within a government, stakeholder organization, or community, the latter focuses on interactions among multiple stakeholders at the same level. Other researchers have highlighted the importance of consolidated, multilevel, ie, vertical, and broader, ie, horizontal, collaboration in a hierarchical structure as a precursor to successful decentralized disaster management.[43] In the US, the National Incident Management System (NIMS) serves as a multiagency disaster collaboration model by incorporating a well-defined command and coordination structure to facilitate multiagency collaboration and enable consistent and unified nationwide approach for federal, state, and local governments to prepare, respond, and recover from disasters.[44] Other examples include the Australasian Interservice Incident Management System, the Coordinated Incident Management System in New Zealand,[45] and the Joint Emergency Services Interoperability Program in the UK.[46]

Communication, in particular, is often hampered by the impact of flood conditions on communication lines and the large volumes of information being produced during a disaster scenario.[47,48] Both disaster response actors and citizens face communication difficulties when typical communication lines, such as telephones or the internet, are inoperable or damaged.[27,48] As a result, innovative techniques are often needed to maintain communication with those working in emergency response scenarios, eg, phones, walkie-talkies, social media, and specialized applications.[14,16,27] To complicate matters further, a wide variety of information is provided to first response actors from various sources, all of which need to be collected, organized, and disseminated to inform response decisions.[5,47] This includes information about local physical conditions, weather reports, government correspondence, rescue requests, and more.[49]

Rescue operators, information sources, and decision support tools

Rescue operators encompass many professionals and volunteers who participate in search and rescue operations and emergency response efforts at the onset of a disaster. This group includes healthcare professionals, local police, firefighters, volunteer rescue organizations, and local government officials.[50,51] While rescue operators play a critical role in flood emergency response efforts, limited research identifies their critical information needs or what information sources they often rely on during disaster scenarios.[49] Some research has assessed the role of big data in accessing flood depth information in urban areas. For example, social media images of submerged vehicles[52] and other submerged objects were analyzed using artificial intelligence (AI) for estimating flood depth.[13] More recently, AI-based image analysis was applied to crowdsourced photos of traffic signs submerged in floodwater to estimate flood depth at the street level.[32,53]

GIS provides technical solutions to help the decision-makers understand interactions between spatial and cultural factors in spatiotemporal decision-making processes. However, many conventional GIS approaches lack knowledge sharing and communication support among decision-makers, which weakens their ability to deal with conflicting objectives.[30] On the other hand, knowledge is often derived from various geospatial data sources with different formats, modalities, and scales, leading to tremendous computational challenges. Spatial decision support systems (SDSSs) help automate or guide decision-making by analyzing and synthesizing available information from the field.[54] SDSSs come in various shapes and sizes, including privately developed applications, open-source software, and commercial systems designed for particular organizations.[54]

The advanced cyberinfrastructure powered by high-performance computing, advanced data storage systems and instruments, and geo-visualization tools has enabled a new way of thinking about systematic approaches to spatial decision-making.[33,55] Cyberinfrastructure consists of computing systems, data storage systems, advanced instruments and data repositories, visualization environments, and people, all linked by high-speed telecommunication networks.[56] CyberGIS represents a complex geospatial system that connects interdisciplinary fields such as advanced computing and cyberinfrastructure, GIS, and spatial analysis and modeling. CyberGIS been successfully applied in many domains, including emergency management, agriculture, food production, and energy[30,57,58] to create SDSSs for integrating spatial analysis, data management, domain knowledge base, and spatial visualization. Interactive CyberGIS-enabled SDSSs can help capture community priorities and risk perceptions and enhance knowledge elicitation and sharing.

SDSSs can display several data layers—eg, traffic data,[59] Twitter® data,[60,61] census data, infrastructure, and flood risk data–that can be combined for effective spatial decision-making.[60,61] Such tools can help enhance communications between residents and emergency units, provide spatial knowledge representation, and help stakeholders visualize and disseminate flood risk uncertainties received by humans, so they can determine the best ways to use such information for decision-making at the local level. Various components that are related to decision models, data and knowledge relevant to a decision, and geospatial visualization tools can be shared over a cyberinfrastructure platform in a way that they can be seamlessly integrated and used by various decision-makers.

Incorporating community input into disaster management and first responses

A key component of a successful SDSS is the *knowledge base*. The knowledge base uses knowledge acquisition techniques, citizen science, and dynamic interactions between the user and the user interface to provide an ideal space for all users, ie, local governments, communities, and residents, to challenge decision problems in flood hazard and identify and implement solutions. An SDSS must incorporate local knowledge when developing the knowledge base to be effective—failing to do so can often result in systems inappropriate for local evacuation and rescue operations when flood disasters occur.

METHODS

Survey development

The survey for this project was developed to provide a way to incorporate local knowledge and decision-making needs at both the individual citizen and rescue operator level into an SDSS knowledge base. The survey approaches the question development process from a bottom-up approach, where respondents were asked to share their previous experiences with flood disasters and are prompted to discuss different types of information that would be critical for them during flood events. The survey also focuses on two distinct groups of respondents: public citizens and respondents involved in rescue operations during flood events. The general public respondents were asked questions regarding their previous flooding experience, whether they ever needed rescue aid during flooding events in recent years, and what information they wished they had or wanted to help them evacuate. Rescue operators were asked the same questions as the general public. However, they were also asked additional questions regarding their experience with rescue operations, what critical information is needed when trying to get to people in need, different challenges they faced while conducting rescue operations, and what information they wished they had to help carry out rescue operations. Most questions were multiple-choice, select all that apply, or short answers (Appendix). The survey and accompanying consent materials were approved by the Institutional Review Board (IRB2019-1506D).

Survey sampling

We used a convenience sampling method to gather participants who were familiar with the topics of the study. The survey was deployed online through Qualtrics, an online survey administration and sampling platform. Initially, the survey link was shared through email listservs and social media accounts of community partners, namely, Hazard Reduction and Recovery Center, Texas Target Communities, and several disaster relief NGOs, including Cajun Navy Relief and Crowdsource Rescue. A link to the survey was also shared at community meetings, research presentations, and through community partners' outreach networks and campuswide university emailing lists.

Data

Collected survey data were first cleaned by removing incomplete surveys or those who refused consent. Of the 31 questions in the survey, only four short answer questions were selected as the focus of our analysis (Table 1). These questions were split between two groups of people: rescue operators and evacuees. Each group was asked what information was used during a flooding disaster (Q10 and Q20) and what information they wished they had (Q12 and Q22). Responses for each of the four questions were then imported as separate projects into Atlas.ti (version 9).

Survey analysis

Questions were analyzed in Atlas.ti using descriptive coding.[62] Coding involved pattern coding that specified the type of information (evacuation and built environment) and information characteristics (real-time and accuracy) needed.[62] Codes focused on the types of information used/wanted, eg, flood conditions, and the sources of that information, eg, social media. Responses that were not pertinent or blank were not coded. A total of five to six themes were produced for each question.

Table 1. Survey questions and number of responses analyzed

Target audience	Question
Evacuees	What information did you use to help you evacuate?
Evacuees	What information do you wish you had or wanted to help you evacuate?
Rescue operators	What information did you use when trying to get to people in need?
Rescue operators	What information do you wish you had or wanted to help you carry out rescue operations?

RESULTS

Types of information used Citizens. For citizens, the types of information used during a flood were organized into six themes: *notification systems, physical conditions, government sources, online sources, social networks*, and *personal knowledge*. Notification systems encompassed existing systems that contacted individuals to provide information on weather conditions, evacuations, or emergency information, such as a call alert system or local government warnings. For example, respondents indicated they had relied on a "[l]ocal telephone call alert system," an "Emergency Alert from work," or other similar systems.

Respondents also relied on *physical condition* information, which we defined as specific information about a region's weather conditions or flooding status. Respondents indicated utilizing "storm surge estimates," "flood gauge[s]," and "weather predictions," among others. While similar to physical conditions, as many *government sources* include weather data, this category included information specifically identified as being sourced from the government, such as a "county website" or "the National Hurricane Center/NOAA." Government sources included federal and local resources for weather conditions and evacuation instructions.

Online sources also shared some potential overlap with government sources; therefore, they were defined as sources of information online that did not originate from the government, such as the news or social media. Some examples included general responses like "[i]nternet [w]arnings" as well as more specific examples like "spacecityweather.com." *Social networks* were defined as individuals within a social support network, like family or friends, who provided information regarding a flooding event. This was less commonly referred to but included receiving information from individuals, like "mom" or their "parents," as well as "word-of-mouth from neighbors." Finally, *personal knowledge* was also identified as a source of information, which we defined as information from previous experience or local knowledge to aid in disaster response behavior. Responses included "knowledge of the city," "past experience," and personal observations of nearby flood conditions (Table 2).

Table 2. Types of information from previous experience or local knowledge used by survey respondents to aid in disaster response behavior

Citizens	Rescue operators
Notification systems	Physical conditions
Physical conditions	Real-time information
Government sources	Evacuee information
Online sources	Emergency management organizations
Social networks	Local resources
Personal knowledge	Government sources

Rescue operators

For rescue operators, the types of information used during a flood were also organized into six categories: physical conditions, real-time information, evacuee information, emergency management organizations, local resources, and government sources. Many rescue operators identified physical condition information as something they relied on during flooding events. This type of information included similar information to their citizen counterparts, such as "water depth," as well as more specific information, like "road conditions" and whether or not "any debris was blocking the normal route." Real-time information was also crucial to rescue operators, which we defined as types of information or information sources that provided current or immediate conditions. This included having "direct line communication," "GPS" information, and "resources available."

Equally important was evacuee information, which covered the characteristics or conditions of those being rescued. Rescue operators specifically stated needing the "last known location" of evacuees, the "medical needs" of evacuees, and "[h]ow many pets they had." Emergency management organizations, such as the Cajun Navy Relief and local Dispatch, were also listed as key sources of information. This category explicitly encompassed identified organizations that were utilized during the rescue process. Available local resources broadly encompassed any resources specific to the rescue area that did not fall into a previous category. This included "local first responders" and "local news" accounts of those needing evacuation. Finally, government sources were also identified by respondents as a key source of information, which included specific government entities such as the National Oceanic and Atmospheric Administration (NOAA) (Table 3).

Types of information wanted Citizens. For individual citizens, the types of information wanted were organized into five broad categories: *accurate information, advanced warning, evacuation destination/resources, evacuation process*, and *safety*. Respondents indicated a desire for more *accurate information* during a flood scenario, for both flood/weather predictions and conditions. This included specific information like the "accurate forecast of the severity of the flood," "better rainfall predictions," and "[l]ocal flood conditions," as well as information about the vulnerability of the structural environment, such as "how structurally sound the drainage was."

Having an *advanced warning* was also important to respondents. This included flood conditions/extent, such as when flooding would occur and when dams would release water, and warnings through social media or government agencies, such as when to evacuate. Several respondents indicated that they lacked advanced warning during major floods. One individual stated, "Harvey was unprecedented in comparison to other storms. We were not given enough time to prepare and to leave." Another respondent mentioned that it "[w]ould've been nice to get a warning as the dam release was about to occur."

Table 3. Types of information survey respondents want or would like to better aid them in making disaster response decisions

Citizens	Rescue operators
Advanced warning	Rescue information
Accurate information	Real-time information
Evacuation destination/ resources	Evacuee information
Evacuation process	Physical conditions
Safety	Accurate information

Additional information about *evacuation destinations and resources* was identified as important to citizens. This theme included information on evacuation destinations, shelters/evacuation sites, and resources for specific populations such as the elderly or pet owners. Several respondents identified not knowing where to evacuate. For example, one respondent stated: "I know to evacuate to a higher elevation at least, but besides what I could visibly tell was higher ground, I did not know of any specific evacuation routes or evacuation sites." There were also concerns about available evacuation resources, such as whether there were pet-friendly evacuation sites, transportation options to shelters, and if there were low-cost or discount evacuation sites.

Respondents also stated they wanted more information about the evacuation process for future events. Unlike evacuation destinations and resources, this theme covered information about how to evacuate safely, such as the time of day to leave or how to evacuate via personal vehicle. Respondents indicated wanting additional information on evacuation routes, mainly whether or not routes were flooded or contained traffic. For example, one respondent wanted to know about the "[e]vacuation routes and congestion on those routes."

Finally, respondents raised concerns about a lack of information on *safety* during previous flooding events, specifically regarding electrical power during a flood event. While not as common, a few respondents stated they had concerns about electrical power safety in flooded homes and power reconnection during a flood.

Rescue operators

Responses from respondents identified as rescue operators were examined separately. The results were grouped into different categories: *rescue information, real-time information, evacuee information, physical conditions*, and *accurate information. Rescue information* was commonly identified by respondents and was classified as logistical information required to perform rescue operations during a flooding event. For example, respondents mentioned wanting "[b]etter coordination [between] all levels of the response to get the response assets into the areas of need effectively" and a way to know "which areas needed the most help." *Real-time information* was also key to rescue operators and contained some overlap with rescue information. This category expressly referred to information provided to rescue ops as soon as it was made available. Respondents specifically mentioned wanting to know the "real-time placement of other rescuers" and "[k]nowledge that those needing rescue had already been rescued."

Evacuee information, or information about those requesting rescue operations, was also identified as critical. Information about the evacuees themselves, such as date of birth or "how many pets they had," and "more information on shelters, longevity of shelter, and transportation for victims" was cited. Information about the *physical conditions* impacting rescue operations included impediments to rescue routes and water conditions. Water depth was the primary concern, although water speed and physical obstacles, like downed powerlines, were also mentioned. Finally, respondents identified a need for *accurate information* in general. While not as specific as previous categories, one particular respondent recognized why this was of great concern:

> Can't think of how we would have gotten more information. Sometimes there is too much information. Less information that is more accurate would be better than a ton of information.

DISCUSSION AND CONCLUSIONS

Disasters provide enormous amounts of data that can help gain insights to better prepare for future events if properly mined. Contrary to common belief, the problem is not whether enough data and information exist. The advent of modern sensing and instrumentation technologies, handheld devices with mobile connectivity, and social media tools have enabled people and authorities to collect large volumes of data every day. However, in the absence of a systematic approach and organized ways to analyze that information, such as data organized in an SDSS, more data may not necessarily lead to better response.

When developing a SDSS to address these challenges, the *knowledge base* must be developed with the users' information, and accessibility needs at all stages of development. Flood disasters create tumultuous situations for rescue operators and citizens alike, with large volumes of information produced and collected from various sources. This information directly informs rescue operators and citizens' decisions as a flooding event unfolds. This information can be conflicting, confusing, or of questionable accuracy. To improve future rescue and response efforts, identify citizen needs, and develop needs-based SDSSs, it is critical to understand what types of information are relied on and how they can be improved.

Information used versus information needed

From our survey results, it is clear that both citizens and rescue operators rely on government information and physical conditions during a flood. Aside from these, however, they rely on very different types of information to navigate a flood scenario. The themes identified were information on available local resources, evacuees, emergency management organizations, and real-time data. While previous research has focused on understanding how social media can help rescue operators retrieve information that helps direct rescue efforts,[16] our results identify a key limitation to that method of communication: accurate and real-time data. Rescue operator respondents were more concerned with accurate information necessary to perform rescues than with the volume of information received, reflecting findings from Salmoral et al.[5] and Blum et al.[56] There is a distinct difference in the benefit of general disaster information that takes hours to process versus real-time data. Salmoral et al.[5] stated that rescue operators need accurate information *when they receive it*, so they are less likely to use data that are not real-time. However, the desire for less, more accurate data contradicts studies that found that timeliness of data is often more important than high levels of accuracy. This theme could have arisen because data timeliness and accuracy are often prioritized differently during versus after the event. Depending on how the respondents interpreted the survey, they may have talked about rescue operations that occurred at different points on the disaster timeline.[63]

In comparison, citizens relied on information from less formal channels or specifically designated for the public. Themes included existing notification systems, social networks, online sources, and personal knowledge. Our results suggest that citizens generally feel they need more dependent and accessible communication from various sources, including government organizations. Increasing the use of social media is one potential method for increasing information access. However, as seen in Mihunov et al.,[16] disparities in social media do exist and are more likely to be used by populations less vulnerable to disaster impacts, eg, well-educated, employed, higher income, etc. As seen in previous research by Ali et al.,[27] effective information communication through reliable networks is critical for disaster impacts, especially for vulnerable populations whose communication networks may already be minimal, eg, no access to phones. Therefore, communication networks must be developed using various information structures, including social media, government information sources, and community networks, such as community groups or friends and family.

While some information needs overlap between groups, it is evident that information is driven by accessibility, the needs of the groups, and the nature of the recipient's response action (rescue versus evacuee).[64] Both rescue operators and citizens require more accurate information and rely on evacuation information. However, rescue operators require information on the evacuees and rescues, while citizens are more concerned with information on evacuation destinations/resources and the evacuation process. Rescue operators also want more information that is real-time. On the other hand, citizens desire more advanced warning and greater safety information during flooding events. Similar conclusions have been reached by Gralla and Goentzel.[49]

How can this information be applied?

The information needs of both rescue operators and citizens can inform and improve the decision support system by identifying which types of information should be prioritized when developing SDSSs, big data analytics models, and mapping interfaces. In particular, designing an SDSS that considers community priorities and risk perceptions is critical for enhancing knowledge elicitation and sharing among disaster first responders and coastal community residents. Democratizing the process of data collection and exchange requires a systematic outreach approach and community engagement that can be sustained across time (before, during, and after a disaster) and space (regardless of location).

This study attempts to systematically solicit the input and experiences of citizens and rescue operators alike by incorporating key information priorities and risk perceptions regarding flood hazards into flood management SDSS models and mapping interfaces. First, findings of this work underline the importance of seeking alignment between SDSS designs and the information environment in which people and rescue teams operate and interact. The context and complexity of a large-scale disaster event, eg, flood, and surrounding circumstances may lead to increased levels of cognitive and mental workloads for those involved, which, in turn, can hinder their decision-making ability. An inadequate SDSS design may worsen this problem by leading to information overload or underload and failing to properly aggregate and present various disparate information elements at the right time and in the right place. An SDSS design that is informed by the information needs of its intended users, on the other hand, is more likely to help them navigate and contextualize just the right amount and type of information, ultimately improving the quality and efficiency of the resulting decisions.

Second, more accurate information is wanted by both rescue operators and citizens. Real-time information related to evacuation information, such as evacuation mandates and accessible evacuation routes, is critical for both evacuations by citizens and rescue operators alike. An SDSS that addresses these needs could enable users to access various information in real-time. Information might include road closure data to assist in evacuation planning or wayfinding and tools to predict and track flood risk during rescue operations, eg, average water level, vehicles submerged, days before floodwater receded, tweets asking for help, and timing of rescue missions. When further personalized based on the characteristics and preferences of the end user, such data could also provide accurate information about the physical conditions and accessibility of evacuation routes that are critical both to rescuers and the general public.

Finally, greater advanced warning and accurate, real-time information are needed by all parties to enhance disaster response and recovery efforts. Most existing flood warning systems are reactive, expensive, and often use proprietary data and communication protocols to communicate flood risk (at fixed locations). For example, Texas's Harris County flood warning system consists of 163-gauge stations, and water level measurement sensors are mainly placed in bayous and creeks. The coverage of these sensors is not dense enough to provide accurate (or timely) flooding information to create real-time flood depth maps.[65] There is a need for SDSSs to incorporate flood risk estimation models that can be easily deployed at the community level at no cost to conduct flood analysis and risk calculation using existing urban elements and ubiquitous mobile devices.

A well-designed big data analytics model and mapping interface are needed to equip coastal communities and planners with means for ubiquitous disaster impact data collection and exchange and synthesizing accurate information that characterizes disaster risk. Big data analytics can augment existing capacities by identifying weaknesses in disaster management plans and expediting damage assessment.

CONCLUSION

The novelty of including user-identified information when developing decision support systems lies in integrating advanced cyberinfrastructure, decision science, and GIS to enable optimal decision-making in flood emergency management.

This research does have some notable limitations. First, our sample size does not accurately represent the Houston Metropolitan Area, which inhabits over 7 million people. Due to the COVID-19 pandemic, we could not conduct in-person community meetings in the early stages of our research, so surveys were deployed online using a convenience sampling methodology. As a result, we received a limited number of responses to analyze. Second, there is also an imbalance between respondents from the two groups (citizens and first responders). While this does not necessarily negate our results, this should be noted, especially if the SDSS being developed is targeted explicitly toward rescue operations. Third, none of these responses are analyzed based on respondent demographics, as the short answers were simply a source of information to guide the SDSS development. For a more robust method for determining whether interests vary between demographic groups, a greater sample size would be necessary for such a large metropolitan area. Finally, this survey does not necessarily include moderating factors such as the frequency of respondents using social media, accessibility to the internet, and/or income level.

The findings of this survey will be used in future work to ensure an SDSS is developed for this project, but this information is also critical for any community-based SDSS. Including local input in SDSS development is crucial for improving higher-resolution flood risk quantification models, as it democratizes data collection and analysis, creates transparency and trust between people and governments, and leads to transformative solutions for the broader scientific community.

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APPENDIX: SURVEY QUESTIONS

Q1. What is your date of birth? (Note: you must be 18 years or older to participate in this study.)

Q2. Please enter your 5-digit ZIP code.

Q3. Do you have prior first-hand experience with any type of flood event, ie, hurricane, tropical storm, flood, storm surge, etc.?

- ○ Yes
- ○ No
- ○ Don't know

Q4. How many flood events have affected you personally, eg, place of residence or work was impacted by flood waters, in the last 5 years?

- ○ None
- ○ 1
- ○ 2
- ○ 3
- ○ 4
- ○ 5 or more

Q5. Please describe those flood events in more detail, eg, name, date, location, etc.? (short answer)

Answer the following questions for the most recent major flood event you have personally experienced in the last 5 years.

Q6. Did flood waters enter your home?

- ○ Yes
- ○ No
- ○ Don't recall

Q7. How high was the flood waters in your home? Estimate as best as you can recall.

- ○ 2 in. or less
- ○ 3-6 in.
- ○ 7-12 in.
- ○ 1-2 ft.
- ○ 3-4 ft.
- ○ 5-6 ft.
- ○ More than 6 ft.

Q8. Did you evacuate?

- ○ Yes
- ○ No

Q9. How did you evacuate?

- ○ I evacuated on my own with no further help
- ○ I evacuated, but required rescue assistance

Q10. What information did you use to help you evacuate? (short answer)

Q11. Did any of the following limit your ability to evacuate from a flood event, whether you did or did not evacuate? Please select all that apply.

- ☐ I didn't have enough time to evacuate to a safer location
- ☐ I didn't have enough money to evacuate to a safer location
- ☐ I didn't know where to go
- ☐ I didn't know what evacuation routes existed
- ☐ I had no access to transportation
- ☐ I felt I could ride out the storm
- ☐ I stayed to protect my home or business
- ☐ My pets/animals made it difficult to evacuate
- ☐ A medical condition or disability (either myself or a family member) made it difficult to evacuate
- ☐ I had to stay for work
- ☐ I stayed to care for family members
- ☐ Other. Please specify.
- ☐ No, I didn't experience any of these barriers

Q12. What information do you wish you had or wanted to help you evacuate? (short answer)

Q13. Did you need help from first responders or volunteers to evacuate? Please check all that apply.

- ☐ Yes, I asked for evacuation information or directions
- ☐ Yes, I eventually required rescue assistance after evacuation
- ☐ No

Q14. How fast did rescue personnel arrive after you sent out a message or called for help?

- ○ Less than 30 minutes
- ○ Between 30 minutes to 1 hour
- ○ 1-2 hours
- ○ More than 2 hours
- ○ More than 6 hours
- ○ More than 12 hours
- ○ 1 day or more
- ○ Never

Q15. How long did it take for your household to recover from physical damages to your property?

- ○ No major damages
- ○ 1 week or less
- ○ 2-4 weeks
- ○ 4-8 weeks
- ○ 2-3 months
- ○ 4-6 months
- ○ 7-12 months
- ○ 1-2 years
- ○ More than 2 years

The following section will ask you about your involvement (if any) in disaster rescue operations during a major flood event in the last 5 years.

Q16. In the last 5 years, did you participate in rescuing others during the flood event either by boat or car?

- ○ Yes
- ○ No

Q17. Please describe those flood events in more detail, eg, name, date, location, etc.?

For the most recent *flood event during which you participated in rescue activities, please answer the following questions.*

Q18. How high was the flood water in the areas of rescue operations you were involved in? If floodwater depths were variable, please check all that apply.

- ☐ 2 in. or less
- ☐ 3-6 in.
- ☐ 7-12 in.
- ☐ 1-2 ft.
- ☐ 3-4 ft.
- ☐ 5-6 ft.
- ☐ More than 6 ft.

Q19. How did you get to people in need? Please check all that apply.

- ☐ On foot
- ☐ Surface drive boat, eg, gator tail
- ☐ Flat or short bottom boat
- ☐ Air boat
- ☐ Car
- ☐ Truck
- ☐ SUV
- ☐ Other (please specify)

Q20. What kind of information did you use when trying to get to people in need? (short answer)

Q21. What challenges did you experience while conducting rescue operations? Check all that apply.

- ☐ Didn't know depth of floodwaters
- ☐ Didn't know where to take people after rescue
- ☐ Had trouble finding exact locations of victims
- ☐ Ran into debris
- ☐ Other (please specify)

Q22. What information do you wish you had or wanted to help you carry out rescue operations? (short answer)

Q23. Do you have a mobile device (smartphone or tablet computer)?

- ○ Yes
- ○ No

Q24. Is it Android or iPhone? If you have multiple mobile devices, please check all that apply.

- ☐ iPhone (Apple)
- ☐ Android (such as Samsung, LG, Motorola)

Q25. Do you have access to data without WiFi (only through cellular network) on your mobile device?

- ○ Yes
- ○ No

Q26. On a daily basis, how much time do you spend on your mobile device?

- ○ Less than an hour
- ○ 1-2 hours
- ○ 3 or more hours

Q27. In your opinion, what features should a mobile/computer app have in order to be useful for personal evacuation navigation or flood rescue operation? (short answer)

The following section will ask you questions about your demographic background.

Q28. What is your gender?

- ○ Male
- ○ Female
- ○ Nonbinary
- ○ Do not wish to answer

Q29. What is your race/ethnicity? Please select all that apply.

- ☐ White or Caucasian
- ☐ Black or African American
- ☐ Hispanic
- ☐ Asian
- ☐ American Indian or Alaska Native
- ☐ Native Hawaiian or other Pacific Islander
- ☐ Other
- ☐ Do not wish to answer

Q30. What is your annual household income?

- ○ Below $14,000
- ○ $14,001-$31,000
- ○ $31,001-$63,000
- ○ $63,001-$113,000
- ○ $113,001-$184,000
- ○ $184,001-$248,000
- ○ $248,001-$475,000
- ○ Above $475,000
- ○ Do not wish to answer

Q31. What is your highest level of education?

- ○ High school graduate
- ○ Some college
- ○ Associate's degree (2-year college)
- ○ Bachelor's degree (4-year college)
- ○ Master's degree
- ○ Doctoral or other professional degree
- ○ Do not wish to answer

Notes

JEM

Manufactured housing communities and climate change: Understanding key vulnerabilities and recommendations for emergency managers

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ABSTRACT

Manufactured housing communities (MHCs), commonly referred to as mobile home parks, provide an estimated 2.7 million American households with largely unsubsidized, affordable housing. Climate change threatens those who call these communities home by exacerbating known structural and social vulnerabilities associated with this housing type—including but not limited to increased risks to flooding, extreme temperatures, high winds, and wildfires. Climate change requires emergency managers to understand the diverse, integrated, and complex vulnerabilities of MHCs that affect their exposure to climate change risk. This article presents findings from an integrative literature review focused on the climate-related vulnerabilities of these communities described at three levels of scale: household, housing structure, and park community. It then draws on 15 years of engagement and action research with MHC residents and stakeholders in Vermont, including several federally declared flooding disasters, to distill key recommendations for emergency managers for assisting MHCs to prepare for and respond to emergencies. As climate change accelerates, emergency managers can increase efficacy by learning about the MHCs in their jurisdictions by leveraging the best available data to characterize risks, integrating MHCs into planning and mitigation activities, and engaging in conversations with stakeholders, including MHC residents and their trusted partners.

Key words: climate change, vulnerability, affordable housing, manufactured housing, mobile home parks, Vermont, flooding; DOI: 10.5055/jem.0845

INTRODUCTION

As the effects of climate change are felt across the United States (US), manufactured housing communities (MHCs) are often among those experiencing the greatest impact, disruption, and loss from climate-related hazards. The increasing and intensifying impacts of climate change require emergency managers to characterize risks, identify vulnerabilities, engage stakeholders, develop plans, and mobilize resources to enhance resilience at all scales.[1,2] Significant research has established that climate-related hazards, such as floods, severe storms, wildfires, drought, and extreme temperatures, have disproportionate impacts on communities of color, low-income households, and households with compromised health conditions that exacerbate structural social inequities that previously existed in the community.[3-6]

MHCs, commonly referred to as mobile home parks, provide an estimated 2.7 million American households with affordable housing.[7] This is the largest source of unsubsidized housing that is affordable for low- and moderate-income households—especially given the current national housing affordability crisis that is impacting homeowners and renters alike.[8] The affordability of MHC living has led to greater access for low-income households to obtain similar privileges as single-family homeowners, such as privacy and independence, along with lower maintenance costs and less upkeep.[9-13]

This article provides emergency management professionals and scholars with an overview of this critically important housing type in the context of climate change. Part 1 synthesizes findings from the scholarly literature about key vulnerabilities associated with each level of scale within the typical MHC—the household, the housing structure, and the park community. Part 2 describes experiences from an action research program dedicated to MHCs in the state of Vermont. The article concludes with recommendations for emergency managers that can inform response, recovery, and mitigation efforts in the face of climate change.

METHODOLOGY

We used an integrative literature review method of the scholarly literature related to the three dimensions of vulnerability of MHCs to synthesize recommendations for emergency managers regarding MHCs and their vulnerability to climate-related hazards. Such an approach is useful for exploring dynamic relationships.[14] This review focused on peer-reviewed scholarship published between 1990 and 2023. The article then presents a case study of climate-related hazards impacting MHCs in Vermont with actionable connections to emergency management. We propose five recommendations supported by experiences gained from an action research-based program[15] between university researchers and a statewide nonprofit organization that spans more than 15 years of collaboration. The University of Vermont partnered with the Champlain Valley Office of Economic Opportunity's (CVOEO) Mobile Home Program, a statewide advocacy organization for Vermont's MHC residents. This partnership has conducted several surveys on MHC residents, led emergency planning activities with MHCs, and completed two statewide flood risk assessments since 2007.

Understanding MHCs in a climate-changed world: Vulnerabilities at different scales

Emergency management professionals equipped with a working understanding of MHCs are better positioned to identify how vulnerability manifests at different levels of scale nested within these communities. This section discusses how vulnerabilities can be analyzed at the three scalar levels of a MHC: the household, the housing structure, and the park. This understanding can allow emergency managers and planners to recognize potential opportunities for actions that can increase resilience to climate-related hazards within these communities.

Vulnerabilities at the household scale. MHCs have long been the largest form of unsubsidized housing for low- and moderate-income Americans at a time when affordable housing is increasingly difficult to obtain.[7,11,12,16] MHCs provide residents with the potential to enjoy privacy, a sense of agency within their homes, and lower maintenance responsibilities compared to other forms of housing.[9,10]

Recent studies examining the demographic composition of MHC households highlight the intersections of class, gender, race, and ethnicity with manufactured housing.[7,17-19] Kear et al.[18] identified four different subgroups of MHC households that face greater likelihoods of housing insecurity: (1) noncitizens, Latino households with children, (2) extremely low-income female-headed households, (3) low-income female-headed households in older homes, and (4) fixed-income older adults. Clusters of manufactured housing units are an indicator of informal residential subdivisions, known as "colonias," along the southern border region of the US.[17] These communities have higher rates of poverty, immigrants, and noncitizens relative to all Census Places.[17] Colonias also face health and safety concerns stemming from substandard or nonexistent water infrastructure, which are exacerbated when disasters occur.[19] To address disparities recognized during the response and recovery efforts from Hurricane Katrina, the National Council of La Raza, an organization dedicated to supporting the health and wellbeing of Latino communities, designed a toolkit to inform emergency managers and first responders about the unique needs of Latino communities during and after disaster events.[20] Additionally, manufactured housing is a significant source of affordable housing in tribal communities, accounting for an estimated 12 percent of the total housing units on tribal lands.[21]

Vulnerability resulting from social stigma associated with MHCs has been studied by various scholars.[10,13,22-24] MacTavish[23] found that while two thirds of MHC residents reported their communities as being "somewhat desirable" or "very desirable" places to live, many families experienced social disadvantages stemming from their park residence, including exclusion from other resources and programming in the municipality. This stigma can limit the ability of park residents to form valuable social capital and networks with organizations and agencies serving the municipality or the broader region, which can be detrimental for community-building opportunities, especially in times of crisis.[13] The social vulnerability of manufactured housing residents exposes them to a variety of hazards, including flooding, hurricanes, tornadoes, wildfires, and extreme heat.[5,25-28]

A key challenge for owners is the classification of manufactured homes as personal property in most states,[11,12] which largely denies them the full benefits of homeownership, creating barriers to accessing financing and home improvement loans.[12] MHC residents are in a precarious financial and legal position due to owning oftentimes immobile housing units, while renting the land underneath them on a month-to-month or yearly lease. Sullivan[22] investigated park closures within investor-owned MHCs and documented the socio-emotional costs of losing one's home to park closures. With the current trend of large real estate investment companies purchasing more parks, more MHC residents are likely to face similar circumstances.[29]

Vulnerabilities at the housing structure scale

Construction practices and materials. Manufactured housing units that predate the 1976 HUD Code, which regulates construction methods and materials, are particularly vulnerable as their structural integrity is likely to be inferior to that of newer HUD Code homes.[30] These housing units (also disparagingly referred to as "trailers") have long been associated with images of depression-era recreational trailers being pulled behind vehicles.[24,30] After World War II, the industry moved to producing the larger, rectangular forms that most people would recognize as a manufactured home, moving from a factory on a trailer to be installed on a site—either on a rented lot in a park or on privately owned property.[11,31] As production increased, concerns grew about lax and dangerous construction practices, including unhealthy indoor air quality, faulty electrical wiring, poor insulation, and the unreliable integrity of roofs and floors.[30] Tragically, due to their small size and highly flammable construction materials, fires can be especially devastating in mobile homes compared to other single-family housing units.[32] Federal legislation was enacted in 1976 to regulate an industry producing a massive number of housing units with variable construction practices by establishing important safety and quality standards.[31] However, an estimated 26 percent of the total manufactured housing stock, or nearly two million pre-1976 mobile homes, remain in the nation's housing stock.[30]

Structural vulnerabilities to high winds remain a significant issue for manufactured housing units, even for those constructed after 1976.[27,33-38] Despite fatalities from tornadoes declining in the US over the past century, residents of manufactured housing are estimated to be 10-15 times more likely to die from a tornado compared to residents of conventional housing.[27] In fact, although manufactured housing represents 8 percent of the total US housing stock, 40 percent of tornado fatalities occur in manufactured housing units.[39] Despite amendments to the HUD Code in 1994 to strengthen wind resistance and other protective actions once installed, these units remain vulnerable. A recent study investigated wind damage to mobile homes during Hurricanes Irma and Michael in Florida.[38] While they found that homes built prior to 1976 sustained damages at even the lowest wind speeds, homes built between 1976 and 1994 had a large range of damages. Unfortunately, even homes constructed after 1994 subjected to the highest wind speeds suffered complete losses.

Energy burdens. In a study exploring the nexus of energy, poverty, and climate change, Jessel et al.[40] found manufactured housing units were likely to be poorly insulated—particularly those built prior to the HUD Code of 1976. The use of low-cost construction methods led to significantly energy-inefficient structures, which, while lowering the upfront purchase price of the unit, led to annual energy costs that were double those for a conventional single-family home on a per-square-foot basis.[41] MacTavish et al.[32] found residents of older manufactured housing units in Oregon were spending 70 percent of their monthly incomes on energy costs during the winter months. The consequences of these hidden and disproportionate energy burdens that residents must shoulder from these structural deficiencies can be tragic and of particular concern as temperatures rise in the coming years. A study of heat deaths in Maricopa County, Arizona, from 2006 to 2018 found that manufactured housing residents suffered a disproportionate share of indoor heat-related deaths; 27 percent of fatalities occurred in mobile homes while representing less than 5 percent of the county's total housing stock.[28] These residents were also less likely to have air conditioning and more likely to have had their electricity turned off—lacking the financial means to afford energy costs to maintain their homes at a safe temperature.

Not so mobile homes. The often-used term "mobile home" is outdated and contrary to the current reality of manufactured home ownership. The 2011 American Housing Survey found that the vast majority of these housing units, 79 percent, had remained in place at their original site.[30] While surprising, financial barriers and structural concerns oftentimes make relocation unfeasible. Residents moving their home need significant financial resources—assuming another location is available. Costs to relocate a home range between $4,900 and $10,350, depending on the size of the home and the length of the move.[42] This is only possible *if* the home is in stable enough structural condition to withstand a move. An additional implication is that when parks are close to being redeveloped or homeowners face eviction, it is likely they cannot move their home. Pointing to the prohibitive costs of moving and the precarity of land-lease placements, MacTavish et al.[32(p110)] summarize, "these insecurity factors, inherent to mobile home life reinforce housing vulnerability. Families have homes but can lose them in a heartbeat if a park is sold, rents for a lot or home are raised, or if park management reneges on their rights to live there."

A hidden financial burden that further complicates the transition from older to newer homes is the cost of disposing of older manufactured homes when these units are no longer habitable. Whether for park owners who are looking to rehabilitate a formerly occupied lot with an abandoned home or an existing homeowner who wants to upgrade to a newer home, the cost of removal and disposal of abandoned or uninhabitable homes is substantial, ranging from $5,000 to $10,000. Furthermore, testing and removal of hazardous waste, particularly asbestos, can add thousands of dollars in additional costs.[41]

Vulnerabilities at the community scale

Siting concerns. The risks to MHCs are exacerbated by exclusionary zoning practices at local levels that discriminate against this housing type. Manufactured homes were found to be located further away from positive public facilities such as hospitals and schools when compared to other types of housing units, including single-family homes, condominiums, and apartment buildings.[43] The American Planning Association policy guide addressing equity issues related to manufactured homes urges practitioners to revise outdated and exclusionary regulations and to uniformly apply design standards to all housing types and not solely manufactured homes.[44] Zoning issues may limit options for relocating parks out of hazardous areas. Many parks were built prior to land use regulations, and multiple studies have found that parks are more likely to be placed in places prone to flooding compared to other housing types.[25,45,46] A 2022 study by Pierce et al.[47] examined the risk of households living in manufactured housing units to heat and wildfire hazards in the state of California. They found that 81 percent of the state's manufactured housing park lots were in areas within the wildland-urban interface and determined the lots to be disproportionately located in areas with greater exposure to extreme heat.[47]

Infrastructure concerns. MHCs have been relatively low-investment housing developments. In earlier times, only the most basic elements were required to start a park—reasonably level ground, basic roads, electrical hookups, and access to water and sewer systems. Given the lack of land tenure and relative immobility of most mobile homes, this housing type is vulnerable when ownership transfers occur and residents could be displaced.[22] With few to no new MHCs being built today, the supply of lots is highly constrained in many places, including in rural states like Vermont.[48] One driver of this supply limitation is the very expensive infrastructure required (both installation and maintenance) for a park relative to the low rents that would be generated. Pierce and Jimenez[49] found unreliable potable water access to be three times as likely for MHC residents compared to residents in all other housing types. Ehrenfeucht[50] identified aging infrastructure, including water systems, insufficient electrical service, and inadequate fire protection as disincentivizing for the development of new MHCs.

EXPERIENCES IN VERMONT: MHCS AND CLIMATE-RELATED HAZARDS

Vulnerability to climate-related hazards can manifest in different ways at each of the three levels of scale within MHCs: the household, the housing structure, and the community. This section builds on the scholarly literature to share how these vulnerabilities have been observed within MHCs in Vermont. For a state that is often seen as a refuge from the worst climate-related hazards,[51] its MHCs have repeatedly experienced significant damage from flooding.

The State of Vermont has 238 MHCs registered with the Agency of Commerce and Community Development that offer 7,094 lots for manufactured homes, with 90 percent of those units owned by residents.[52] Demand for lots in MHCs is high, as demonstrated by the low statewide vacancy rate of 4.6 percent in 2022, with rates even lower in the state's most populated areas in northwestern Vermont.[52] High demand has been amplified by the loss of lots due to park closures. Since 1999, Vermont has experienced the closure of 28 MHCs, resulting in the loss of approximately 350 lots.[48] Fewer than 10 new MHCs have been built in Vermont since 1995, with no new MHCs developed since 2014.[52] Over 95 percent of Vermont's MHCs are greater than 20 years old and were established prior to current land use planning regulations. Many are located on land marginally suitable for housing developments, including poorly drained soils in areas with flood.[53,54] These older MHCs are often in need of major infrastructure upgrades, and park owners often choose to sell or close to redevelop the park for a different use when faced with such significant expenses.

The Vermont Climate Assessment[55] found that extreme events, including heavy precipitation storm events and flash flooding, have become increasingly common due to climate change. Vermont's MHCs are already no stranger to impacts from severe storms and flooding. In 1998, a severe storm led to the flash flooding of a small community with nine homes along the New Haven River in Bristol, Vermont. All homes were deemed to be a total loss, and the park was subsequently closed. The property was bought out through hazard mitigation funding and transformed into a green space, providing local residents with access to the river and walking trails.[56]

The "Memorial Day Storms" of 2011 (DR-4001) brought flash flooding to a 35-lot MHC along the Stevens Branch of the Winooski River. This park was later re-opened with only 10 lots; notably, many of the manufactured homes placed on these remaining lots were destroyed in the recent flooding event described later in this section. Just 3 months after the "Memorial Day Storms," Tropical Storm Irene brought more devastating flooding to the state. An unusually rainy summer in central and southern areas contributed to the widespread extreme flooding after 4-8 in. of rain fell within a 24-hour period.[57] President Obama declared a major disaster (DR-4022), providing Individual Assistance funds for four counties, which led to over $23 million in total funding from the Individuals and Households program.[58] This storm serves as a significant focus event, highlighting climate change impacts on MHCs, catalyzing policy debates about vulnerability and resilience at the state and local levels.[59,60] While manufactured homes in parks represented just 7 percent of the state's overall housing stock, they were 15 percent of the total flooded housing units receiving assistance from the US Federal Emergency Management Agency

(FEMA).[54,61] Twenty parks (6 percent of the state's parks) with over 200 homes were directly affected by flooding, leaving many homes destroyed and two parks with a total of 17 lots permanently closed.[42,48] Ten lots located in the floodway were permanently closed at an age-restricted (residents aged 60 or greater) MHC as a hazard mitigation action. All lots were eventually re-opened at Weston's Mobile Home Park in Berlin, Vermont, where 80 of the park's 83 lots sustained some level of flood damage.

Questions quickly emerged about what to do with numerous flooded manufactured home units. Homeowners were faced with the burden of disposing of their destroyed homes while navigating recovery resources to find their next stable housing option. The State of Vermont, with the support of philanthropic funds, coordinated a deconstruction process that removed 68 destroyed homes.[62] Many older homes were found to have asbestos, lead paint, and mercury issues that required remediation—adding further complications and costs to this process. This experience raised critical questions about processes related to accessing FEMA Individual Assistance funds and condemnation authority within Vermont, resulting in the creation of a "condemned to be destroyed" order process that is led by the State Fire Marshal Office.[42] Eighty-two homeowners from MHCs were able to receive additional funding from FEMA as a result of that process—yielding $1,029,997.70 to support the homeowners' recovery.[42]

Most recently, the July 2023 flooding events damaged or destroyed over 70 manufactured homes in four MHCs in three counties. The 32-dwelling Berlin Mobile Home Park is nearly 100 percent in the designated floodway, with very few homes properly anchored and elevated.[63] While this community had largely been spared widespread flooding during previous storm events, it was severely flooded on July 10, 2023. A nearby MHC with a known history of flood damage also experienced severe flooding, resulting in seven of its nine remaining homes being destroyed and leaving two additional homes unhabitable because of severe damage to park infrastructure. Local emergency officials visited the two communities to encourage residents to evacuate, and several households required technical rescues as the flooding worsened.[64]

Similar to the circumstances during Tropical Storm Irene, many impacted households were older adults with health conditions and mobility concerns. Some households have been challenged with remaining within a reasonable distance of employment or health services. Families with school-age children faced significant challenges in finding affordable temporary or permanent housing solutions that would enable them to remain in their community, close to family, friends, and schools. The State Division of Fire Safety provided condemnation orders due to severe health and safety concerns resulting from the flooding of 52 manufactured housing units in the four impacted MHCs.[65] The fates of these impacted MHCs remain uncertain as recovery efforts were continuing at the time of this study. The lack of available lots in existing MHCs located outside of flood hazard areas is a major constraint for long-term recovery prospects. Similar to past flooding disasters, it is unlikely that new MHCs will be developed to offset any potential permanent lot closures, given the expensive land acquisition and infrastructure development costs.

Table 1 provides a summary of how vulnerability emerged at each of the three scalar levels during the various flooding events impacting MHCs. Compounding vulnerabilities across scales presented challenges for MHC residents, emergency management professionals, and other key stakeholders as they navigated response and recovery processes. These experiences are further complicated by the extreme shortage of affordable housing in Vermont, driven by many factors, including high property prices, low vacancy rates, a shortage of new construction, and strong demand, across the state.[66]

DISCUSSION

Experiences from Vermont's flood-impacted MHCs underscore the opportunity for emergency managers to engage in efforts to build community resilience within these unique communities to address vulnerabilities before, during, and after an event. This section offers five recommendations that emergency managers should consider as climate change further

Table 1. Summary of vulnerability by level of scale in Vermont MHC flooding experiences	
MHC scale	**Common observations from Vermont MHC flooding events**
Household level	• Households with limited financial resources
	• Older adults (65+ years or older)
	• Pre-existing health or mobility concerns
	• Households needing to remain close to employment, education, and other critical community resources
Housing structure level	• Housing units unable to be feasibly repaired from flooding damages
	• Older units contained hazardous materials
	• Lack of homes properly elevated and anchored
Community level	• Parks sited in FEMA-mapped floodways, 100-year and 500-year floodplains
	• Aging infrastructure concerns
	• Threat of whole community displacement
	• Challenges with finding temporary and permanent housing alternatives

heightens the risks facing MHCs. These recommendations ask emergency managers to address vulnerabilities to climate-related hazards across all three scalar levels of these communities.

While these recommendations are discussed within the context of Vermont, these are generally applicable in other regions across the US where MHCs are prevalent, such as in Colorado, where several parks were affected by extreme flooding in Colorado, 2013,[25] severity and frequency of hurricanes in Florida,[67,68] or extreme heat in Arizona.[28] Emergency managers may need to advocate for policy changes, program development, or resources to be able to put these recommendations into action with actual communities on the ground.

1. Emergency managers should know where MHCs are located within their jurisdictions and identify risks relative to climate hazards

To effectively engage, plan, and respond to climate-related emergencies affecting MHCs, emergency managers and planners need to know where they are located within their jurisdictions.[25,54,69] Far too often, MHCs are, in practice, invisible—either due to their separate physical location or social stigmas that can leave these communities isolated from the rest of the community.[13,23,24,43,70] This is an opportunity for emergency managers to leverage existing datasets to improve their understanding of where MHCs may be located in their communities and use these data to assess risks to climate-related hazards at the community scale.[54] Beyond identifying locations, it is important to engage with households about specific risks within their community. In addition to facilitating the identification of MHCs at risk from climate change, having a registry of manufactured housing can be invaluable for identifying dense, vulnerable communities and quickly deploying scarce resources to meet critical needs during disaster events.

The State of Vermont requires all MHCs to be registered on an annual basis with the Department of Housing and Community Development pursuant to Chapter 153 of Title 10 of the Vermont Statutes.[42] Using this annual registry data, which includes information such as park owner, lot numbers, vacant lots, and lot rent figures, the authors and their collaborators have undertaken two Geographic Information System-based analyses.[63] These efforts produced mapping deliverables that created individual park maps for each community in the state and overlay analyses of MHC locations relative to the best available FEMA flood maps and other state-generated flood risk data.[54,63] These maps are important tools in conversations with MHC residents about flood risks within their communities and with emergency managers, nonprofit housing organizations, and regional planners.

In states that do not require MHC owners to register, it can be challenging for emergency managers to access address information at the park scale. In these instances, geospatial tools and datasets, such as

Enhanced 911 building locations, can be helpful with identifying clusters of manufactured housing units across the landscape. Reaching out to local planners and housing stakeholders could help close the information gap. Emergency managers can also advocate for policy changes to collect MHC location data and for resources to support risk assessments to learn more about the household demographic characteristics and conditions of the housing structures within MHCs.

2. Emergency managers should be familiar with the household social vulnerability and community dynamics of the MHCs within their jurisdictions

Social vulnerability at the household scale can pose a challenge for emergency managers when working with MHCs to address climate-related hazards at the community scale.[25] Prior to engaging with parks, emergency managers should consider the contributions parks make to their communities, such as offering affordable housing and dense development, and consider sensitivities, including social stigmas surrounding MHCs. In our work in Vermont, municipal officials and emergency managers were often unaware that residents in MHCs may not have an established body that represents them, making it difficult to identify who represents the concerns of residents.[10] Resident organizing takes time, trust, and sensitivity—issues that can benefit from establishing partnerships with trusted local organizations that work closely with park residents. Attempting to engage with MHC residents about risks and preparedness without investing in community organizing can lead to wasted efforts and disappointment for all involved.

Tropical Storm Irene impacted MHCs disproportionately relative to other forms of housing.[54] Given that these communities provide the most affordable unsubsidized housing to low- and moderate-income Vermonters, it was not surprising that MHC residents faced challenging circumstances during the response and recovery efforts. In some cases, multiple generations of the same family were displaced from their homes since they were living in the same park, and this led to limited temporary housing options in the days and weeks following the initial impact of the storm. A statewide survey of MHC residents found that over 40 percent of residents had a health condition or mobility challenge that would be a concern in an evacuation situation.[71] Having a sense of who lives in local parks enables emergency managers to make informed decisions before, during, and after emergency events—whether in response to forecasts for severe storms and flooding concerns or enacting plans for cooling shelters in communities enduring extreme heat waves. It is also important for emergency managers to understand the ownership structure of the MHCs in their communities. Are the MHCs owned and operated by a regional housing organization or by a large, out-of-state investment company that may not have a good understanding of local conditions or processes? In some cases, the park may be cooperatively owned by the residents themselves, and the cooperative board could serve as a key point of contact.

3. Emergency managers should develop long-term partnerships with trusted organizations and networks to strengthen connections with MHCs

MHC residents may not have a history of positive experiences with local officials, given persistent social stigmatization and marginalization—or just simply a lack of time to engage in local matters due to work and family obligations.[10,13] Emergency managers should develop partnerships with trusted organizations that already engage park residents to accelerate building connections. In Vermont, the CVOEO Mobile Home Program is the statewide mobile home advocate and is well-positioned to bridge connections between university researchers and community members. Nonprofit-owned parks may have a trusted manager, a functioning resident association, or a resident-owned cooperative park board that may be an interested partner in emergency planning efforts.

One of the greatest vulnerabilities of MHCs stems from being communities of largely homeowners with relatively immobile homes on land owned by another agent. This limits the mitigation measures that homeowners could pursue due to a lack of financial investment or support from park ownership. Promoting resident-owned cooperatives and providing resources as a resilience-building strategy can not only protect the residents living in the park at the time but also preserve the park for future homeowners.[70,72] Supporting resident organizing as a resilience-building strategy, especially in MHCs without a formal board, can be a resource-intensive effort but can be an important asset to draw upon in emergencies. There is a wealth of scholarship that supports social capital as a key driver of resilience within communities.[73,74] Even small steps to promote engagement between park residents and other community resources can yield benefits for resilience when a crisis occurs.

4. Emergency managers should be aware of the unique vulnerabilities and needs of MHCs during the response and recovery phases when hazards occur

Emergency managers familiar with how vulnerability exists at the different levels of scale within MHCs can better support households during response and recovery efforts. Understanding that climate-related hazards will exacerbate residents' social vulnerability can inform emergency managers' initial steps when hazards occur. For example, older MHC residents may not be able to use smartphones or computers to access critical resource information. Emergency managers may need to provide printed information or collaborate with local organizations to ensure that all impacted residents have equitable access to information. This may require providing translated printed materials in languages other than English or developing outreach tactics that engage residents in culturally appropriate ways.[20] During federally declared disasters, there are important processes that need to be followed to maximize benefits and support MHC residents' long-term recovery. Ensuring that residents can access these resources is vitally important for their resilience—especially in regions where there may be a severe lack of affordable housing options available.

The housing unit itself presents vulnerability considerations that differ from conventionally built homes. Damages from climate-related hazards, such as severe storms or flooding, can leave manufactured housing owners with the challenge of how to remove and dispose of their destroyed homes.[62] Manufactured housing units can quickly develop severe mold issues if not effectively treated immediately when damaged by flooding. If floodwaters twist the steel frame of a manufactured home, it is considered a total loss. MHC residents may then find themselves with the triple challenge of dealing with their destroyed home and securing a safe temporary housing situation while trying to determine their permanent housing solution.

Emergency managers can play an important role in identifying opportunities that exist within current state and federal programs to address MHC-specific concerns. For example, in the aftermath of Tropical Storm Irene, over 100 manufactured homes were effectively destroyed beyond reasonable repair. The State of Vermont and several partner organizations developed a program to remove destroyed homes at no cost to the affected homeowners. Baker et al.[62] calculated the cost per home deconstructed and disposed of to be nearly $3,000 in 2011. This figure includes the necessary asbestos testing and abatement as well as the disposal of any household flood debris left inside the homes, offset by the revenue created by salvaging scrap metal.

5. Emergency managers can seek opportunities for hazard mitigation and emergency preparedness activities within MHCs

Climate-related hazards pose risks to MHCs across the US, while the human toll and economic costs of leaving these communities in harm's way continue to increase. Emergency managers are well-accustomed to assessing risks and identifying potential opportunities to mitigate those risks. Identifying parks that may be at the highest degree of risk and investing in hazard mitigation funds to harden infrastructure or even relocate households out of the riskiest areas could prevent future repetitive losses.[54,70] This could be as straightforward as developing new

Table 2. Recommendations for emergency managers to increase MHC resilience		
Recommendations	**Primary scales**	**Potential action steps to consider**
Emergency managers should know where MHCs are located within their jurisdictions relative to climate hazards	Community	• Develop a database with MHC locations relative to known hazards within the community
		• Conduct assessments with local MHCs and relevant stakeholders
Emergency managers should be familiar with the household social vulnerability and community dynamics of the MHCs within their jurisdictions	Household, community	• Increase understanding of the specific MHCs, including the number of households, ownership structure, and presence of resident organizing efforts
		• Ensure that local emergency response officials are aware of MHCs located in their communities
		• Advocate for resources to reduce social vulnerability of MHC households
Emergency managers can develop long-term partnerships with trusted organizations and networks to strengthen connections with MHCs	Household, housing structure, community	• Establish connections with MHCs during noncrisis times
		• Encourage residents to develop park-specific emergency plans
		• Support partnerships with local emergency responders to engage park residents in emergency preparation activities specific to the community context of the park—especially those in known hazardous areas
Emergency managers should be aware of the unique vulnerabilities and needs of MHCs during the response and recovery phases when hazards occur	Household, housing structure, community	• Ensure that recovery officials understand that park residents have different needs than conventional homeowners and renters
		• Assemble local resources to support park residents in a "one-stop shop" format to meet the urgent needs of socially vulnerable households
		• Provide case managers and volunteers with specific knowledge of MHCs to support residents in the recovery process
		• Determine if there is an opportunity for creating a large-scale deconstruction program of destroyed homes
		• Seek opportunities to creatively leverage recovery funds or philanthropic sources to support residents in purchasing more energy-efficient mobile homes
Emergency managers can seek opportunities for hazard mitigation and emergency preparedness activities within MHCs	Household, housing structure, community	• Gather as much data as possible about MHCs—location, risk exposure, resident demographic characteristics
		• Educate local and state hazard mitigation planners about the unique vulnerabilities of MHCs
		• Align with resources, like weatherization programs, to make tangible improvements in the housing stock that reduce resident vulnerability to extreme temperatures
		• Identify MHCs to co-develop hazard mitigation projects
		• Identify MHCs most "at risk" (including hazard exposure, infrastructure concerns, and ownership status) to consider strategic park buy-outs with an emphasis on resident preferences for relocation

lots in existing parks where it is possible to do so within property boundaries, or as complex as relocating whole communities to new sites with less risk exposure where feasible.

Emergency managers can develop programs for creating MHC-specific emergency preparedness plans and activities that span all three scales of MHCs. Having a tangible emergency plan template that MHC households and whole communities can use, test, and revise is an important step toward building resilience. The plan could be used as the basis to call for mitigation activities, like installing a generator at a park's community center. Beyond the quality of the plan, the process of planning is an opportunity to build relationships and facilitate connections to community resources. Emergency planners can seek opportunities in the process to foster connections with local and regional resources to open lines of communication and reduce the invisibility of these communities to external partners. For example, collaborations between emergency managers and weatherization organizations could lead to concrete actions that address energy burdens that strain the financial wellbeing of MHC households while making physical improvements at the housing structure level—reducing risk of extreme temperatures.

Table 2 provides emergency managers with recommendations and specific actions that could be pursued to address MHC vulnerabilities, deepen engagement, and build resilience. Whether advocating for policy changes, seeking new resources, identifying potential projects, or simply initiating conversations with MHC residents, emergency managers have an opportunity to contribute to the overall sustainability of these important communities.

CONCLUSION

Emergency managers who understand the relationship between climate-related hazards and the three dimensions of vulnerability of MHCs will be better positioned to effectively engage in planning, responses, recovery, and mitigation activities. While this article has largely focused on the impact of flooding on MHCs, emergency managers can identify the climate-related hazards that are most relevant to their jurisdictions—hurricanes, wildfires, or extreme heat. As climate change impacts increase and the housing affordability crisis continues, the social vulnerability of MHC residents is amplified, and there may be few housing alternatives when disaster strikes. Emergency managers can identify policies and resource gaps that impede efforts to increase resilience-building efforts with MHCs. Future research can identify factors that positively influence the success of efforts undertaken by emergency managers and MHCs in the context of a climate-changed world. Using an equity lens to guide future work is critical to addressing vulnerability at different scales. For example, given the high prevalence of manufactured housing in tribal communities across the US, emergency management practitioners and researchers alike could pursue opportunities to engage with tribal leaders to explore potential resilience-building activities. This work is required to address the heightened vulnerabilities of this vitally important affordable housing stock.

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Notes

JEM

Rapid assessment of public interest in drought and its likely drivers in South Africa

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ABSTRACT

The monthly search volumes for drought were extracted from Google® for South Africa using the Keywordseverywhere.com *plugin from January 2004 until June 2022. To identify the potential qualitative drivers for such public interest the following data extracted by the plugin were investigated and analysed: the drought-related keywords, the long-tail keywords similar to drought, and the "people also searched for category" from the South African users. The Google Trends monthly score was extracted for South Africa and the Eastern Cape Province, and specific local municipalities/towns/cities in the province. The aim was to assess the relative significance of the drought interest in comparison to public interest in other search terms. The results of the Kruskal–Wallis analyses of variance by ranks showed that there was a statistically significant difference between individual values of the monthly search volumes for drought in South Africa, as a function of time of data extraction (5 percent level of significance; p-value $\leq 4.7 \times 10^{-14}$). The monthly search volumes increased with time, which is based on the results of the Mann–Kendall test at a 5 percent level of significance (p-value ≤ 0.0092). Analyses of the Google Trends scores indicate that the relative interest in drought in South Africa and the Eastern Cape Province increased with time between January 2004 and June 2022 (the Mann–Kendall test at a 5 percent level of significance; p-value = 0.0011). The population's searches for drought were relatively low when compared to other search terms on Google. Drought adaptation of the South African community could be considered a driver of the Google searches for drought, but it is a marginal topic compared to other topics in Google searches. It might be necessary to increase this significance by investigating the "Google-search patterns for droughts" in the areas of Tshikaro, Mafusini, Cofimvaba, and Nxotsheni in the Eastern Cape Province of South Africa.*

Key words: search volumes, Google Trends, monthly volumes, statistical analyses; DOI: 10.5055/jem.0834

INTRODUCTION

Climate change, coronavirus disease 2019 (COVID-19), and the need for humanity to deal with these challenges and disasters have become a set of ever-present, nagging, and probably even "obsolete" terms in the context of human existence in the 21st century.[1] Impacts of these nagging terms, the real-life outcomes of the intersection of climate change, socioeconomic vulnerabilities, and concurrent disasters have been seriously disrupting the nature of individual and collective existence of humanity.[1] The relevant disaster outcomes and impacts vary in type, extent, and the scope and nature of possible cascading effects. Practically, climate change and the reality of the 21st century are both results of the collective action of humanity over the centuries. This can be related to the increasing concentrations of greenhouse gases in the atmosphere and the reality of the impact of humanity's current actions at the species and individual levels.[2] Disasters often take place at the interface between climate change and human reality in the 21st century.[3] This interface is dynamic, as are the outcomes/impacts of the resulting disasters. In particular, the impacts can vary with the geographical location of the human population in question,[4] the level, and the planned nature of human development in an area.[5] Additional factors of influence will include the status of the resilience of the population and the disaster risk management systems at the community, national, and international levels.[6] Global specifics and country-level examinations are often done, but narrower geographical examination often lacks the detail necessary for effective planning.[7]

South Africa is a country with a diverse disaster profile, and examples from recent years include dwelling fires in low-income settlements.[8] Floods have occurred in the provinces of North-West and Gauteng in late 2022.[9] Periodical and recurring droughts have unfolded in the provinces such as the Eastern Cape since the 1940s, and the recent drought has been in place for about 7 years.[10] As humans are at the center of evaluating and mitigating disaster impacts, it is important to study the temporal trends in the public interest of specific disasters and their outcomes. The relevant disaster hazards in South Africa are associated with the vulnerability of the country's population, which is strongly affected by the history of unequal distribution and access to resources prior to 1994.[11] Drought has an impact on the important drivers of economic activity and livelihoods in many provinces of South Africa.[12] In 1992, a massive drought led to a 40-50 percent drop in cattle prices and the deaths of 500,000 bovines in one of the former homelands of South Africa, namely, Transkei.[13] In 2007 and 2008, socio-economic status and access to land determined whether farmers decided to reduce the livestock herd size as a mitigation and adaptive measure in the face of disaster. Archer et al.[14] showed that the 2015-2019 drought in the Eastern Cape was comparable to past droughts in the province. Edossa et al.[15] showed that there were on average 13-20 droughts across South Africa between 1952 and 1999. The most severe droughts occurred between 1973 and 1995,[15] followed by the 2015-2019 one with a significant impact on agriculture.[14] The grassiness index was higher in 2020, as compared to 2019, and so a partial recovery of the vegetation was underway in 2020. However, the combined effects of the drought and COVID-19 had profound impacts on agriculture, which is the mainstay of livelihood for many people in the Eastern Cape Province.[14,16]

At the same time, the long-term nature of the ongoing drought in the Eastern Cape Province of South Africa could signal profound impacts on the lives of the Eastern Cape population. Therefore, it is likely that South African and the Eastern Cape population are interested in drought as a major livelihood impact and a disaster hazard. Based on the information from the previous paragraph, such a long-term nature of disaster impacts could also signal a more prolonged duration of the state of drought-triggered disaster, which impacts the well-being of the population and agricultural production. In the Eastern Cape and similar rural areas, agriculture is the backbone of economic activity in many and areas provinces of South Africa, eg, the Alfred Nzo District Municipality in the Eastern Cape Province (based on the 2011 census data).[17] Therefore, public interest in drought and related topics will need to play an important role in disaster risk management in the country. As shown by Shao et al.,[18] multiple sources of information must be evaluated and used to understand how drought as a slow-onset hazard/disaster is perceived by the affected population. Against this background, the authors look here to use internet search-engine data to examine the trends in the interest in the drought by the South African population for the January 2004-August 2022 time period. The working hypothesis of this article is that there is an increasing interest in drought among the South African population and that the drivers/contributing factors will include topics and elements of climate change adaptation.

METHODOLOGY

Public interest can be examined using surveys and various online tools. One of the most flexible and near-real-time tools is the examination of social media and analysis of online searches.[19] Online and anonymized data can be here of particular interest, eg, the volumes of search engines used to recover information. Kam et al.[20] showed that Google Trends volumes spiked among California residents when the

Governor of California declared a drought emergency in January 2014. In addition, further spikes were recorded when the El-Niño struck and when floods struck California between 2014 and 2017.[20] Google had issues with the email privacy of its Gmail® platform users,[21] but it also contains the most extensive record of search data history from all big tech companies. That record stretches from January 2004 to the present. In addition, Google search engines can be augmented with additional plugins, which can be used to extract more detailed data from Google searches.[22] An example of such a plugin is *Keywordseverywhere.com*, which allows users, eg, disaster risk management professionals, to obtain reverse-extracted values of the total monthly and/or weekly search volumes for a specific search term in a specific geographical area (Burilová et al., 2018; see *https://keywordseverywhere.com/frequently-asked-questions.html*, website accessed on January 8, 2023). Analyzing these data can lead to the examination of the fluctuation and temporal variability in the public interest in disaster information, eg, drought-related topics.

Here, the *Keywordseverywhere.com* plugin was installed on their personal computers by the authors, and the search credits were obtained via the online portal (see *https://keywordseverywhere.com/credits.html* for details, website accessed on January 8, 2023). The authors attempted a quick search for the drought interest among the South African public from 2004 until June 2022, based on the factors from the "Introduction" section.[10] A single keyword of drought was used, and the monthly search volumes for South Africa were extracted for the time period from January 2004 until June 2022. This was done in two ways. First, the data were extracted from the *Keywordseverywhere.com* plugin on three separate occasions between September and November 2022 (referred to as the first data extraction in the further text of the article). Second, one additional dataset of monthly search volumes was extracted, for the January 2004 to June 2022 time period, in January 2023 (referred to as the second data extraction in further text of the article). Search volumes for a particular month were ordered in ascending order from 1 for January 2004 up to 222 for June 2022. The extracted data from the first and the second data extraction were analyzed in the following way. First, the three datasets from 3 to 5 months after June 2022 were analyzed for statistically significant differences using the Kruskal–Wallis analysis of variance by ranks at a 5 percent level of significance (Past 3.0).[23] Then, the analysis was repeated to include all four datasets from both data-extraction occasions. The separate extractions were done to examine whether the time of the data extraction had any impact on the actual values of the monthly search volumes for drought in South Africa. The arithmetic averages and the standard deviations were tested for statistically significant differences between the first extraction and the second data extraction. That analysis was done using the Mann–Whitney test at a 5 percent level of significance (Past 3.0).[23] The search volumes from the first data extraction were then averaged and evaluated for statistically significant differences in the measures of central tendency. That was done against the data from the second data extraction using the Mann–Kendall test at a 5 percent level of significance (Past 3.0).[23]

Next, the authors attempted to obtain a sense of the relative interest in drought and the significance of the interest in South Africa and the Eastern Cape Province of South Africa, which is the area of interest to the authors' research. For this, the authors extracted the Google Trends scores for South Africa and the Eastern Cape Province from *trends.google.com* (see *https://trends.google.com/trends/explore?date=all&geo=ZA&q=drought* for details, website accessed on January 8, 2023). The time period for the Google trend scores was the same as for the *Keywordseverywhere.com* plugin data. Only one data extraction was done from Google Trends, as the point in time of the data extraction was not found to influence the Google Trend scores, contrary to the *Keywordseverywhere.com* data. The Google trend scores were analyzed using the Mann–Kendall test (trends in the Google Trends scores as a function of time) and the Mann–Whitney test (the average Google Trends scores between South Africa and the Eastern Cape). The cities or local municipalities in the Eastern Cape with the highest relative interest in drought by the public were also identified using the municipal-level Google Trends data. Finally, to gain an understanding of the qualitative drivers of the Google searches for drought in South Africa, the following information was also obtained and analyzed using the *Keywordseverywhere.com* plugin: the drought-related keywords, the long-tail keywords similar to drought, and the "people also searched for" category from the South African users.

To provide more insight into the climate change drivers and the potential policy changes/policy dimensions, several additional analyses were performed. The authors attempted to see if there was any increase in the average surface air temperature (ASAT) in South Africa over the study period from 2004 until 2022. For this, data on the ASAT values were extracted from the World Bank Group Climate Knowledge portal.[24] The data were processed for significant trends with time using the Mann–Kendall test at a 5 percent level of significance in the Past 3.0 statistical software package.[23] At the same time, the seasonal number of searches was averaged for the following months in 2004-2020: December-January-February, March-April-May, June-July-August, and September-October-November. Then, the average seasonal ASAT values were extracted from the World Bank Climate Change portal for the 1991-2020 time.[24] Whether the interest of the South African public in drought might have been driven by the increase in the ambient and perceived temperature, then there would be a correlation between the search volumes on Google and the mean ASAT values in individual seasons was investigated by the running Spearman correlation coefficient (see *https://www.socscistatistics.com/tests/spearman/default2.aspx* for details, website accessed on November 26, 2023). The average temperatures and search volumes per season were arranged in this way based on data format and availability. Finally, the policy considerations and dimensions of drought management in South Africa were analyzed using the South African parliamentary briefs and discussion documents. For this, the keyword drought was searched for using the internal search engine of the Parliamentary Monitoring Group (see *pmg.org.za* for details, website accessed on November 26, 2023). The total number of occasions when the drought was discussed in a parliamentary committee or setting in a given calendar year was extracted. This variable was considered a proxy indicator of the intensity of the policy discussion about drought in South Africa. The number of drought-related discussions per year was correlated with the annual search volumes for drought in South Africa, based on Google search data (the yearly volumes were calculated as shown by Tandlich et al.[25]).

RESULTS AND DISCUSSION

The monthly search volumes for drought in South Africa are shown in Figures 1 and 2. For the first data extraction, the distribution of the monthly search volumes for drought had the following measures of central tendency and spread of data (Microsoft Excel, Inc., Johannesburg, South Africa): the arithmetic mean was equal to 6,971, the median was equal to 4,595, the mode was equal to 3,097, and the standard deviation was equal to 10,142. In the second data extraction, the distribution of the monthly search volumes for drought had the following measures of central tendency and spread of data (Microsoft Excel, Inc., Johannesburg, South Africa): the arithmetic mean was equal to 29,908, the median was equal to 17,600, the mode was equal to 17,600, and the standard deviation was equal to 46,817. Results of the Kruskal–Wallis analyses of variance by ranks showed that there was a statistically significant difference between the monthly search volumes for drought in South Africa (p-value = 1.7×10^{-19} for the first data extraction and p-value = 4.7×10^{-14} for the combined data from the first and the second data extraction). The averages or medians of the monthly search volumes from January 2004 until June 2022 were statistically significantly different between the first and the second data extraction (p-value = 0.0001). For the first data extraction, the results of the Mann–Kendall test at a 5 percent level of significance indicated that the monthly search volumes, by the South African population on Google, increased with time between January 2004 and June 2022 (p-value = 0.0092).

For the second data extraction, there was also an increasing trend in the South African population interest in drought, based on data analysis using the

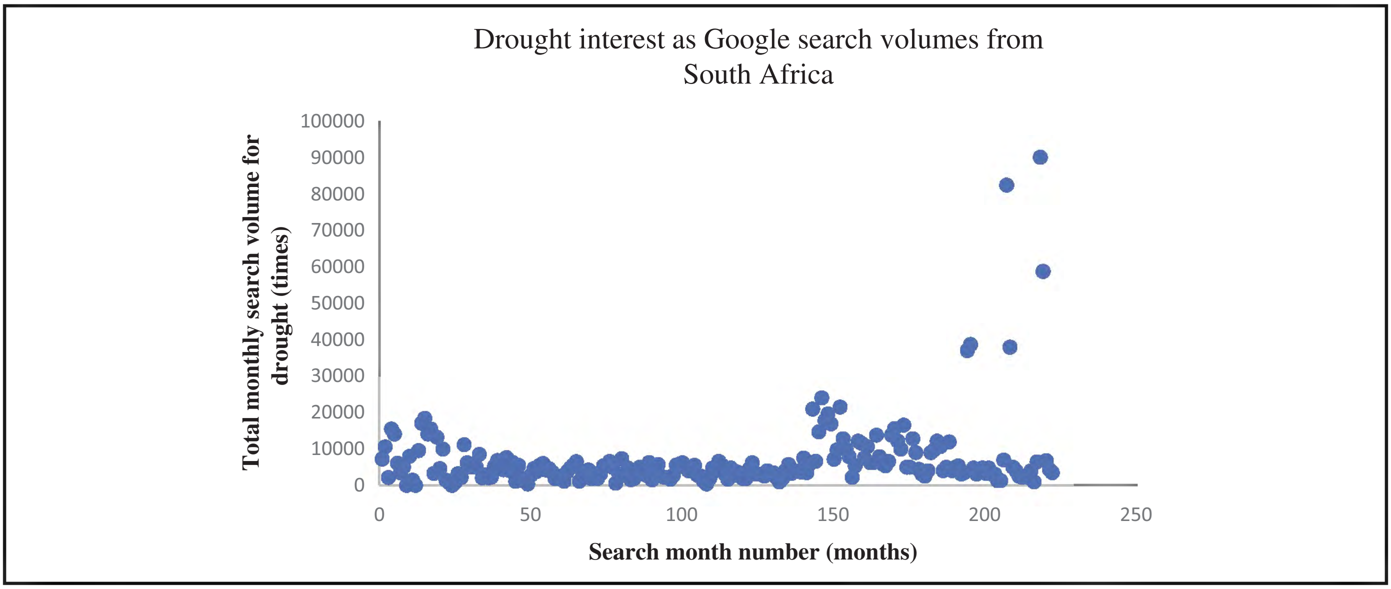

Figure 1. The monthly search volumes for drought in South Africa for the time period from January 2004 until June 2022. These data were extracted between 3 and 5 months after June 2022 (the first data extraction).

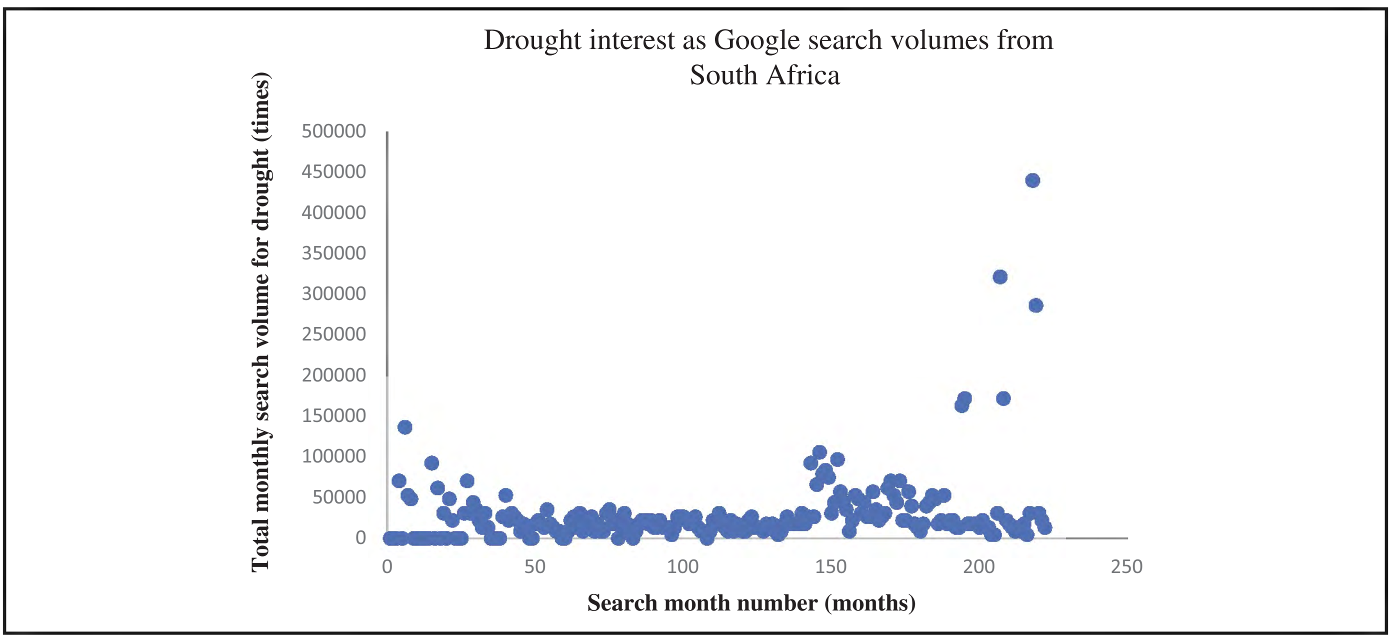

Figure 2. The monthly search volumes for drought in South Africa for the time period from January 2004 until June 2022. These data were extracted 7 months after June 2022 (the second data extraction).

same test as for the first data extraction (p-value = 9.9×10^{-6}). Therefore, the time of the monthly search volume extraction from the *Keywordseverywhere.com* plugin does strongly influence the drought interest by the South African public. These seem to vary by about one order of magnitude. The Google Trends data for the entire South African population are shown in Figure 3. The relative significance of the searches for drought is highly variable on a temporal scale, in comparison to other search terms in South Africa, between January 2004 and June 2022. This trend was, however, increasing with time based on the

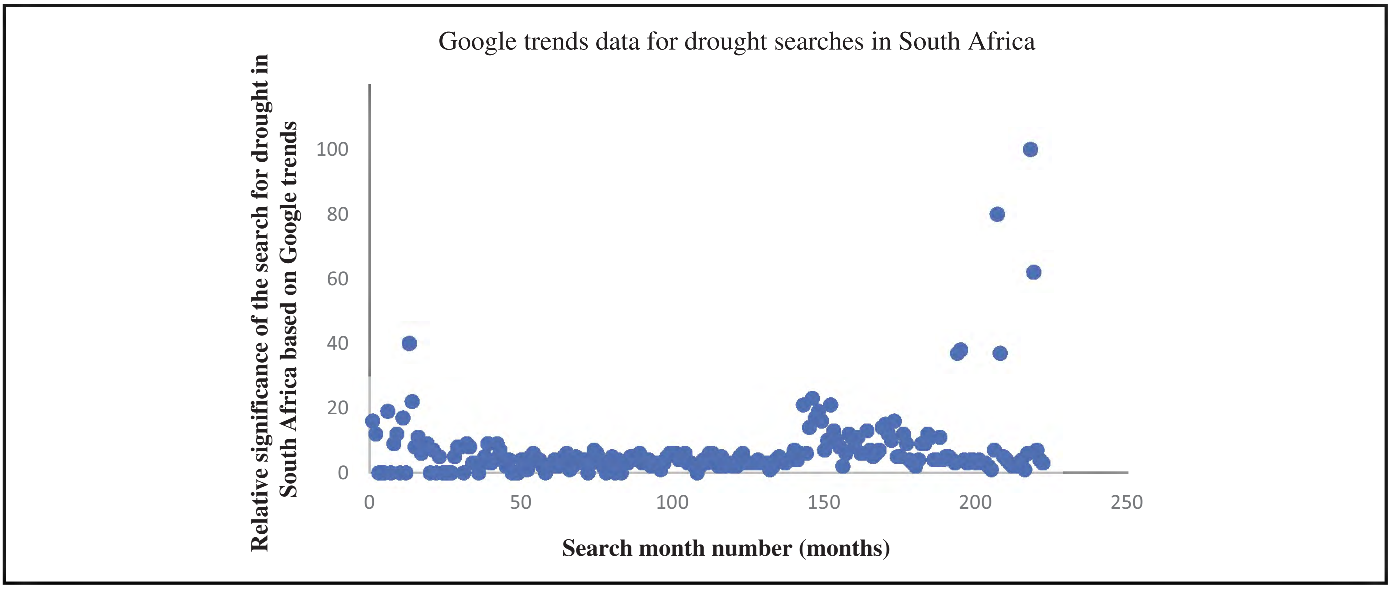

Figure 3. The monthly Google Trends data that indicate the relative significance of the South African population's interest in drought compared to other topics between January 2004 and June 2022.

results of the Mann–Kendall test at a 5 percent level of significance (p-value = 0.0011). For the distribution of the monthly Google scores, the following measures of central tendency and spread of data were obtained (Microsoft Excel, Inc., Johannesburg, South Africa): the arithmetic mean was equal to 7, the median was equal to 4, the mode was equal to 4, and the standard deviation was equal to 11. A Google Trends value of 0 indicates no interest of the South African public in a given term, as it is not searched for in a given month. On the other hand, search terms that capture the majority of the searches or are the most commonly searched for terms in South Africa would have a Google Trends value of 100.[22] Therefore, the South African population's searches for drought indicate that the topic was not a commonly searched term on Google between January 2004 and June 2022.

The relative interest was further investigated in the province of Eastern Cape, South Africa, and the data are discussed in Figure 4 and in the text. In this province, the relative significance of the searches for drought was highly variable with time, in comparison to other search terms, increased with time between January 2004 and June 2022. This is confirmed by the results of the Mann–Kendall test at a 5 percent level of significance (p-value = 0.0007).

For the distribution of the monthly Google scores, the following measures of central tendency and spread of data were obtained (Microsoft Excel, Inc., Johannesburg, South Africa): the arithmetic mean was equal to 3, the median was equal to 0, the mode was equal to 0, and the standard deviation was equal to 12. A Google Trends value of 0 indicates no interest of the Eastern Cape public in a given term, as it is not searched for in a given month. On the other hand, search terms that capture the majority of the searches or are the most commonly searched for terms in the Eastern Cape Province would have a Google Trends value of 100.[22]

Therefore, the Eastern Cape population's searches for drought indicate that the topic was not a commonly searched topic compared to other search terms on Google between January 2004 and June 2022.

The results of the Mann–Whitney test, for any statistically significant differences in the medians at a 5 percent level of significance, indicate that the Google scores were higher for the whole South African population vs the Eastern Cape one (p-value = 0.0001).

The related keywords and the qualitative drivers under the category of "people also searched for," indicated from the *Keywordseverywhere.com* plugin, indicated that the drought interest of the South African public was driven by an interest in the

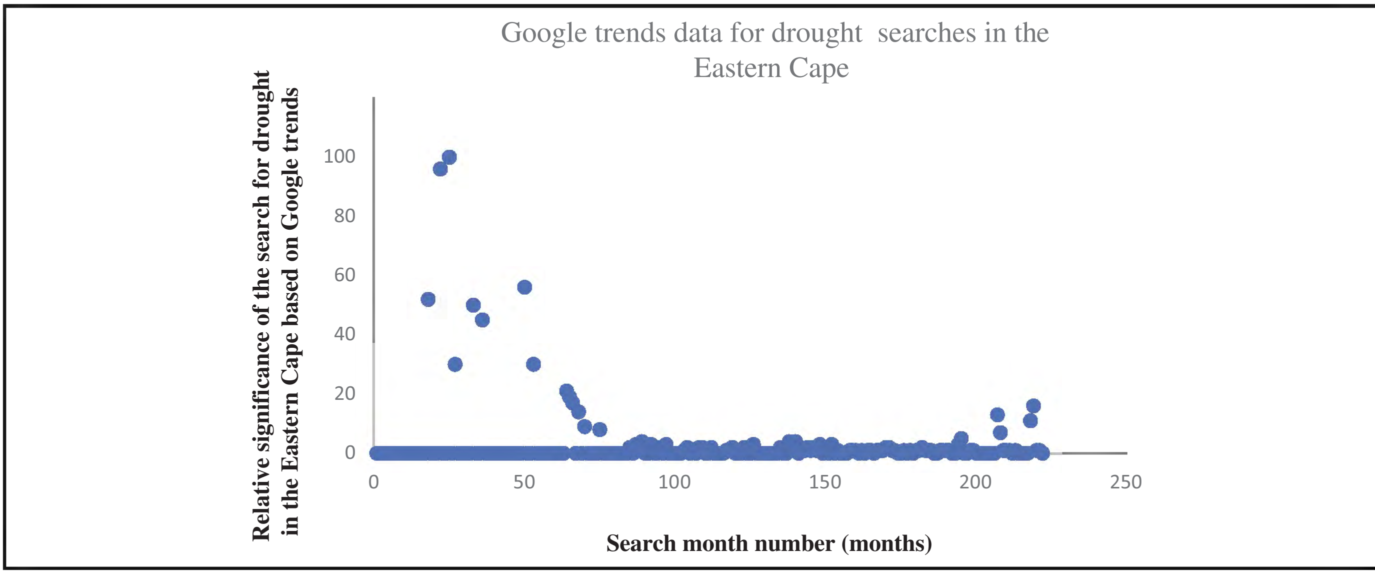

Figure 4. The monthly Google Trends data that indicate the relative significance of the Eastern Cape population's interest in drought compared to other topics between January 2004 and June 2022.

following topics: "types of drought, effects of drought, what causes drought, how to prevent drought, drought map, drought facts, drought pronunciation, what is a drought disaster." This would indicate that the drivers of the drought interest of the South African public were the triggers of the drought, the nature of the drought impacts, and the way that a drought would be declared. The long-tail keywords, which were related to the drought, were as follows: "types of drought, causes of drought, drought definition, drought meaning, drought in Europe, China drought, drought pronunciation, drought California/California drought, drought tolerant plants, drought UK, drought monitor, US drought monitor, lake mead drought, drought Mississippi River, drought meaning in Hindi, Mississippi river drought, Texas drought map, Colorado River drought, Texas drought." The long-tail keywords indicate that interest or related keywords might indicate the following drivers of the South African population's interest in drought. They could be interested in seeking information from the developed world about references for mapping and dealing with drought. The individual topics could also be a sign of the population searching for tools and information to help them adapt to the drought and to increase their individual or household resilience. Taking the Google Trends scores and the qualitative driver analyses together, drought adaptation of the South African community could be considered a marginal topic in Google searches. However, there will be hotspots of interest in drought, ie, these are areas that have a Google trend *g*-score of >25. In the Eastern Cape, those hotspots are likely the following locations/municipalities: Tshikaro (Google Trends score 60), Mafusini (Google Trends score 52), Cofimvaba (Google Trends score 51), and Nxotsheni (Google Trends score 40).

The ASAT values as a function of time between 2004 and 2022 are shown in Figure 5. There was no statistically significant trend of the ASAT with time between 2004 and 2022 (based on the results of the Mann–Kendall test at a 5 percent level of significance with p-value = 0.2623). The temperatures in individual years ranged from 17.97°C in 2006 and 19.27°C in 2019. The ASAT values remained constant over the study period, so it is unlikely that the mean ambient temperature was a driver of the public's interest in drought. The potential link between the average seasonal ASAT values and the average seasonal search volumes for drought in South Africa was analyzed for possible correlation. The graphical representation is shown in Figure 6, and the results of the correlation analysis are presented later. The Spearman

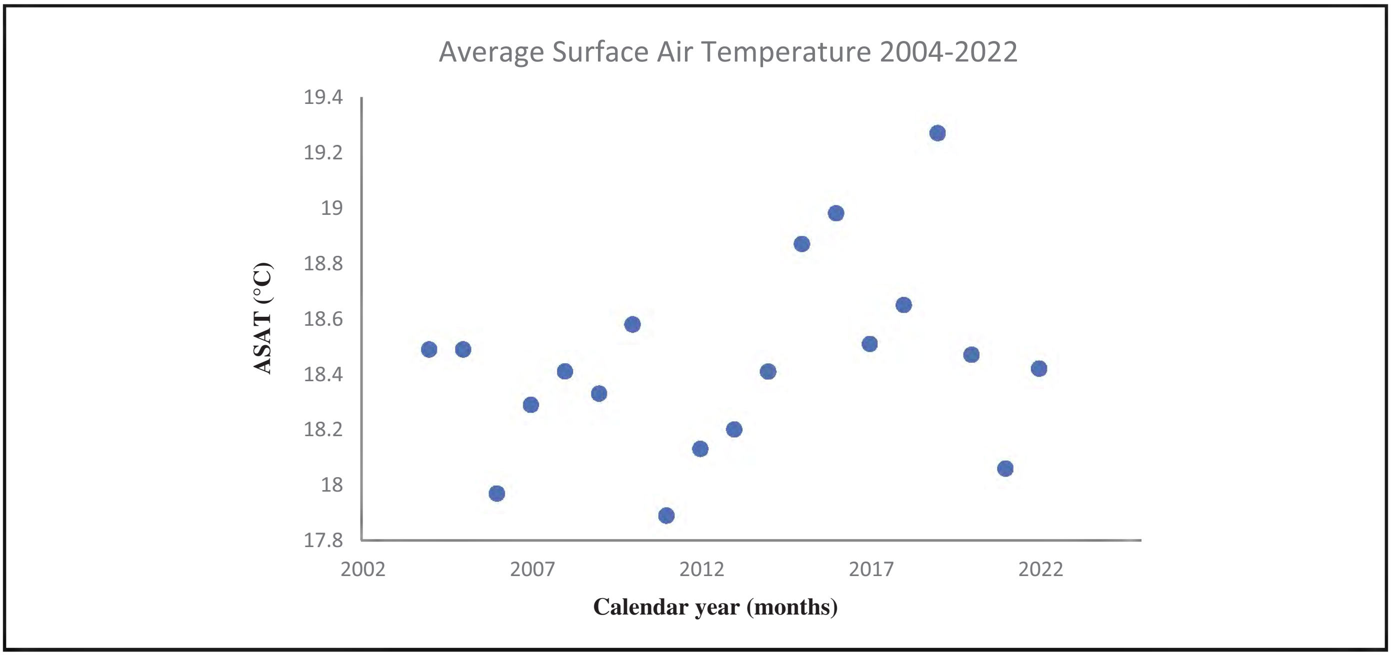

Figure 5. The average surface air temperature (ASAT in °C) in South Africa as a function time between 2004 and 2022.

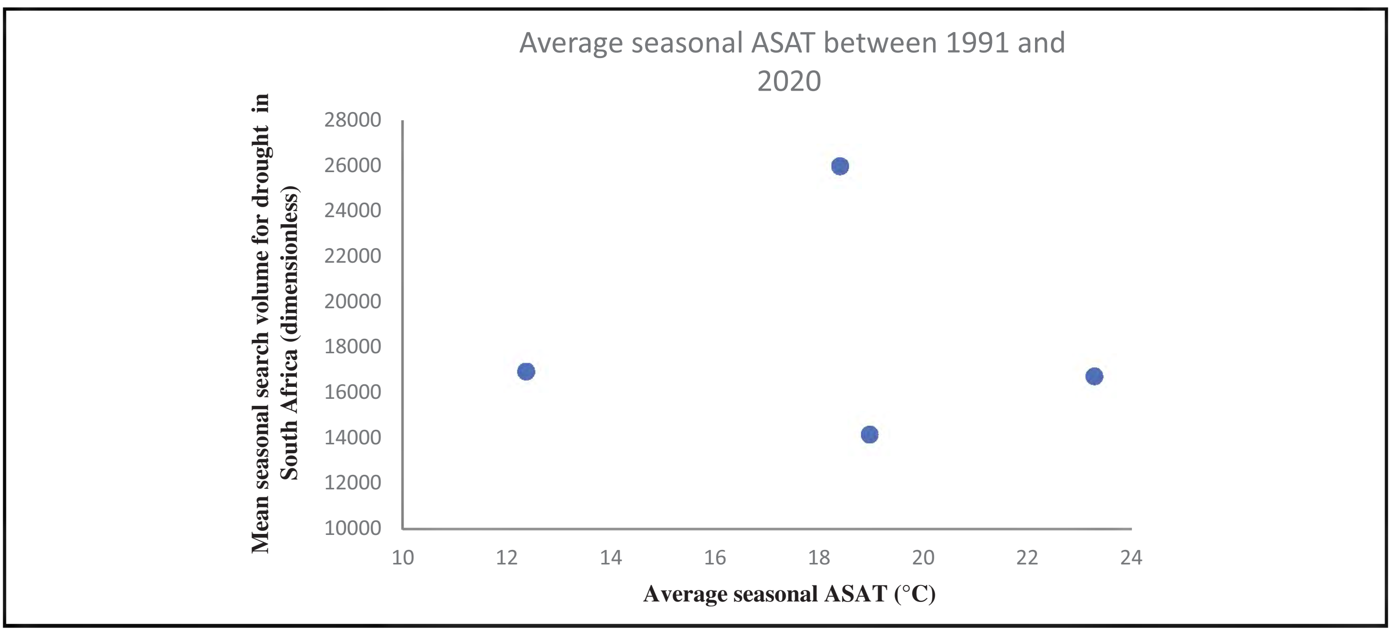

Figure 6. The mean seasonal search volumes for drought from Google in South Africa between 2004 and 2020 are plotted as a function of average surface air temperature (ASAT in °C) for seasons in South Africa as 1991 and 2020.

correlation coefficient for the variables in Figure 6 was equal to –0.6, and the p-value was equal to 0.4, ie, there was no statistically significant relationship between the mean seasonal surface temperature and the search volumes for drought by the South African population on Google. Therefore, the results from this correlation analysis and Figure 6 prove that temperature was not a driving factor behind the South African population's interest in drought, as per the Google search volumes.

The policy discussion intensity, as measured by the parliamentary briefs and meetings focused on drought in South Africa, was assessed by graphical and correlation analysis. The results are shown in Figure 7 and presented later. The Spearman correlation coefficient calculations resulted in the conclusion that the number of parliamentary briefs or meetings focused on drought was not a statistically significant determinant at a 5 percent level of significance, as $r_S = 0.1334$, p-value = 0.5862. It is, however, clear that the number of briefs and meetings increased from 121 to 305 between 2015 and 2019 (the period of the most recent drought).

This was higher than the predrought levels of about 10-40 annual briefs (Figure 7). At the same time, South Africa has made strides to adopt a national drought management plan.[26] Therefore, the long-term and recurrent nature of drought occurrence in South Africa and its impact on the population, along with increasing access to the internet between 2004 and 2022,[27] were likely one of the main driving forces of the observed Google-derived search volumes for drought. The recent policy focus mirrors the significance of drought to South African society at the parliamentary level of discourse.

The trends in the Google-derived data from their search volumes indicate that the interest in the drought increased with time in South Africa. Similar trends were indicated for the Eastern Cape Province, which has been impacted by drought since 2015.[10] An increased level of adaptation to drought has been documented in countries of the Southern Hemisphere, eg, Australia where farmers have adapted to long-term droughts.[28] Self-assessment or self-reporting indicators of drought by the subjective farmer perception have shown a high degree of correlation with standard indicators such as rainfall and soil moisture levels (Hughes et al.; Figure 5).[10] The long-term adaptation would, at least implicitly, indicate ongoing interest in drought, eg, through specific topics such as farm profits and the (mental) well-being of farmers.[28] Such conclusions could be observed in South Africa, especially in agricultural provinces such as the Eastern Cape Province. In South Africa, urbanization and climate change overlap and place very strong requirements and additional performance pressure on local government.[29] It seems that many times crisis management or a reactive approach to drought risk management

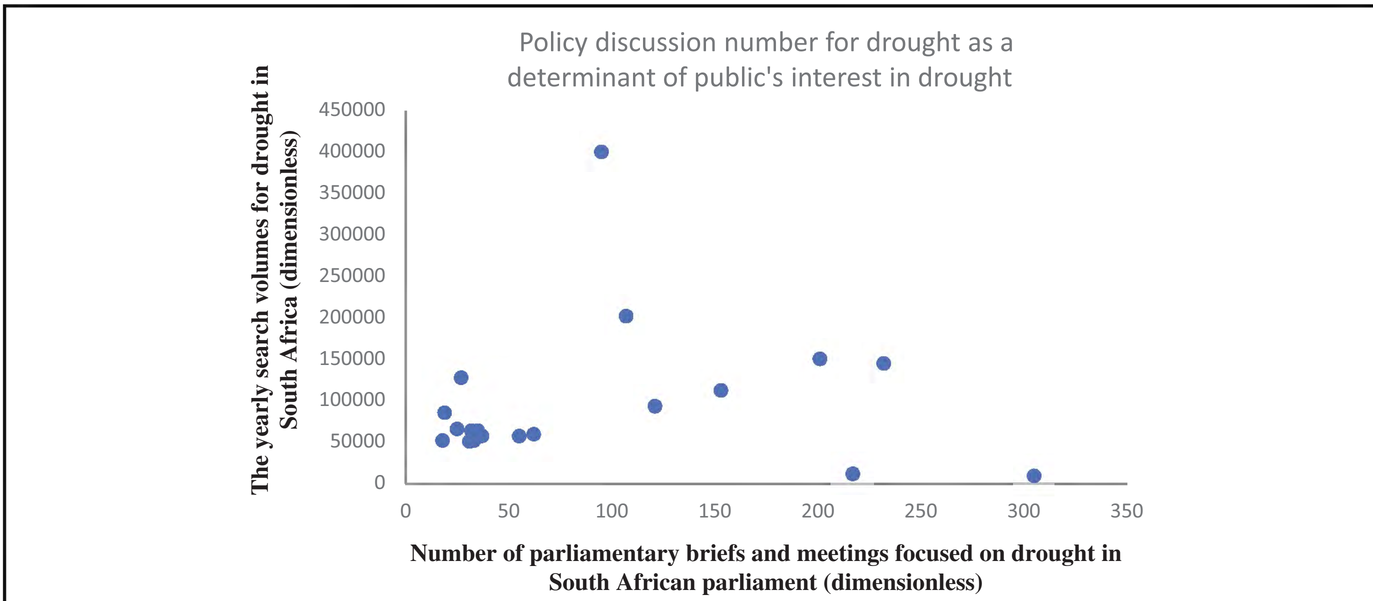

Figure 7. The yearly search volumes for drought from Google in South Africa between 2004 and 2022 and the number of parliamentary briefs and meetings focused on drought for the same time period.

is the modus operandi of government structures.[30] The *Keywordseverywhere.com* plugin does allow for almost real-time evaluation of the public's interest in drought, so proactive and permanent evaluation of this interaction is possible. The software tool is relatively inexpensive, as 100,000 searches are charged at a rate of 160 ZAR or 10 USD (see *https://keywordseverywhere.com/credits.html* for details, website accessed on January 10, 2023). Therefore, the drivers of the public interest can, thus, be evaluated for drought and other topics. At the same time, general temporal trends of this interest and the potential qualitative drivers of the drought interest can be evaluated on an ongoing basis.

The plugin and Google Trends can, thus, be used as a tool to implement, as a rapid evaluation tool to monitor the drivers of public interest in drought, and from a disaster risk management and crisis management point of view. Google Trends is a freely available tool, so qualitative evaluation and the relative evaluation of the interest in drought, as compared to other topics, are possible with just an internet connection. The low-cost plugin *Keywordseverywhere.com* is behind a paywall, but investment in it could be possible and ethically justified. The ethical justification can be based on the ability of disaster risk and crisis managers to target information campaigns, about hazards such as drought, to the specific needs and drivers of the population in a particular geographical area. There is one problem with possibly putting a procurement of the plugin out for a tender, as *Keywordseverywhere.com* is the only such product on the global market at the time of the writing of this article. As of the 2021/2022 financial year, there were 257 metropolitan, district, or local municipalities in South Africa.[31] A grant from the National Treasury could facilitate the purchase of the licenses to all municipalities via a conditional grant.[32]

The findings of this study indicate that drought is a fact of life, and there is marginal interest in it among the South African and the Eastern Cape population. The overall interest of the South African and the Eastern Cape population in drought was found to increase with time for the 2004-2022 period. This correlates with the occurrence of drought in South Africa and the Eastern Cape, indicating that the plugin provides reliable estimates of public's interest in drought. The low relative interest in drought in comparison with other terms, as based on the Google Trends data, might indicate that the drought interest is implicit and might rather be focused on individual terms related to drought, eg, water conservation.[33] The current communication points to the single-keyword paradigm of research that could be seen as a first instance or quick assessment of the public interest. More robust approaches, eg, based on nine locally relevant terms for water, sanitation, and hygiene,[27] might provide a more academic approach. In this way, the variability of the extracted search volumes might be reduced. However, this study provides an indication of the usability of the *Keywordseverywhere.com* as a low-cost and near-real-time tool to assess the public interest in disaster-related topics. The findings will have to be validated by comparing the qualitative drivers and quantitative data with real-life research on drought perceptions of the Eastern Cape population. However, the data seem to be correct as the main hotspot of Google search was in the North to Northwestern part of the Eastern Cape—a highly rural and agricultural area. The findings of this study show the potential for low-cost methods of monitoring data on public interest in disaster and emergency management topics. The methodological approach and the evaluation of results in this study provide a potential near-real-time tool to monitor, evaluate, and adjust/modify disaster risk communication and messaging to the target or affected population.

CONCLUSIONS

This study indicates the usability of *Keywordseverywhere.com* as a low-cost and near-real-time tool to assess the public interest in disaster-related topics. The findings will have to be validated by comparing the qualitative drivers and quantitative data with real-life research on drought perceptions of the Eastern Cape population.

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JEM

Climate change and health: The case of mapping droughts and migration pattern in Iran (2011-2016)

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ABSTRACT

Introduction: *Migration and mobility of population have been reported as a common reaction to drought. There is historical evidence to suggest the health effects of droughts and human migration linkage in Iran. This study aimed to map the drought and migration patterns in Iran in 2011 and 2016 and explore their possible health impacts.*

Methods: *This sequential explanatory mixed-method study was done in two stages of spatial analysis and qualitative study. Data mapping was conducted through the equal interval classification and using drought, migration, and agriculture occupation data based on provincial divisions in Iran in 2011 and 2016. This qualitative study was conducted using the content analysis approach.*

Results: *The in-migration rate was higher in 2011 rather than 2016. Migration to cities was much higher than migration to villages in both years. The frequency of male migrants was higher than females in all provinces in 2011 and 2016. Physical and mental diseases as well as economic, sociocultural, education, and environment effects on health were extracted from the qualitative data.*

Conclusion: *A holistic picture of droughts and migration issues in Iran and their health consequences were achieved by the present research. Further research is needed to explore the determinants of health impacts of climate change in vulnerable groups. Public health problems can be prevented by adaptive and preventive policy-making and planning. This can improve the coping capacity of the population facing droughts and enforced migration.*

Key words: droughts, migrations, geographic information system, climate change, health, Iran;
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INTRODUCTION

Climate change is directly affecting human migration by destroying homes and infrastructure and indirectly by disrupting economic and welfare activities.[1,2] Climate change may increase the likelihood of both internal and international migration due to increasing drought, desertification, and competition for scarce resources.[3] Migration has been used by populations as an adaptive strategy to adverse environmental and climatic conditions or as a forced movement following land degradation and conflict. Change in human migration pattern is considered as one of the potential impacts of climate change on affected communities.[4] In this regard, the World Bank estimated that more than 143 million people will be forced to migrate due to climate change by 2050.[5] The literature suggested a link between the consequences of climate change and migration.[6,7]

Drought is considered as a negative shock affecting livelihood and well-being of people. Migration and mobility of population have been reported as a common reaction to drought.[8] The effects of droughts on migration as one of the important aspects of human population dynamics have been reported in the literature.[9] Migration as an adaptive strategy to cope with the negative consequences of droughts was reported in various regions of the world, especially rural areas of India, Canada, and Africa.[10-13] On the other hand, the impacts of droughts can be magnified by poverty, health issues, political decisions, as well as lack of adaptive capacities.[8]

Drought can exacerbate water- and foodborne diseases, especially in areas where such diseases are endemic.[14] Collective violence and mental health problems can be considered as impacts of climate change on communities, especially vulnerable groups such as children and old and disabled people.[15-17]

The arid and semiarid climates in Iran are characterized by low rainfall and the high potential for evapotranspiration. As a country with arid and semiarid climates, Iran has been hit by abnormal climate conditions.[18] Exacerbated with mismanagements, it was challenged by droughts, floods, shrinking lakes and rivers, and land subsidence. The situation of groundwater resources is also extremely endangered due to overexploitation, which ranks Iran among the world's top groundwater miners.[19]

The effects of climatic events such as droughts on migration of Iranian population have been reported by a few studies.[20-22] For example, about 30 percent of the population migrated out of the rural regions in the southern Iran after the 2008 drought event. In addition, ~1,100 people with a mean monthly income of ~30 United States dollars migrated and annually posed a loss of ~400,000 in the southern Iran after the 2008 drought occurrence.[20] Migrations and droughts were reported as the important factors of unemployment or joblessness in the affected regions, especially more agricultural sector employment.[13] Iran, as a highly drought-prone country, has experienced two major drought events before 2010. The first drought occurred over the whole country between 2000 and 2004, and the second one took place in 2008. Following the droughts, important consequences were observed, including community disintegration and conflicts of water access in rural areas, higher rate of rural unemployment and reduced household income, reduced access to education, psychological and emotional impacts, as well as rural to urban migration.[23,24]

The spatial distribution of population in Iran may be a more concerning challenge rather than its population growth.[25] While there is historical evidence to suggest linkage between climate and human migration in Iran, the issues of drought and migration patterns have been underpinned only by a small body of scientific research. On the other hand, Madani's[26] analysis showed that Iran's water crisis consists of depleting groundwaters, drying lakes, shrinking water supply, and extreme water-related occasions. Iran faces serious challenges regarding both prolonged droughts and floods likewise. The latter alone has affected 11 million people's lives and resulted over 2,600 casualties.[26]

Thus, this study aimed to, first, map the drought and migration events in relation to employment and agricultural occupations in Iran's provinces (2011-2016) and, second, conduct a qualitative study in order to explain the health effects of droughts and migration on the affected people.

METHOD

Design

A mixed-method sequential explanatory design was applied for conducting the research. This design consists of two phases: quantitative followed by qualitative. At the first stage, the quantitative data are gathered and analyzed. Then, the researchers collect and analyze qualitative data in the sequence to help explain the quantitative findings obtained from the first stage.[27] Accordingly, the quantitative stage consisted of mapping migration (migration to provinces) and drought based on provincial divisions in Iran in the years of 2011 and 2016. Then, the qualitative phase was designed to explain the health effects of droughts and migration on the affected people using the conventional content analysis approach.

Study area

This study was conducted in Iran. Iran, one of the Middle East countries, is located in the arid and semiarid belt. Iran is divided into four areas: Caspian, Central Plateau, Zagros, and the Southern coastal plain.[28] Drought is experienced by various regions. Iran may encounter a difficult situation due to severe rainfall fluctuations and scarcity of water resources.[29]

Quantitative phase: Drought and migration mapping

Database generation. Drought data (Drought Severity Index) were obtained from the Iranian Meteorological Organization as well as the National Research Center of Drought. Furthermore, the migration data were collected from the Statistical Center of Iran and the Ministry of Cooperative Labor and Social Welfare. Since the national census is carried out every 5 years in Iran, migration data in 2016 were the most recent ones to be applied in the current research. Datasets were created in Excel® sheets and analyzed by ArcGIS software at the next step. All drought and migration data were entered and categorized based on provinces for both time series.

Data analysis and mapping. In this stage, data analysis, data zoning, and mapping were conducted by exploring the relationship between migration and the drought phenomenon. For this purpose, the map of Iran was coded separately for each time, and five types of codes including very high, high, medium, low, and very low were assigned for the drought data. Then, the numerical values of migrations based on gender and agriculture occupation were entered into the information layer of the map for each province. The quantity symbology and drought code were used separately in order to zone and prepare the drought map in each time series, respectively, with separate color assigned to each type. Chart symbology was used for the migration data. Furthermore, the assigned values were entered on maps for both male and female migrants. The ArcGIS software version 10.2 was applied for geographic data analysis.

Qualitative phase. The qualitative stage was conducted to explain the direct and indirect effects of droughts on the affected communities.

Participants. Participants were selected using the purposeful sampling method. Farmers and scholars who had the knowledge and experiences regarding droughts and health were approached for the interview. Farmers were considered due to their personal experiences on droughts' effects. Experts and scholars of climatology, those who had knowledge on health in disasters and public health, and a 1-year experience of research and scientific works in the fields of droughts and migration were considered for interviews. The number of participants was determined based on achieving data saturation, which means no new concepts emerged by further interviews.

Data collection. Data were collected using in-depth semistructured interviews. Data saturation was achieved after six interviews, and one interview was conducted to assure no new concepts emerged. All interviews were recorded by the voice recorder and immediately transcribed verbatim. Each interview lasted between 30 and 120 (average 90) minutes. Confidentiality and the right to decline the interview were stated before interviews.

Data analysis. Data collection and data analysis were conducted simultaneously. Data analysis was done using the inductive content analysis approach. Transcription was performed immediately after each interview session, and the whole text was read several times to extract the meaning units. The condensed meaning units were labeled as codes. Afterward, the codes were categorized into subcategories and categories based on their similarities and differences.

Trustworthiness. Data trustworthiness was determined based on credibility, confirmability, dependability, and transferability.[30] Credibility was achieved by member check and peer check. Member check was done by sending a summary of initial results to the participants to check whether results were compatible with their experiences. To conduct the peer check, the initial codes and categories were examined by independent researchers. To ensure confirmability and dependability, authors included different perspectives using the triangulation method. Data transferability was achieved by providing details of the research process.

RESULTS

Quantitative phase: Mapping droughts and migrations

The total number of provinces affected by severe droughts reduced from 19 provinces in 2011 to 13 provinces in 2016. The provinces having low or very low droughts increased from 1 province in 2011 to 10 provinces in 2016. Tehran province, the capital of Iran, had the highest rate of migrants, and Ilam province had the lowest number of migrants in the years of 2011 and 2016. The rate of migrations into the western provinces of Iran, which were affected by low or very low droughts, increased in 2016 when compared to 2011 (Figures 1 and 2).

Gender

Based on the gender-disaggregated data, the frequency of male migrants was higher than females in all provinces in both 2011 and 2016 years. Tehran province was the first destination for male and female migrants, and Ilam province had the lowest rates of male and female migrations in 2011 and 2016.

Employment

About 12.16 and 12.42 percent of populations who could be economically active were unemployed in 2011 and 2016, respectively. Tehran province had the highest numbers of employed populations, and at the same time, it was the first destination for in-migrant populations in 2011 and 2016. On the other hand, Ilam and Semnan provinces had the highest rates of unemployed populations and the lowest numbers of in-migrant people in both 2011 and 2016.

Agriculture

Several provinces located in the northwest of Iran had a growth in the agricultural occupation along with the decrease in severe and very severe droughts in 2016. On the other hand, occupations in the agricultural sector were found to be increased in several

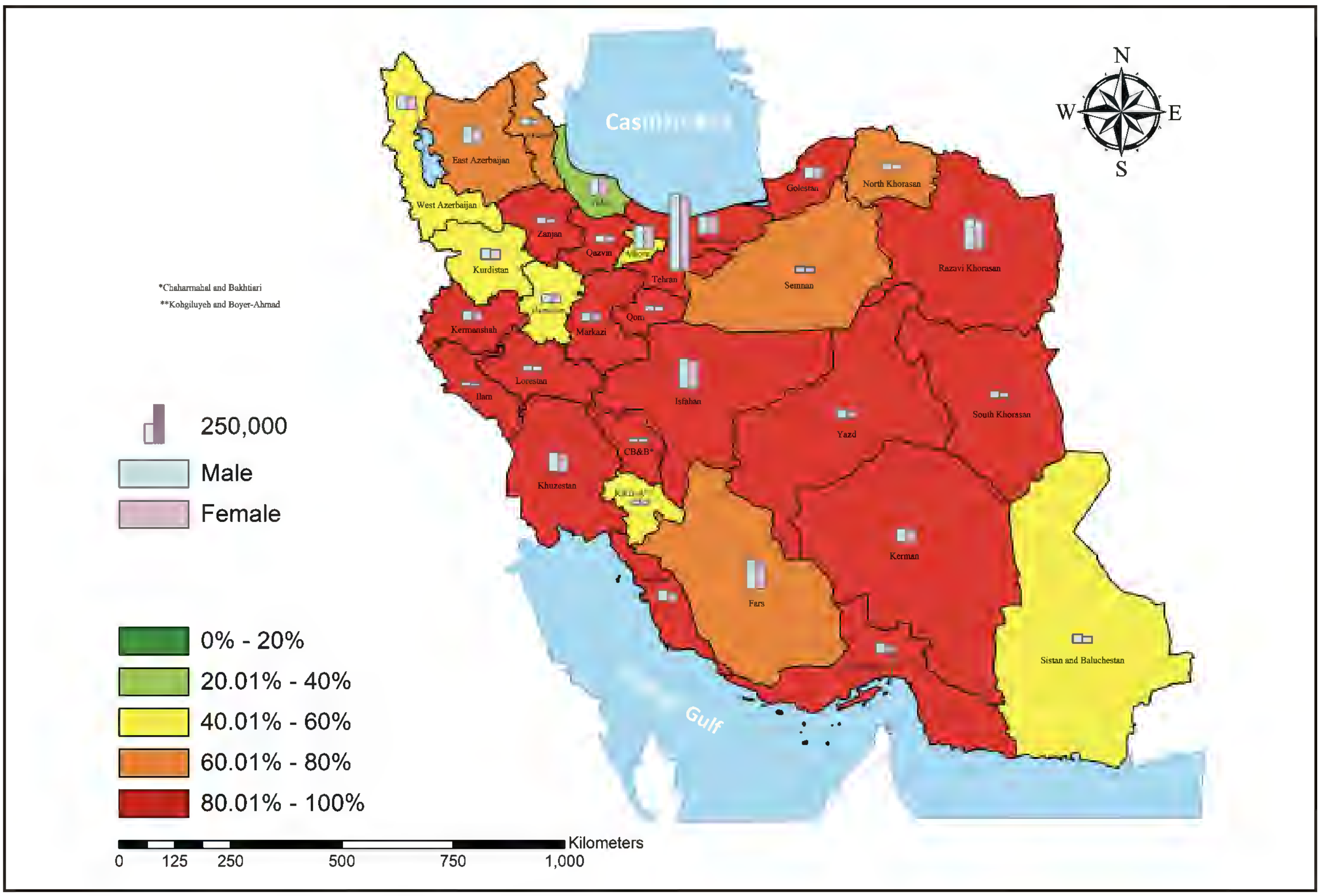

Figure 1. Drought and migration in Iran, 2011.

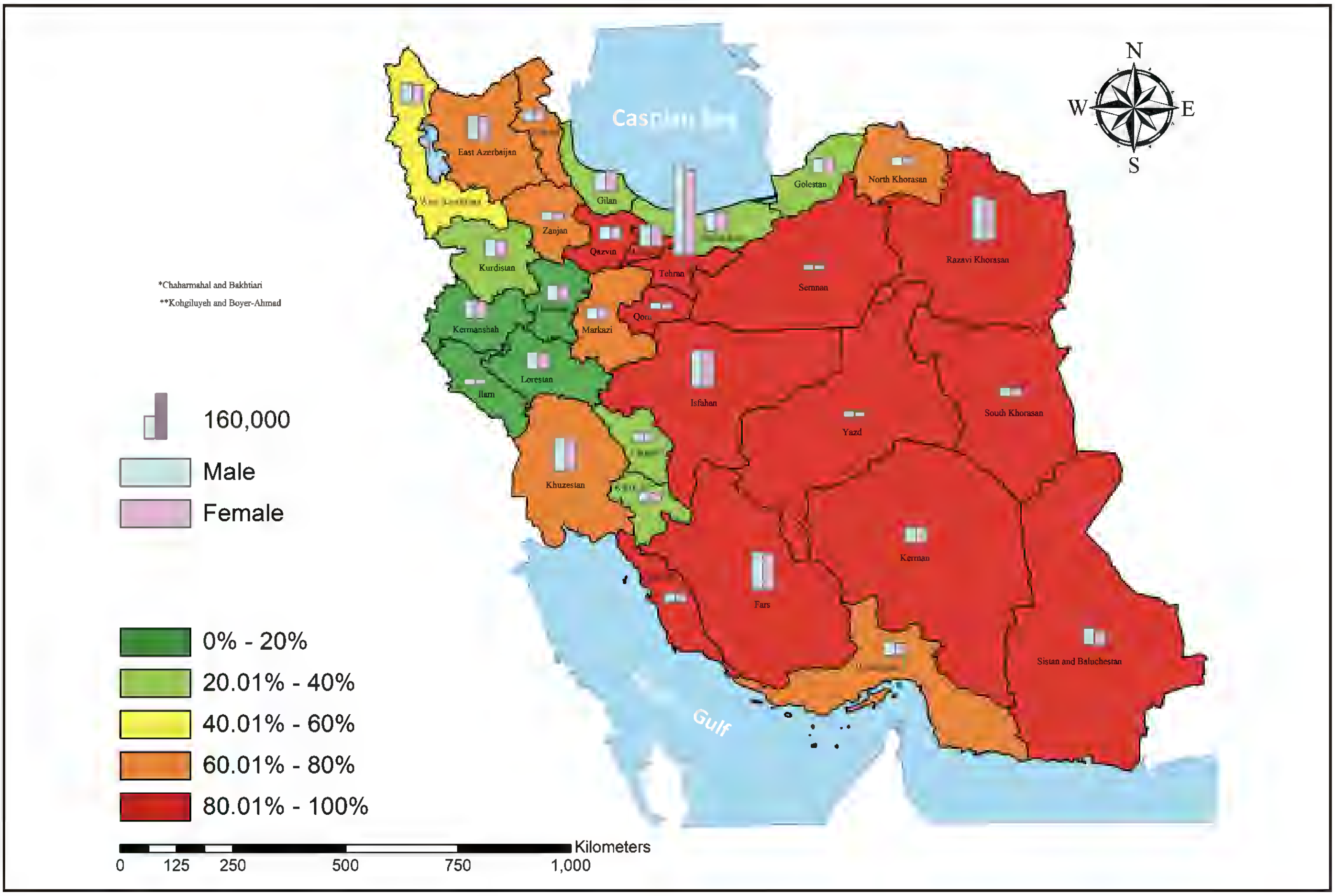

Figure 2. Drought and migration in Iran, 2016.

provinces, eg, Tehran, Isfahan, and Alborz, which had a reduction in their in-migration rates between 2011 and 2016. The provinces of Razavi Khorasan and Fars had the highest rates of agricultural sector occupation as well as in-migrations in both 2011 and 2016 (Figures 3 and 4).

Qualitative phase. A total of seven participants were interviewed. Of these, 71 percent were male, and the remaining were female (29 percent). The educational levels of participants included diploma and academic education (Table 1).

A total of 180 primary codes were identified during data analysis. Two categories of direct effects of droughts on health (subcategories of physical and mental diseases) as well as indirect effects of droughts on health (subcategories of economic, sociocultural, educational, and environmental factors) were extracted from data (Table 2).

Direct effects on health

Physical and mental diseases were two subcategories extracted from data. Direct health impacts of droughts on people living in drought-affected areas can force their migration and threaten the public health status of communities.

Most participants stated different forms of physical diseases, including diarrhea, due to the droughts' consequences in the affected regions. Diseases were chronic and acute ones, which sometimes require urgent control and management. On the other hand, mental health problems were experienced by people affected by droughts, especially farmers who lost their crops and livestock. Mental health disorders

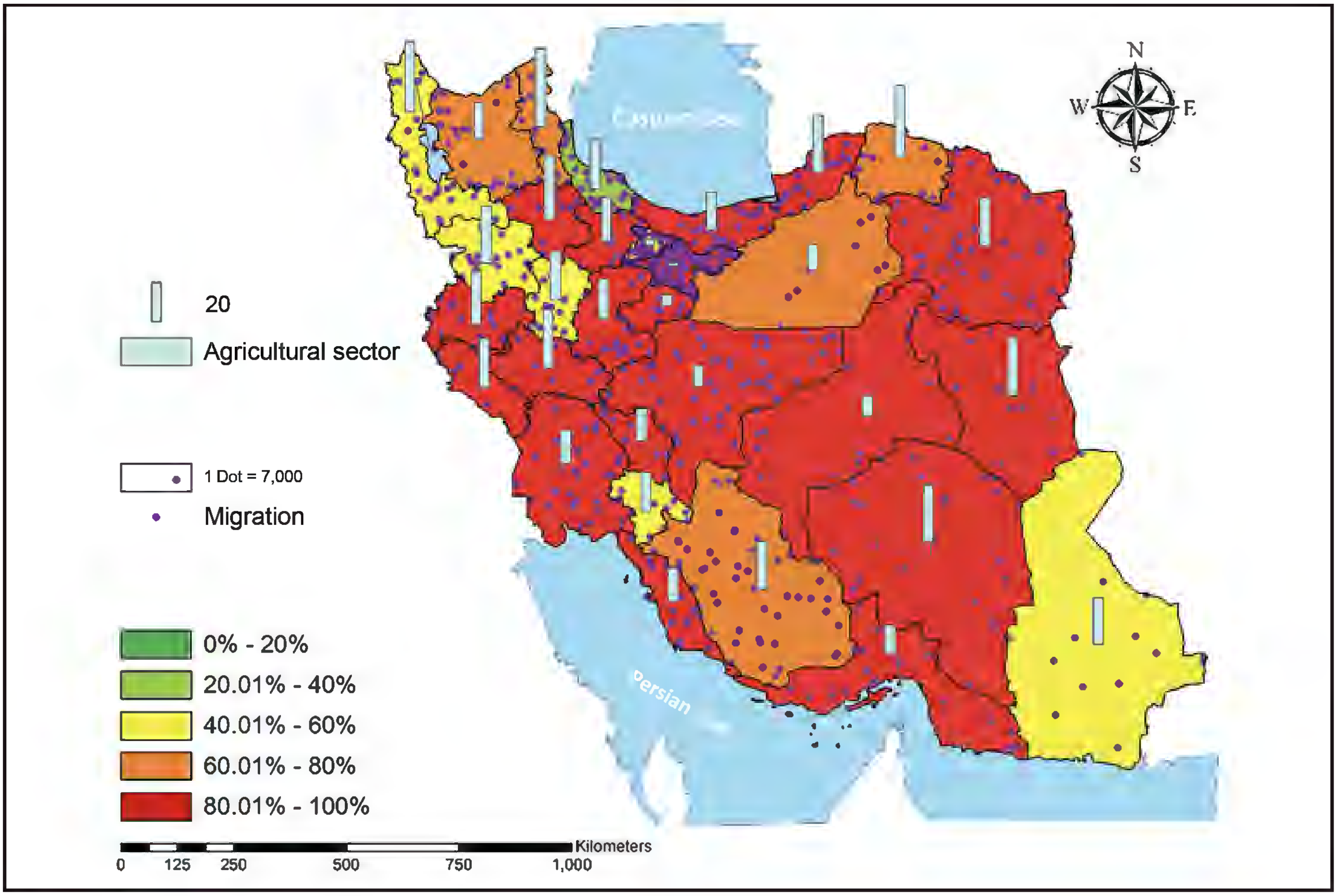

Figure 3. Drought, migration, and employment in agricultural sector, 2011.

such as stress, anxiety, and depression need to be considered through psychological plan and consultant in the drought-stricken regions as well. A participant stated:

> There were many destroyed families. The addictions happened. The murder happened in Ivaneki because the most important thing is water. The conflicts on water have lasted and intensified for a long time caused a lot of murders and injury.

Indirect effects on health

According to the participants' experiences, the subcategories of economic, sociocultural, environmental, and educational factors resulting from droughts can indirectly influence the health status of communities affected by droughts. Such factors can be possibly considered as the main causes of migration as well.

Participants stated that economic challenges caused by droughts resulted in poverty, higher prices for water, and leaving the current agricultural job. Furthermore, sociocultural impacts of droughts were described as the important indirect effect of droughts on the population's health. Marginalization, crime, conflict, divorce, as well as inappropriate quality of life were the reflections of droughts' effects, which can lead to forced migration and finally affect people's health. Based on the participants' experiences, forced migration and poverty resulting from droughts can limit people's access to educational facilities such as schools and universities. Public health education can be negatively affected by droughts and then migration phenomenon. Lack of access to health education and training can increase the prevalence of communicable diseases

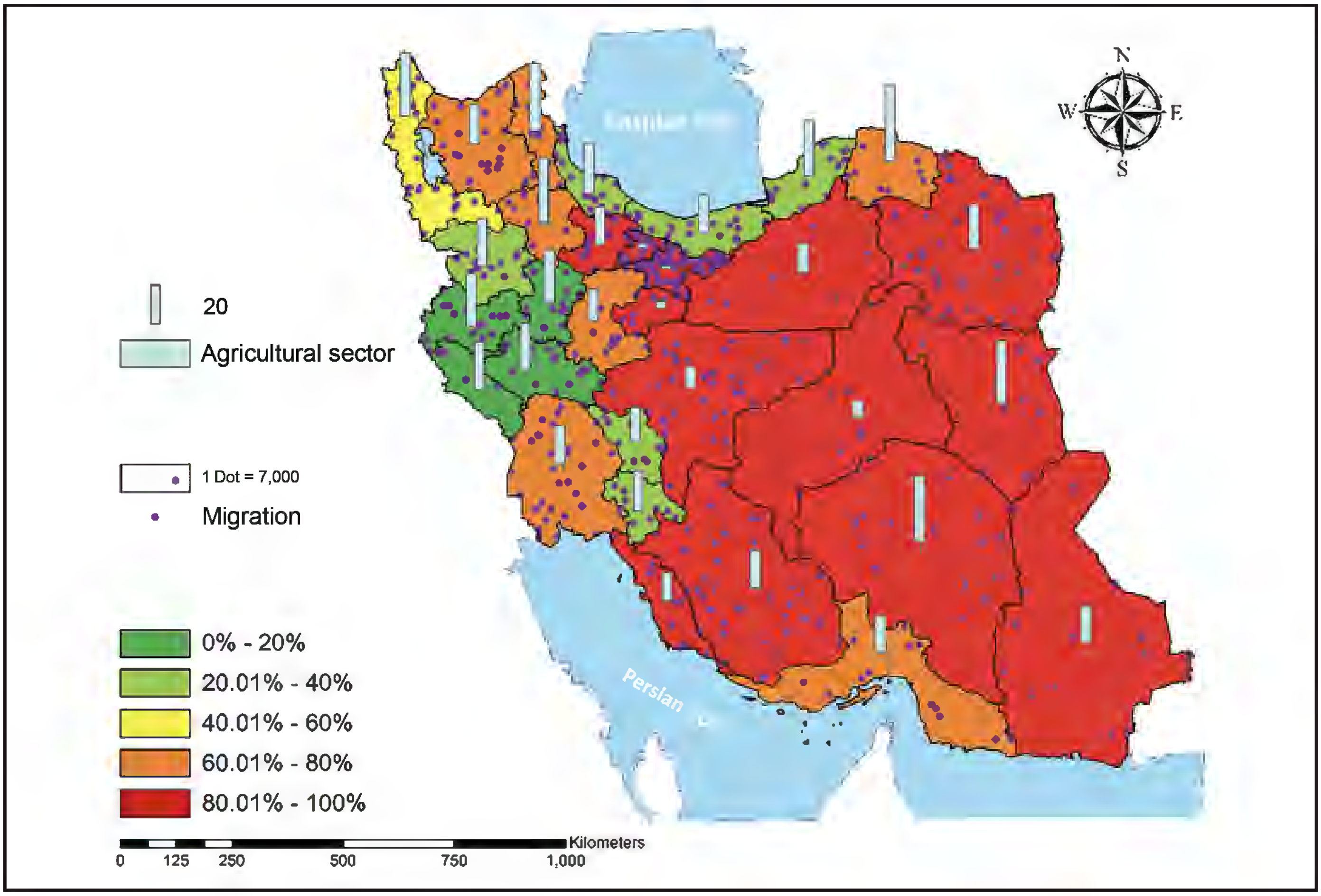

Figure 4. Drought, migration, and employment in agricultural sector, 2016.

Table 1. Demographic characteristics of participants

Variables		Frequency	Percent
Gender	Male	5	71
	Female	2	29
Education	Diploma	2	29
	Bachelor and master	2	29
	Doctoral	3	42
Age	35-44	4	58
	45-55	2	29
	>55	1	13

Table 2. Categories and subcategories extracted from data

Category	Subcategory
Direct effects on health	Physical diseases
	Mental diseases
Indirect effects on health	Economic factors
	Sociocultural factors
	Educational factors
	Environmental factors

in the affected regions. The participants mentioned that the environmental effects of droughts indirectly influence the population health. For instance, water pollution can cause vector-borne and water-borne diseases such as malaria and diarrheal diseases. Such challenges can be the triggers of forced migrations due to serious health problems. A participants stated:

> When the level of environmental pollution increases, causing the increase of entering particles into lung and finally cancer, which is one of the diseases caused in drought-affected regions.

DISCUSSION

According to the findings, agriculture occupation had a growth in the northwest of Iran in parallel with the decrease in severe and very severe droughts in 2016. In addition, in-migration was reported in the provinces with the highest rates of agricultural sector occupation in 2011 and 2016. In-migrant populations increased in provinces with high employment opportunities, eg, Tehran. The findings of qualitative study explained that the droughts and migration can have direct and indirect effects on communities' health in Iran.

Based on the findings, in parallel to the increase in provinces with low or very low droughts in 2016, the rates of in-migrations into these provinces increased in 2016 when compared to 2011. Similarly, the evidence showed that climate change is a very influential factor on rural migration.[20] A significant association between drought and migration was reported in Northern Ghana, Africa. About 51 percent of the population who experienced migration mentioned drought as the important cause of their migrations.[31] In Iran, a positive relationship between rural youth migration and drought consequences was reported.[32] Droughts and migrations are important events, which closely relate to each other in Iran and in other similar contexts.

Consistent with our findings, the study of climate-induced migration in Iran between 1996 and 2011 showed that the migration pattern was significantly affected by levels of annual precipitation and temperature.[33] The study of climate change and migration in the US revealed that education, healthcare, work, and family visits were primarily mentioned as the migration drivers rather than climate change.[34] Thus, further studies need to be defined to investigate all determinants of internal migration patterns focusing on droughts in Iran.

Based on the findings, the provinces with the highest rates of unemployment had the lowest numbers of in-migrations in 2011 and 2016. Accordingly, previous research on climate-related migrations showed that migration was used as an adaptive decision for people who were looking for appropriate livelihood status.[11,34,35] Migration was chosen as an adaptive strategy by populations living in the regions suffering from the low rates of employment and tried to cope with their livelihood and economic issues.[36]

Our findings suggest that provinces that had a growth in the agricultural sector benefited from the decrease in very severe droughts in 2016, and some provinces with high rates of agriculture sector occupation had high in-migration. This is in line with similar research, which reported that drought influenced the employment rates in the agricultural sector.[37] To this end, the government with medium- and long-term planning can reduce the geopolitical challenges caused by the effects of climate change in these areas.[38] It can be discussed while Iran has the rural populations as an important agriculture-dependent group, the climate-related events such as droughts can play major roles in migration patterns.[33]

Based on our findings, drought and migration can directly and indirectly affect the health of people in the community. In some cases, drought effects are the combination of droughts and the risk of infectious diseases, especially water, food, humans and livestock, and disease.[39] Similarly, in developing countries, young children and infants are at higher risk of malnutrition and mortality, suffering from deteriorating livelihood conditions and finally getting more vulnerable to drought and climate-driven migration.[40,41] It seems that both direct and indirect effects of droughts and migrations on health have insufficiently been investigated. Lack of long-term mitigation and adaptation policy-making can worsen the negative effects of droughts on people's health in Iran.

LIMITATIONS

In-migration data were collected and published by the Statistical Center of Iran, and the out-migration data were not available to be included in our research.

Furthermore, health-related data were not available for mapping and analyzing along with droughts and migration. Furthermore, the qualitative study was a small context-based research to explain the health effects of droughts on health. As with any qualitative study, the results of this part may not be applicable to other contexts or disasters.

CONCLUSIONS

The drought and migration data were adjusted and mapped in the present study. Accordingly, migration cannot be considered as a simple response to a singular climatic event. Climate change directly and indirectly affects people's health, especially where communities influenced by forced migration. Although a picture of climate-migration issues in Iran was illustrated by the present research, exploring the determinants and consequences of climate-related migrations is highly suggested. Public health problems can be prevented by adaptive and preventive policy-making and planning in the drought-affected regions.

The cooperation and collaborations of the Ministry of Health with climate management sectors are recommended to mitigate the health effects of droughts. Further research is needed to explore the determinants of health impacts of climate change in poor and vulnerable groups. Socioeconomic factors affecting the population mobility and migrations can be explored by qualitative studies.

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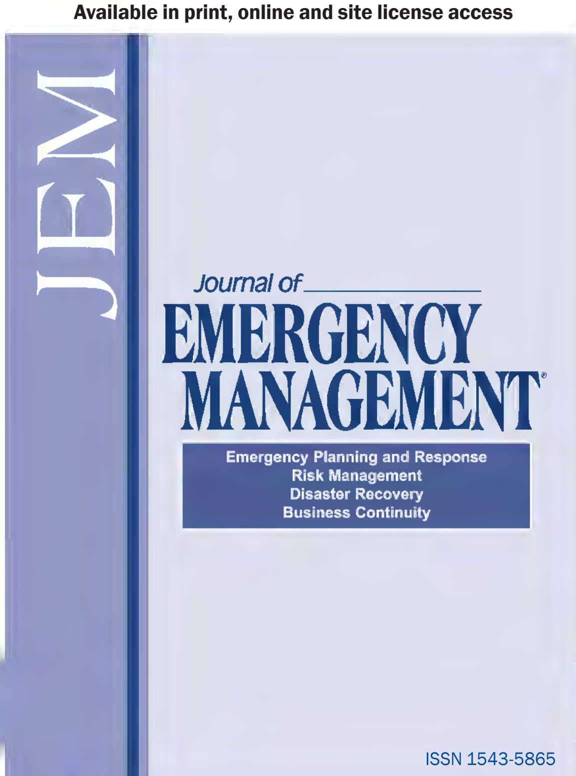

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